

Directed Energy Deposition Simulation Guide

Ansys
ANSYS, Inc.
Southpointe
2600 Ansys Drive
Canonsburg, PA 15317
ansysinfo@ansys.com
http://www.ansys.com
(T) 724-746-3304
(F) 724-514-9494

Release 2026 R1
March 2026

ANSYS, Inc. and ANSYS Europe, Ltd. are UL registered ISO 9001: 2015 companies.

Copyright and Trademark Information

Copyright © 2026 Synopsys, Inc. and ANSYS, Inc. This Synopsys product and all associated documentation are proprietary to ANSYS, Inc. and Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement. All other use, reproduction, modification or distribution of the Synopsys product or the associated documentation is strictly prohibited.

ANSYS, Ansys Workbench, AUTODYN, CFX, FLUENT and any and all ANSYS, Inc. brand, product, service and feature names, logos and slogans are registered trademarks or trademarks of ANSYS, Inc. or its subsidiaries located in the United States or other countries. ICEM CFD is a trademark used by ANSYS, Inc. under license. CFX is a trademark of Sony Corporation in Japan. All other brand, product, service and feature names or trademarks are the property of their respective owners. FLEXlm and FLEXnet are trademarks of Flexera Software LLC.

Disclaimer Notice

THIS ANSYS SOFTWARE PRODUCT AND PROGRAM DOCUMENTATION INCLUDE TRADE SECRETS AND ARE CONFIDENTIAL AND PROPRIETARY PRODUCTS OF ANSYS, INC., ITS SUBSIDIARIES, OR LICENSORS. The software products and documentation are furnished by ANSYS, Inc., its subsidiaries, or affiliates under a software license agreement that contains provisions concerning non-disclosure, copying, length and nature of use, compliance with exporting laws, warranties, disclaimers, limitations of liability, and remedies, and other provisions. The software products and documentation may be used, disclosed, transferred, or copied only in accordance with the terms and conditions of that software license agreement.

ANSYS, Inc. and ANSYS Europe, Ltd. are UL registered ISO 9001: 2015 companies.

U.S. Government Rights

For U.S. Government users, except as specifically granted by the ANSYS, Inc. software license agreement, the use, duplication, or disclosure by the United States Government is subject to restrictions stated in the ANSYS, Inc. software license agreement and FAR 12.212 (for non-DOD licenses).

Third-Party Software

See the legal information in the product help files for the complete Legal Notice for ANSYS proprietary software and third-party software. If you are unable to access the Legal Notice, contact ANSYS, Inc.

Published in the U.S.A.

Table of Contents

1. Introduction to DED Additive Manufacturing 1
2. Implementation of DED in the Mechanical Application 3
2.1. Methodology and Abstractions 4
2.2. Known Issues and Limitations 6
2.3. High Performance Computing 9
3. Preparing the Part for Simulation 11
3.1. Cleaning Up Facets in an `.stl` File 11
3.2. Creating a Part from a Tool Path File: Path to Part 11
3.2.1. Path-to-Part Workflow 12
3.2.2. Path Tool Options 17
3.2.3. Path-to-Part Tool Options 18
3.2.4. Contact Tool Options 19
3.2.5. Using the Script Editor for Customization and Troubleshooting 20
3.3. Slicing the Part for Manual Clustering 21
4. Using the DED Process Wizard 23
5. Workflow Through the Project Tree 27
5.1. Create the Analysis System 27
5.2. Attach Geometry and Launch the Mechanical Application 28
5.3. Insert AM Process for DED and Identify Geometries 30
5.4. Assign Materials 32
5.5. Apply Mesh Controls and Generate Mesh 38
5.6. Create Named Selections for Contact and Clustering 40
5.7. Define Build Settings 57
5.8. Perform Element Clustering 59
5.8.1. Element Clustering: Manual Approach 60
5.8.2. Element Clustering: G-Code Approach 62
5.8.2.1. G-Code Clustering for Multi-axis Printing 78
5.8.2.1.1. Example 78
5.8.3. Editing Individual Clusters for More Precise Control of Process Settings 83
5.8.3.1. Using the Cluster Settings Table 84
5.8.3.1.1. Type and Format of Table Data 85
5.8.3.1.2. Modifying Table Data 87
5.8.3.1.3. Importing and Exporting Table Data 89
5.8.3.2. Editing the Cluster Output File Directly 89
5.8.3.3. Controlling Heating and Cooling During the Build via Parameters (Command Snippets) 90
5.9. Define Connections 91
5.10. Define DED Process Steps 94
5.11. Establish Thermal Analysis Settings 94
5.12. Apply Thermal Boundary Conditions 94
5.13. Solve the Transient Thermal Analysis 95
5.14. Establish Structural Analysis Settings 97
5.15. Apply Structural Boundary Conditions 98
5.16. Solve the Static Structural Analysis 99
5.17. Review Results 99
6. Supported G-Code Commands 103
7. Object Reference 109
7.1. AM Process for DED 109
7.1.1. Details 110

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

7.1.2. Tree Dependencies 111
7.1.3. Insertion Methods 111
7.1.4. Right-click Options 111
7.1.5. Additional Related Information 111
7.2. Build Settings 111
7.2.1. Details 113
7.2.2. Tree Dependencies 116
7.2.3. Insertion Methods 116
7.2.4. Right-click Options 116
7.2.5. Additional Related Information 116
7.3. G-Code Clustering 116
7.3.1. Details 117
7.3.2. Tree Dependencies 120
7.3.3. Insertion Methods 120
7.3.4. Right-click Options 120
7.3.5. Additional Related Information 121
7.4. Manual Clustering 121
7.4.1. Details 121
7.4.2. Tree Dependencies 122
7.4.3. Insertion Methods 122
7.4.4. Right-click Options 122
7.4.5. Additional Related Information 122
7.5. Cluster Settings 122
7.5.1. Details 124
7.5.2. Tree Dependencies 124
7.5.3. Insertion Methods 125
7.5.4. Right-click Options 125
7.5.5. Additional Related Information 125

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Chapter 1: Introduction to DED Additive Manufacturing

Additive manufacturing (3D printing) can be a cost-effective way of producing parts, especially when making use of the design freedoms the manufacturing process enables, such as topological complexity and the ability to print assemblies in one step.

Metal additive manufacturing is used to produce parts for aerospace, automotive, medical, and other industries. These are high-value parts that require careful design and manufacturing, and simulation has long been used to validate the as-built part performance. But the additive process for metals introduces inherent complexities and challenges such that the process itself requires simulation to successfully produce the parts.

Directed Energy Deposition Processes

In a directed energy process (DED)—also known as laser engineered net shaping (LENS), electron beam additive manufacturing (EBAM®), wire arc additive manufacturing (WAAM), or laser deposition technology (LDT)—a laser or electron beam creates a melt pool on previously solidified material where blown powder or fed wire is introduced to add material.

Similar to a powder bed fusion (PBF) process, the DED process produces high temperatures and severe thermal gradients, leading to significant overheating, distortion, and buildup of residual stresses as the layers are deposited. The distortion can be high enough to interfere with the application of the next layer, and the residual stresses high enough to break the part off the build plate or crack the part itself. Additionally, the residual stresses will produce more distortion when the part is removed from the build plate leading to an undesirable final shape.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information
of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Chapter 2: Implementation of DED in the Mechanical Application

Target Users

Target users of additive simulations in the Ansys Mechanical™ application are the engineers involved in the design and analysis of mechanical components, but also manufacturing engineers and technicians tasked with printing the parts on the machine floor as well as the R&D researchers responsible for optimizing the ideal printing process. Current users of the Ansys Discovery™ and Mechanical applications will benefit greatly from running DED process simulations if they plan to use DED additive manufacturing to print their metal parts.

Simulation Goals

The goal of a DED process simulation is to predict the macro-level, temperature-induced distortions and stresses in parts to prevent build failures, and provide trend data for improving designs for additive manufacturing including part orientation and part build order.

The simulation is not meant to provide detailed thermal or structural results needed for prediction of micro-level process phenomena (that is, microstructure). The simulation will also not provide detailed modeling of the welding melt pool. Our complementary offering of Ansys Fluent™ is the product to use to achieve that goal.

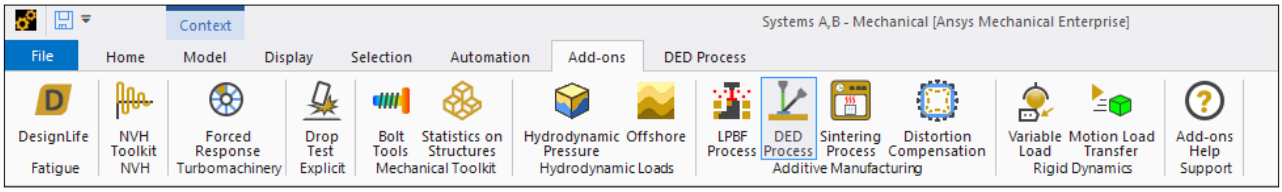
Loading the DED Process Add-on

DED process simulations in the Mechanical application are performed using the DED Process Add-on, referred to as an ACT extension in previous releases.

Important:

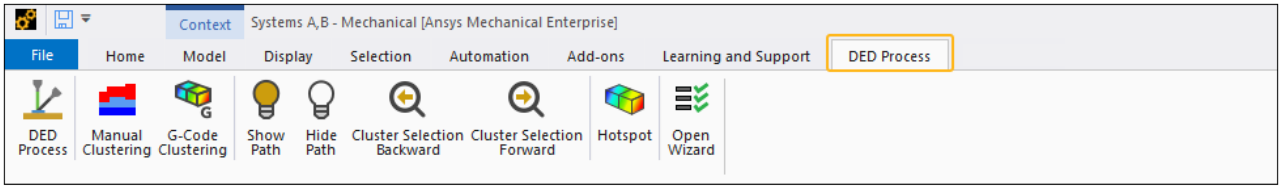
An Additive Suite license is required to use the DED Process Add-on.

Click the Add-ons tab to show the available Mechanical add-ons. Simply click the DED Process button (in the Additive Manufacturing group) to load the DED Process Add-on. When the add-on is loaded, the button is shaded blue and a DED Process tab is available, as shown in the following figure.



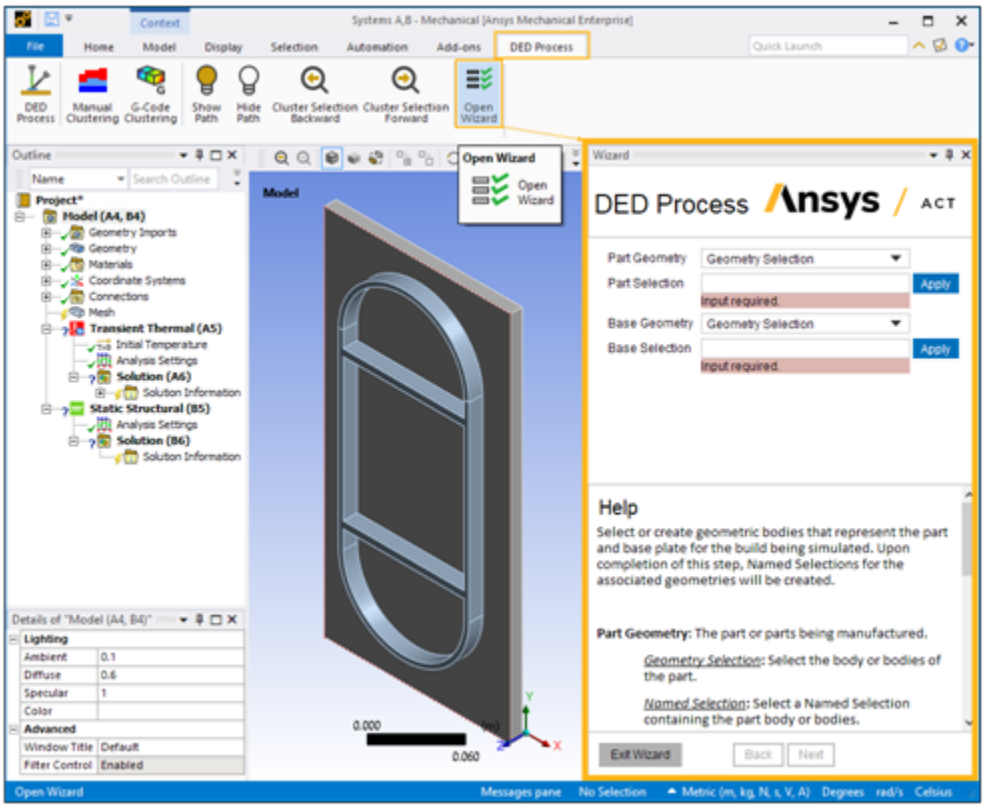
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Click the DED Process tab to access the custom ribbon, which includes a step-by-step wizard as well as custom load and result objects. These and other features will be discussed throughout this guide.



The DED Process Wizard

Included within the DED Process Add-on is an easy-to-use DED Process Wizard (p. 23). We strongly recommend you use the wizard, as it streamlines and simplifies the simulation process. The wizard is an excellent tool to keep you on track and prevent you from missing a task. There are a few capabilities that are not available in the DED Process Wizard but you can easily modify your simulation set-up when you complete the wizard as it does not include a solve.



2.1. Methodology and Abstractions

Simulation of the Direct Energy Deposition (DED) manufacturing process requires that the analysis follows the print process itself: weld track-by-weld track solidification of the part. Since the thermal (temperatures) and structural (distortion and stress) physics are largely uncoupled (that is, a weak coupling), we can simulate the thermal phenomena first and use those temperature results in a following structural simulation.

In a DED process simulation, the model evolves over time, that is, elements are added. We actually mesh the entire part first with Cartesian or Tetrahedron elements and then, using the standard **element**

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

birth and death technique, deactivate all the elements initially and then progressively activate element clusters to simulate the advancing build. Here, a *cluster* is defined as a portion of the weld track. Furthermore, the relevant boundary conditions such as thermal convection surfaces also evolve. The build process reaches completion when all of the element clusters have been activated (made "alive"). See Understanding Element Birth and Death in the *Advanced Analysis Guide*.

The analysis times and time stepping are also driven by the process parameters and are not known a priori. These details are all handled internally during the solution.

Although a DED process and an LPBF process are similar in some ways, the DED process needs to have a much more detailed level of simulation. A layer-wise simulation is not accurate enough since the velocity of energy deposition is higher in a DED process and so simulation along the weld track over time is required. As the LPBF process simulation uses abstractions like super layers (lumping multiple layers for simulation) and layer-by-layer addition (instantaneous element activation for an entire layer), the DED process simulation uses the real welding seams and an abstraction known as element clustering. Clustering is used to split weld lines into smaller pieces of mesh, called *clusters*, that are exposed to one temperature in a time step. Thus, each cluster is a portion of a weld track that is instantaneously activated using the element birth and death technique. Element clusters are deposited following the deformed shape beneath them rather than in the original, undeformed location. You control the size of the cluster with a Cluster Volume (p. 59) option. The solve time and solution fidelity are directly related to cluster volume. The building order of the part can be defined by either applying the G-Code machine file or by using a manual approach of carefully-defined named selections.

A DED process simulation uses the following abstractions to meet the goals in a reasonable compute time, meaning much less than the actual build time:

- **Macro-level welding simulation**: The welding melt pool will not be explicitly modeled. The deposited material, whether it is a wire or powder, will be exposed by activating individual clusters of elements at a defined temperature equal or close to melting temperature. Phenomena such as phase transition will be captured with the capabilities of engineering data or imported materials rather than simulating these phenomena.

- **Clustering**: Clustering is used to split weld lines into smaller pieces of mesh, or clusters of elements, that are exposed to one heat application in a time step. This approach is required to provide the required accuracy while, on the other hand, limit the simulation time. You define the cluster volume to control the simulation.

- **Heat Application**: Heat is applied to new clusters in one of two ways, either an applied *temperature* or a *power-based heat generation*. For applied temperature, the assumption is that the process parameters for the build have been set appropriately so that (1) the developed temperature is always at or above melt temperature (no lack of fusion) and (2) the developed temperature does not substantially exceed melt temperature (no keyholing).

  For power-based heat generation, the laser power and absorptivity are specified and a heat generation load (via **BFE**,,HGEN) is applied to elements in the new cluster. The value for HGEN is determined based on how much energy (power, time for deposition, and so on) goes into each cluster. That energy is then applied over a time span represented by t_a in the following equation:

$$HGEN = \frac{P\eta}{dt_a}$$

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

where P is laser power

η is absorptivity

d is deposition rate

t_a is simulation loading applied time. The time is controlled indirectly using the Thermal Calibration setting (p. 113):

- If Thermal Calibration is Off, t_a=$1e^{-6}$seconds.
- If Thermal Calibration is On, the Cluster Cooling Time Ratio is defined as γ and t_a=$\frac{V}{d}(1-\gamma)$ seconds. V is the volume of the current cluster.

- **Time step size**: Large integration time step sizes are used throughout the simulation. This is sufficient to capture the induced thermal and plastic strains driving the distortion. The localized smooth heating and cooling curves will not be captured in detail.

For a better understanding of some of these concepts, review the information on what happens during the transient thermal solution, under Solve the Transient Thermal Analysis (p. 95).

2.2. Known Issues and Limitations

Review the following issues and limitations for the DED Process Add-on:

General Simulation Setup Limitations:

- The DED Process Add-on is not supported on the Linux operating system.
- The DED Process Add-on requires units to be in millimeters. The add-on checks the unit system and, if necessary, automatically changes it to the Metric system (mm, t, N, s, mV, mA). This check is triggered when you open the DED Process Wizard (p. 23) or when you add the AM Process for DED object into the project tree (p. 30).
- If you use a Mechanical Model from the Workbench application's Component Systems, you must manually create the AM DED Process system. This is due to the fact that AM Custom Systems are pre-populated with AM sample materials in Engineering Data, and linking to the Mechanical Model requires the absence of materials in Engineering Data.
- The DED Process Add-on does not account for symmetry so the entire geometry must be included in the simulation model.
- In order for DED simulations to solve using Remote Solve Manager (RSM), you must first enable the appropriate files to be downloaded. Add the file types *.comps;*.par;*.rth in Solve Process Settings under Advanced > Additional Files to Download. Enter the file types as shown, with semicolons and no spaces between file types.
- When resuming/restoring an archived DED project created with a version of the Mechanical application prior to Release 2023 R1, you will need to clear generated cluster data, generate clusters (p. 59) again, and then solve the simulation anew, even if you don't modify any inputs. This is because, beginning with Release 2023 R1, additional inputs are required on the clustering object and Cluster Settings (p. 84) is automatically added as a child object.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

- DED simulations archived from release 2022 R1 (when the DED extension was first introduced) are incompatible with later releases. We recommend that you start over with those models in a later release.

G-Code Clustering Limitations:

- When using G-Code clustering (p. 62), the placement of the G-Code path must align with the printing part's global coordinate system. We assume a planar tool path in the X-Y, X-Z, or Y-Z plane, normal to print direction.

- A known issue exists in which results may be invalid when G-Code clustering does not follow the G-Code path. Always check the progression of element clusters after performing clustering to be sure the clusters follow the G-Code path as expected. For more information about this limitation, see Known Issue where G-Code Clustering Does Not Follow the G-Code Path (p. 74).

- The G-Code-Reader does not consider the values of the G-Code E-Command. (See Supported G-Code Commands (p. 103)). Only E0.0 is interpreted as no material deposition. The actual value of the material deposition rate is defined in the Build Settings (p. 57).

- Do not change names of element clusters that are automatically generated using G-Code clustering (p. 62). Doing so may cause errors during solution.

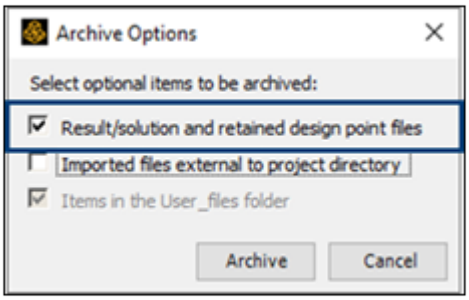
- **Use of Scripting for Additive Manufacturing Add-Ons**: The Additive Manufacturing add-ons (extensions) rely on internal, unpublished APIs that are subject to change between releases. As a result, scripting for these extensions is **not supported**. Although some journal operations may appear to work, subsequent edits or scripted workflows may fail due to internal implementation changes.

  This limitation applies to the following add-ons: **PBF Process**, **DED Process**, **Sinter Process**, and **Distortion Compensation**.

Manual Clustering Limitations:

- We recommend using hex elements rather than tetrahedrons when using manual clustering (p. 60). For some geometries, some tetrahedrons may be excluded from element clusters.

- When archiving a DED project and you used manual clustering, check the option to archive result files with the archived project:



- If you change the mesh size and remesh after manually generating clusters, the existing cluster named selections are suppressed. However, re-generation of the clusters and named selections does not delete previously suppressed named selections, causing an issue when solving the

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

thermal simulation. The workaround is to delete the suppressed named selections manually before running the solve.

- For the very specific workflow related to suppressing a body, the transient thermal portion of the simulation is unable to solve and an error about cluster settings is produced. The workflow is:

  1. Generate clusters for a Manual Clustering object
  2. Suppress a build body
  3. Clear Generated Data on the Manual Clustering object
  4. Generate clusters again on the Manual Clustering object

  The workaround is to clear Generated Data from the AM Process for DED (p. 109) object rather than the Manual Clustering object.

Loads, Solution, and Results Limitations:

- At the assign materials step (p. 32) while working through the project tree, if you leave the Reference Temperature for the print/part geometry set to By Environment, or set to it By Body but leave the Reference Temperature Value at the default of 22°, you may see odd behavior in the structural results, such as missing elements. We recommend that you set Reference Temperature to By Body and set the Reference Temperature Value to the melting temperature of your chosen material. When you use the DED Process Wizard, the Reference Temperature Value is automatically set to the melting temperature of the material, or 1000°C if the melting temperature is not defined for the assigned material.
- A bolt pretension boundary condition is not supported in the DED add-on and will result in an error in the structural solution.
- We recommend you do not add extra thermal boundary conditions/loads on the model other than radiation and the convection boundary conditions described in this user guide—these are, a) the convection from the part to the surrounding gas in the chamber during printing and cooldown that is set in Build Settings (p. 57) and b) the preheat on the underside of the base. Adding extra loads may cause errors, non-convergence, temperatures to be held at unrealistic values, or other unforeseen behaviors.
- Occasionally you may see unexpected, high negative temperatures in the Transient Thermal analysis in the first layer between the part and the base plate. This is a known issue related to severe thermal gradients. These temperature spikes may be reduced somewhat by changing from quadratic to linear elements and/or reducing the element size for the part and/or base plate. These negative temperature spikes may be ignored as they do not have any appreciable effect on the results.

  To change from quadratic to linear elements, re-mesh the model with linear elements or use the **EMID**,REMOVE,ALL command to remove midside nodes from the existing model. A new solution is required.

  Starting with Release 2023 R1, default contact options are set such that these high temperature spikes are usually eliminated.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

- Restarts are not supported in the DED Process Add-on.

Important:

Additional limitations may be identified at the time of release that are not included in this published documentation. Refer to the *Known Issues and Limitations* document for the specific release you are working with. In addition to reviewing AM-specific issues, be sure to review the Mechanical APDL limitations and the Mechanical application limitations.

2.3. High Performance Computing

An AM Process Simulation can be very compute intensive. We recommend you use high performance computing using an Ansys HPC license to take advantage of more than four cores (default).

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information
of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Chapter 3: Preparing the Part for Simulation

There are several Design for Additive Manufacturing (DfAM) considerations to be aware of that affect the cost and quality of your part. Variables include:

- Part design (thicknesses, overhangs, edges, holes, desired surface finish)
- Orientation on the base plate (affects time to print and number of replicate parts that can be nested on a base plate)

The following topics are available:

3.1. Cleaning Up Facets in an .stl File
3.2. Creating a Part from a Tool Path File: Path to Part
3.3. Slicing the Part for Manual Clustering

3.1. Cleaning Up Facets in an `.stl` File

If you have an `.stl` file, be sure to clean it up to eliminate gaps and slivers before importing it into Workbench or the Mechanical application.

In the Ansys Discovery™ application:

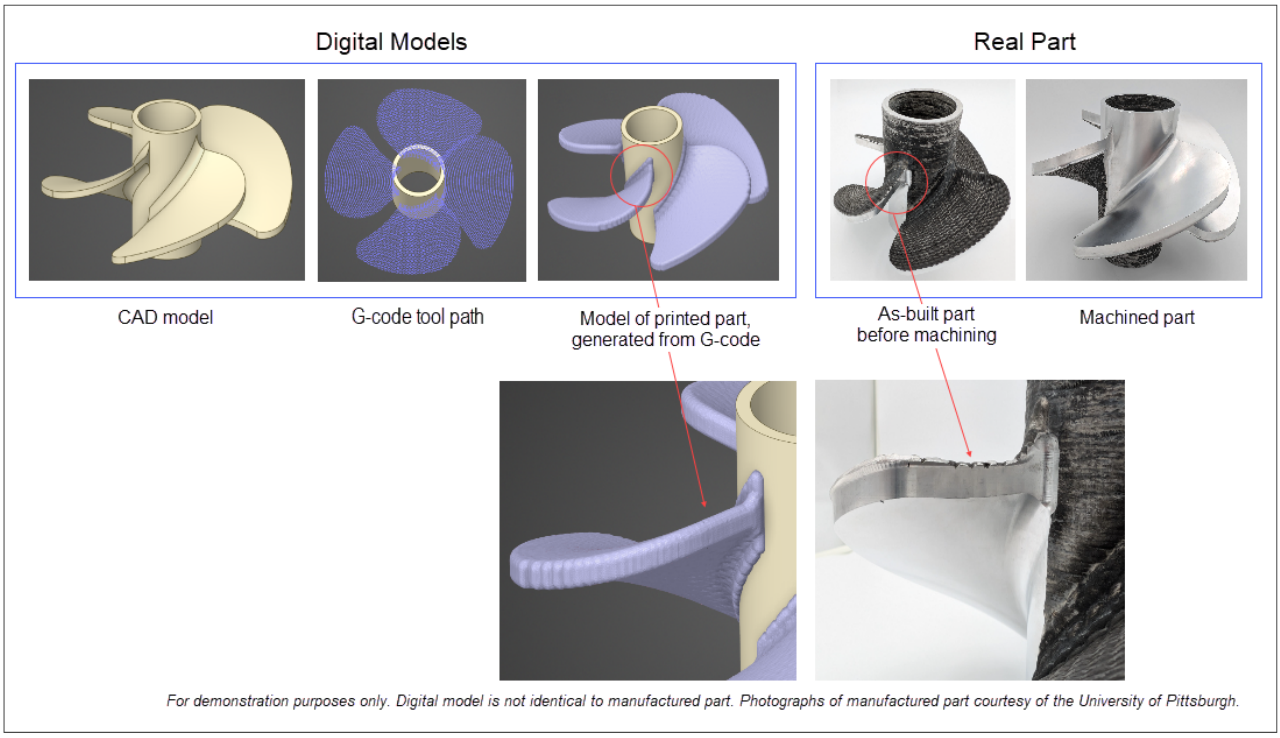
- Under Facets, the Check Facets and the Auto Fix tools are quick ways to find and fix problems in faceted bodies.
- Use Facets > Thickness to identify thin-walled features.

Many additional tools for working with faceted parts are available in the Discovery application. Refer to the Facets section in the Discovery Documentation.

3.2. Creating a Part from a Tool Path File: Path to Part

You can generate geometry models directly from a tool path (such as G-code) for your simulations. This method offers more accurate views of print characteristics, like deposition volume and geometric feature details, compared to using nominal CAD designs. The Additive DED add-in in the Discovery™ application provides a "Path to Part" feature that quickly generates a path-based model with minimal clicks. This model is then automatically attached and prepared for simulation in the Mechanical application.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



The following topics describe the workflow and specifications for the Additive DED add-in's Path-to-Part feature:

3.2.1. Path-to-Part Workflow

3.2.2. Path Tool Options

3.2.3. Path-to-Part Tool Options

3.2.4. Contact Tool Options

3.2.5. Using the Script Editor for Customization and Troubleshooting

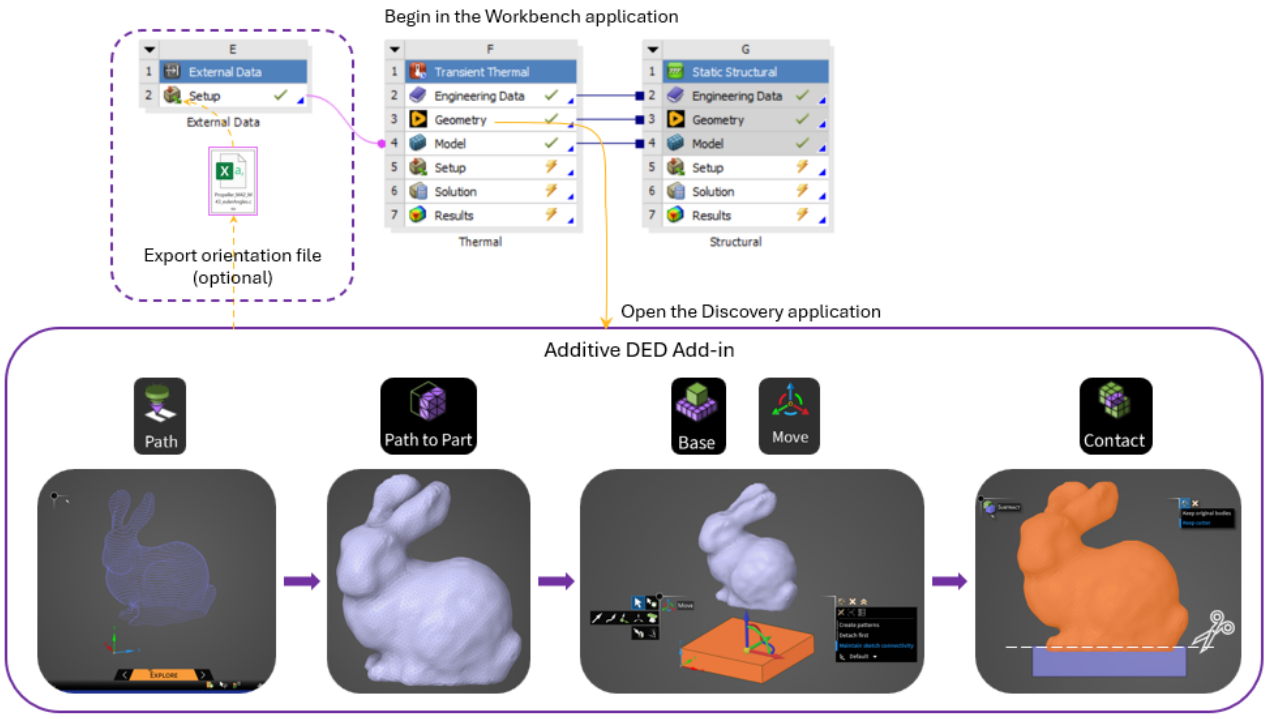
3.2.1. Path-to-Part Workflow

Be aware of the following requirements for using the Path-to-Part functionality:

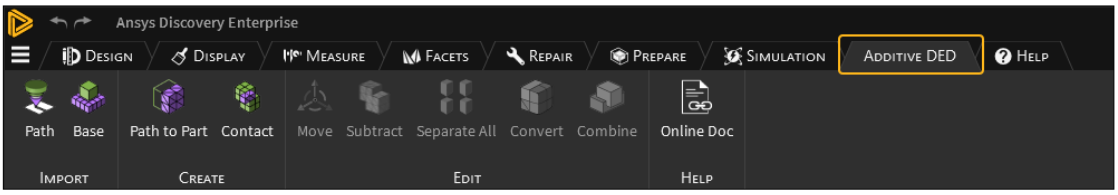
- In addition to Ansys Workbench and Mechanical, the Ansys Discovery application must be installed and available for use.
- You must enable the Additive DED add-in in the Discovery application's settings (**File Menu** > **Settings** > **Add-ins**) and then close the application. This is a one-time step; once the add-in is enabled, it will remain enabled for future Discovery sessions.

The recommended workflow begins in Ansys Workbench, proceeds to Discovery, and ends in the Mechanical application. Data transfer between applications is smooth and many tasks are automated. The workflow is summarized in the figure below and described in the following steps.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



1. Set up an AM DED Process custom system (p. 27) in the Ansys Workbench application.
2. Double-click the **Geometry** cell to open the Discovery application.
3. Click the **Additive DED** tab to view the add-on's custom ribbon that includes Import and Create tools. (Note that the Edit tools are standard Discovery tools and are provided on the Additive DED custom ribbon for convenience in case they are needed.)



4. **Import and Verify the Tool Path**

   Click the **Path** tool and choose the import file format (G-code or CLS (Beta)). For G-code, specify **Laser On** and **Laser Off** commands. Choose **Machine Type**, either Three-axis or Multi-axis, and then specify associated parameters. See Path Tool Options (p. 17).

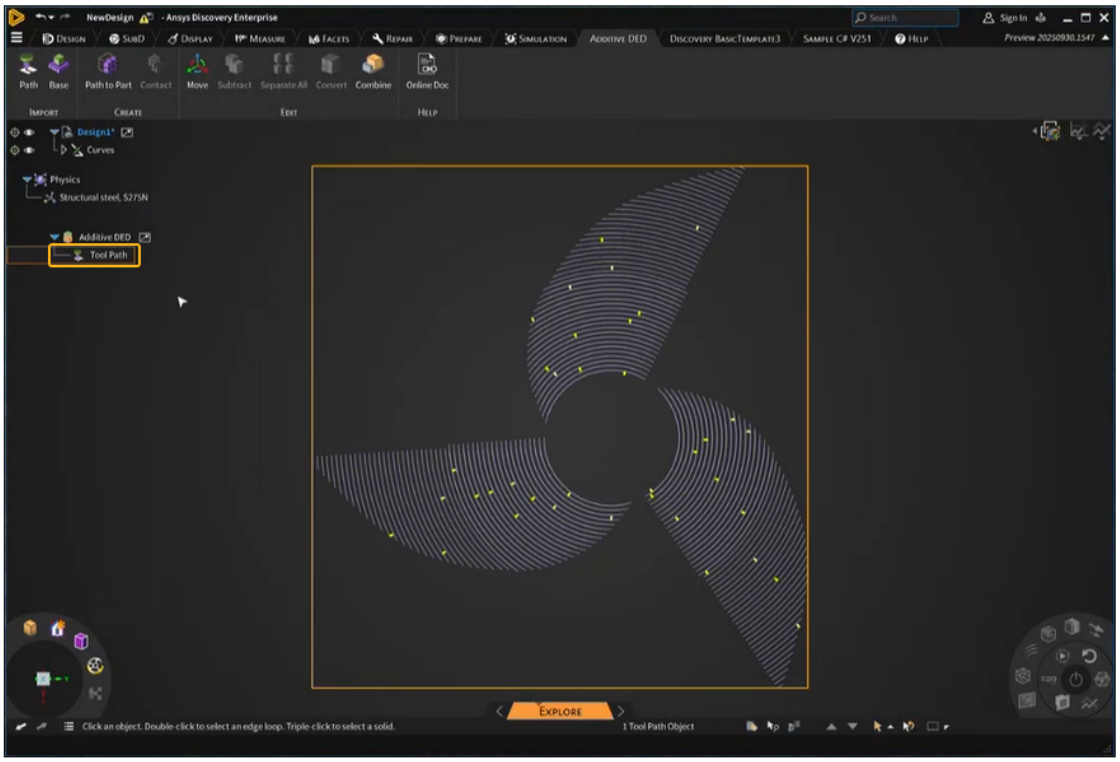
   Click the green checkmark in the heads-up display to import the G-code/CLS file. Navigate to your file and click **Open**.

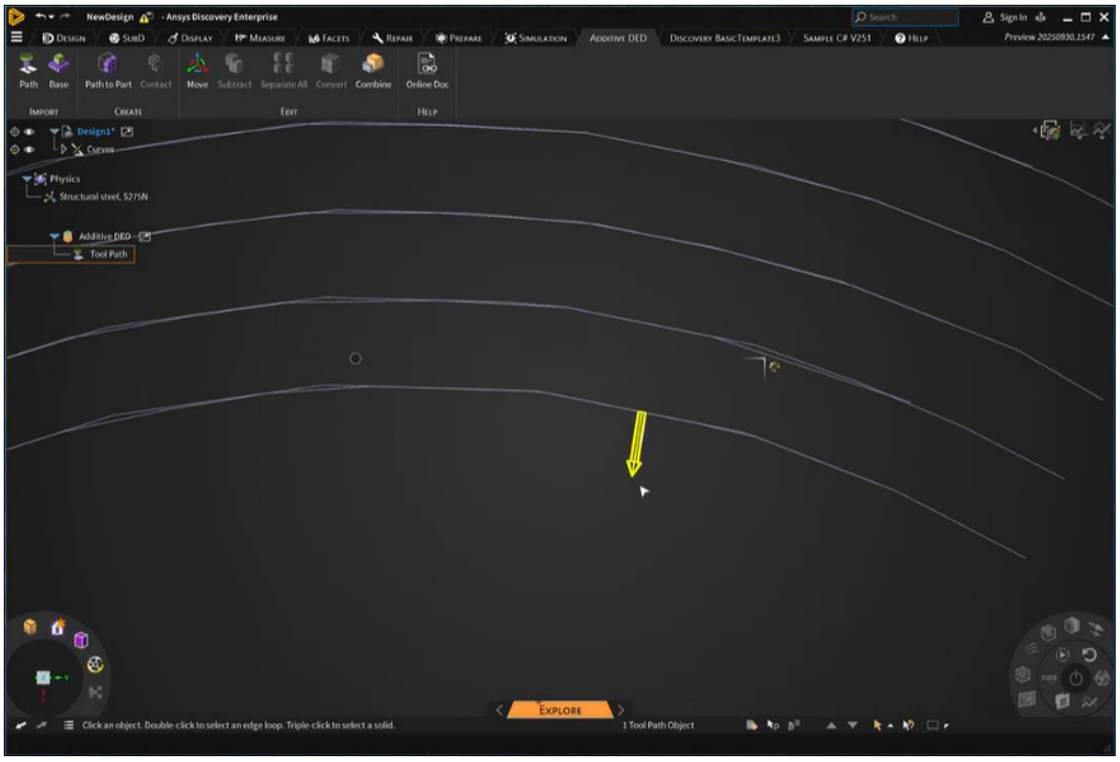
   The following items are added to the tree:

   - A **Curves** object containing the path point locations is added in the geometry area. The tool path displays as purple lines.
   - An **Additive DED** object is added below the Physics area.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

- A **Tool Path** object containing the tool path *orientations* is added under the Additive DED object. Yellow arrows indicate the tool orientations, which are used in defining the bead profile at every location.





Also, a file called `*_polyLines.txt` is added to the folder in which the G-code file is located, where * is the name of your G-code file.

5. **Convert Path to Part**

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Click the **Path to Part** tool and choose your bead profile, either Circular, Rectangular, or Elliptical. Enter the specific dimensions for the bead profile. See Path-to-Part Tool Options (p. 18). Note that you can enter a dimension in any unit system but when you click off the field, it always displays the value converted to meters.

You can specify a custom bead profile by using the Scrip Editor (p. 20) (under Discovery's Design tab). Record a "starting point" Path-to-Part execution and then change the bead profile coordinates in the `profile` line of code in the script.

The next step is slightly different depending on your chosen bead profile. Circular profiles do not require tool path *orientation* information, while Rectangular and Elliptical profiles do. Recall that orientation information is stored in the Tool Path object. To convert the path to a part:

- For a Circular profile: Select either the **Tool Path** object or the **Curves** object in the tree and then click the green checkmark  to complete the Path-to-Part conversion. You can select individual curves (under **Curves**) rather than all curves to specify different bead sizes for different regions of the part. Hint: View the message in the lower, left corner of the status bar for guidance.
- For a Rectangular or an Elliptical profile: Select the **Tool Path** object in the tree and then click the green checkmark  to complete the Path-to-Part conversion. Hint: View the message in the lower, left corner of the status bar for guidance.

The part is rendered and a **Facets** object is added to the tree.

6. Click **Base**, navigate to the base geometry file, and click **Open** to import the base.
7. **Align Part to Base and Create Clean Contact Surface Between Them (if necessary)**

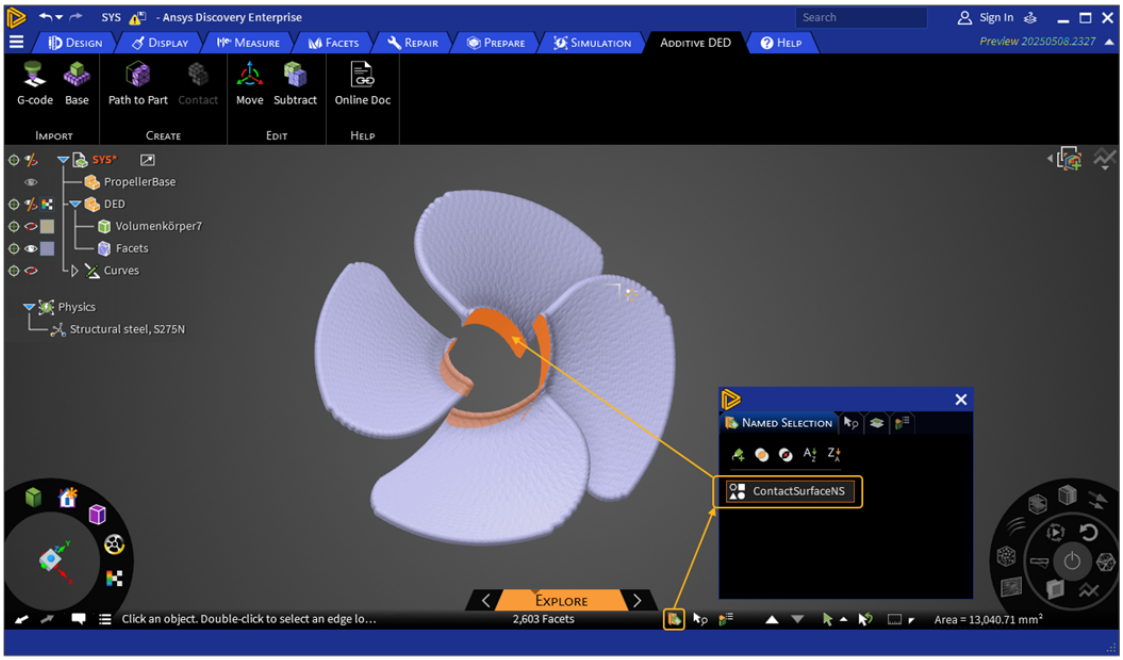
   In case the part and base are not aligned, click the **Move** tool to move the base into alignment with the part. Be sure to align in all directions, X, Y, and Z. Hint: Toggle the Show/Hide buttons on the Curves and Facets objects to see them individually, as desired.

   It is necessary to ensure a clean surface connection between the part and the base with no overlapping regions. The **Contact** tool uses the base as a cutter, subtracts it from the part, and creates a named selection for the contact surface on the part that will be needed for the simulation in the Mechanical application.

   Click **Contact** to start the operation. Then click the facets body in the graphics window. Then click the contact surface on the base. This last selection performs the subtract action automatically and also creates a named selection for contact. See Contact Tool Options (p. 19). While using this tool, remember to view the message in the lower, left corner of the status bar for guidance.

   To see the new named selection, click the **Named Selection** icon in the status bar to open the Named Selection panel. The new named selection, called ContactSurfaceNS, appears in the list. Click **ContactSurfaceNS** to see the orange highlighted surface(s) on the model. It may be helpful to hide the base body.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Note:

If the Named Selection window is open before the Contact operation is performed, the named selection won't be displayed properly. Use Undo and Redo as needed to get back to the state before contact creation. Ensure the Named Selection window is closed before continuing.

At the end of the Contact operation, the part and base are automatically grouped together in the tree as a component called **DED**.

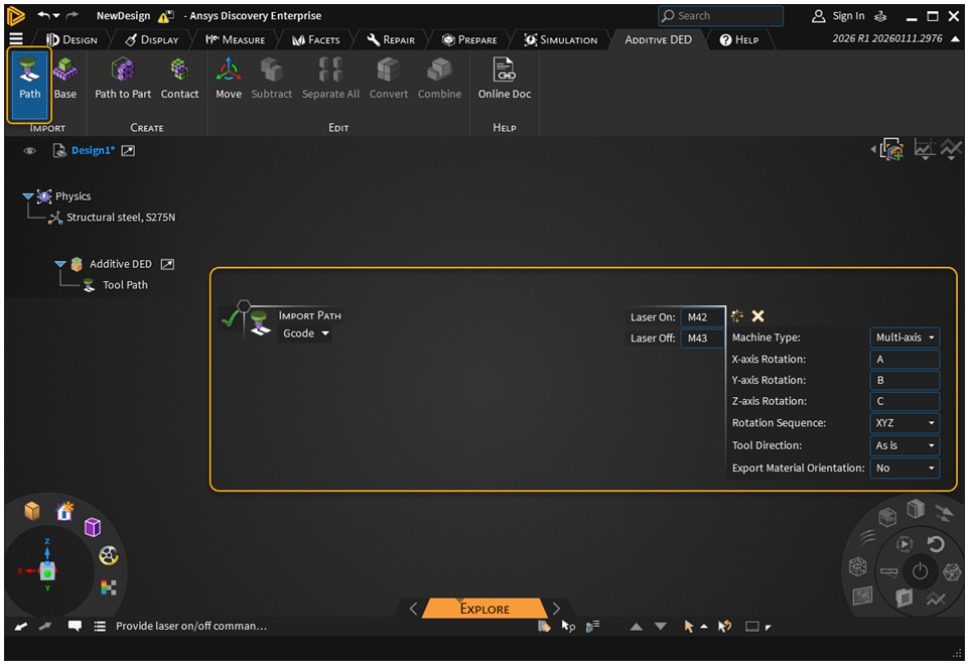
8. The Path-to-Part workflow is done at this point. Back in the Workbench application, double-click the **Model** cell to launch the Mechanical application. Continue your simulation using the DED Process Wizard (p. 23) as usual.

Tip:

If you have a curved base, as shown in the propeller example above, Ansys recommends you set shared topology between the part and the base to force a better connection. You can do this in the Discovery application before you proceed to Workbench. In the tree, select the **DED** component. Click the **Properties** icon in the status bar to open the Properties panel. Click **Transfer Sharing** and then **Share topology**.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

3.2.2. Path Tool Options



The following **Path** tool options are available:

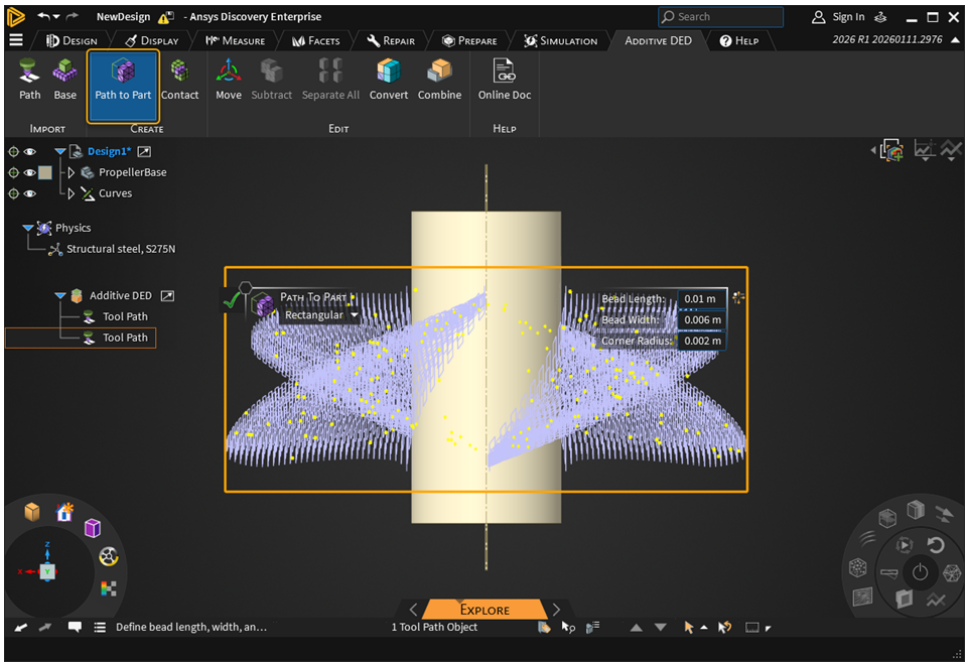
Gcode	Imports a G-code file format.
	Laser On: G-code command used to define laser on.
	Laser Off: G-code command used to define laser off.
CLS (Beta)	Imports a Cutter Location Source (CLS) file format. This is a Beta feature at this release.
	Paint Index: A parameter used to control visualization and attribute grouping during the tool path verification process. The Paint Index acts as a "tag" or "color ID" assigned to specific segments of the tool path.
Machine Type	Your machine type, either Three-axis or Multi-axis.
X-axis Rotation	G-code command used to define the motion about the X-axis.
Y-axis Rotation	G-code command used to define the motion about the Y-axis.
Z-axis Rotation	G-code command used to define the motion about the Z-axis.
Rotation Sequence	The specific order in which your machine's rotational axes are applied to reach a target orientation.
Tool Direction	The orientation of the deposition head in 3D space. Options include As Is and Reverse.
Export Material Orientation	Yes or No (default) to export a CSV file with Euler angles at each point, useful for defining anisotropic material properties. Use the External Data feature to import the data into the Mechanical application.
Interpolation Distance (If Export Material Orientation = Yes)	The distance[a] between *additional* interpolation points that are inserted between the G-code/CLS points defining the tool path.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

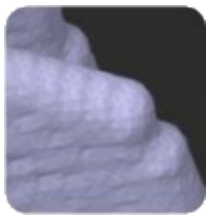
	Euler angles are calculated at the interpolation points that will then be used to define local coordinate systems when exporting to the Mechanical application. Local coordinate systems align the elements of the mesh. A smaller distance forces the insertion of more points and, therefore, generates a higher density of local coordinate systems along the path. This ensures the simulation mesh precisely conforms to the intended tool movement and geometry. The default value (0) means no additional interpolation points will be inserted between defined G-code/CLS points.

[a] You can enter the value in any unit system but when you click off the field, it always displays the value converted to meters.

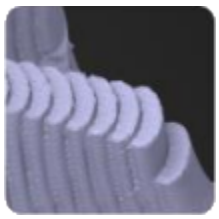
3.2.3. Path-to-Part Tool Options



The following **Path to Part** tool options are available:

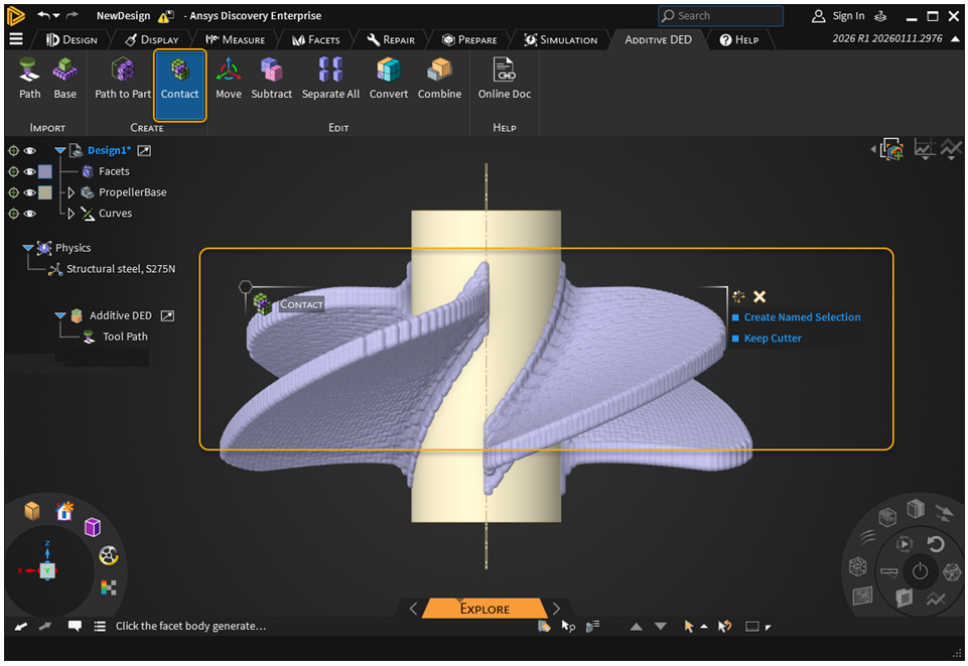
Circular 	Creates a bead profile with a circular cross section.
	Bead Radius: Radius[a] of the weld bead.
Rectangular	Creates a bead profile with a rectangular cross section.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

	**Bead Length**: Length[a] of the weld bead.
	Bead Width: Width[a] of the weld bead.
	Corner Radius: Radius[a] of each rectangle corner.
Elliptical 	Creates a bead profile with an elliptical cross section.
	Bead Major Axis: Radius[a] of the weld bead's larger axis.
	Bead Minor Axis: Radius[a] of the weld bead's smaller axis.

[a] You can enter the value in any unit system but when you click off the field, it always displays the value converted to meters.

3.2.4. Contact Tool Options



The following **Contact** tool options are available:

Create Named Selection	Option (check box) to create a named selection for the contact surface on the part that will be needed for the simulation in the Mechanical application.
Keep Cutter	Option (check box) to keep the cutter surface.

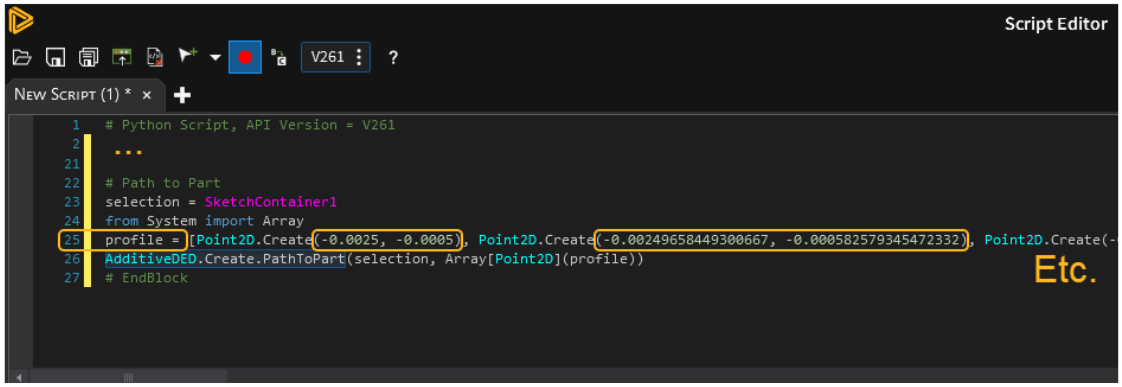
3.2.5. Using the Script Editor for Customization and Troubleshooting

When using the Path-to-Part feature in Ansys Discovery, you can leverage the Script Editor to fine-tune your results beyond the standard UI options. Every tool in Discovery, including Path to Part, generates specific scripting commands that can be modified to handle complex scenarios.

Customizing the script is particularly useful in the following two cases:

A. **Defining a Custom Bead Profile**

While the UI offers standard **Circular**, **Rectangular**, and **Elliptical** bead profiles (p. 18), you can achieve more precise geometries through scripting. To change the dimensions or shape of the bead, locate the `profile` line in your recorded script and manually edit the coordinate values. Note that the coordinate values represent specific inflection points defining the bead's cross section.

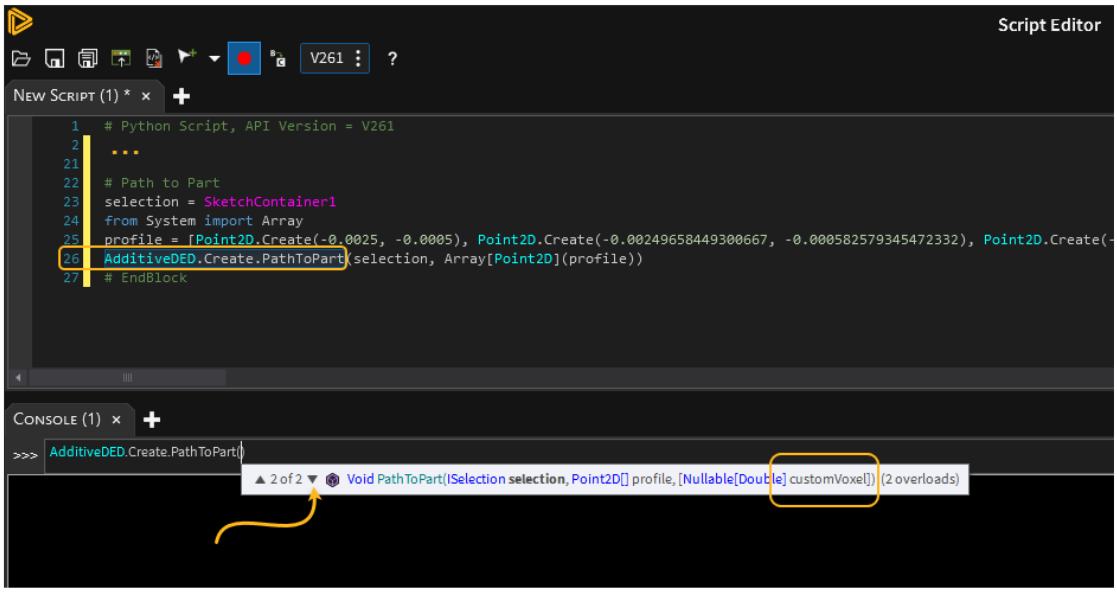


B. **Adjusting Voxel Size for Tool Path Fidelity**

The Discovery application calculates a default voxel size based on your bead's cross-section that affects the tool path resolution. (The default voxel size is automatically calculated as 5% ($1/20^{th}$) of the diagonal dimension of the bead profile's bounding box.) If the generated part doesn't meet your requirements, you can override it using the `customVoxel` optional argument on the `AdditiveDED.Create.PathToPart(...)` scripting command.

- If the generated part has *gaps or is not a closed body*, use finer voxels (a smaller size). A smaller voxel size increases the triangle count to capture intricate path details and ensure the part is a fully closed body.
- When the generated part is *overly complex, has unnecessary detail, or is processing too slowly*, use coarser voxels (a larger size). A larger voxel size reduces detail and simplifies the geometry, resulting in faster generation times.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Recommended Workflow

To implement these customizations, follow these steps:

1. **Record:** Perform an initial Path-to-Part action with the Script Editor open to capture the baseline code.
2. **Edit:** Modify the `profile` coordinates (case A, above) or the `customVoxel` argument (case B, above) within the script.
3. **Replay:** Run the edited script to generate the refined part.

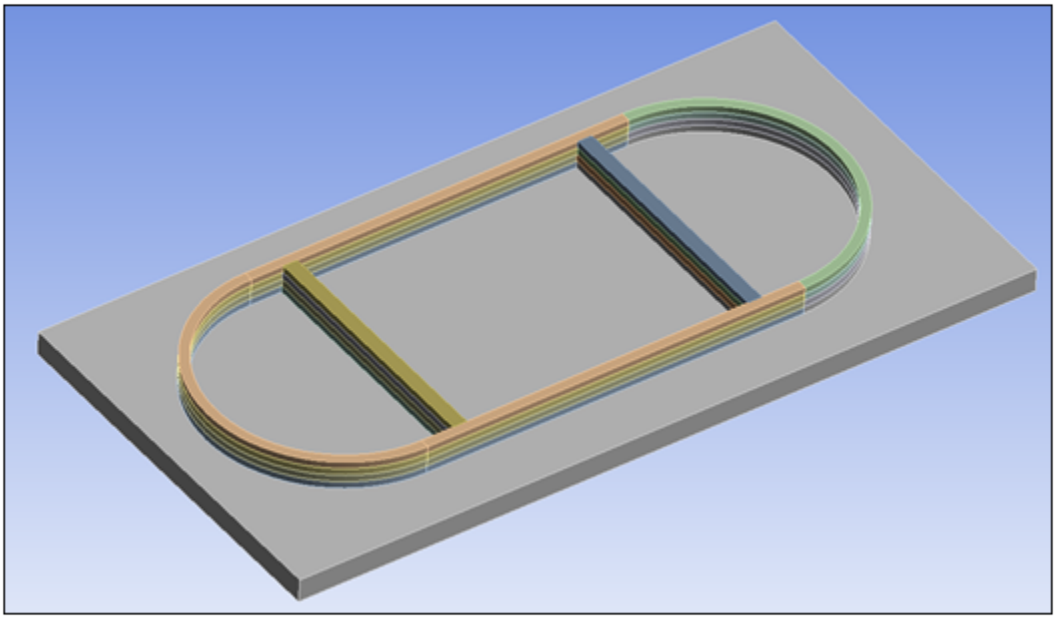
3.3. Slicing the Part for Manual Clustering

Slicing the part into an appropriate weld path is required if you use manual clustering (p. 60) in the DED process simulation. Slicing requires geometrical manipulation, using the Ansys Discovery™ application, perhaps. Consider these hints for slicing:

- Slicing height must follow layer height.
- Closed loops must be split since each weld line requires a defined start face.

Here is the sliced geometry for a racetrack example.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information
of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

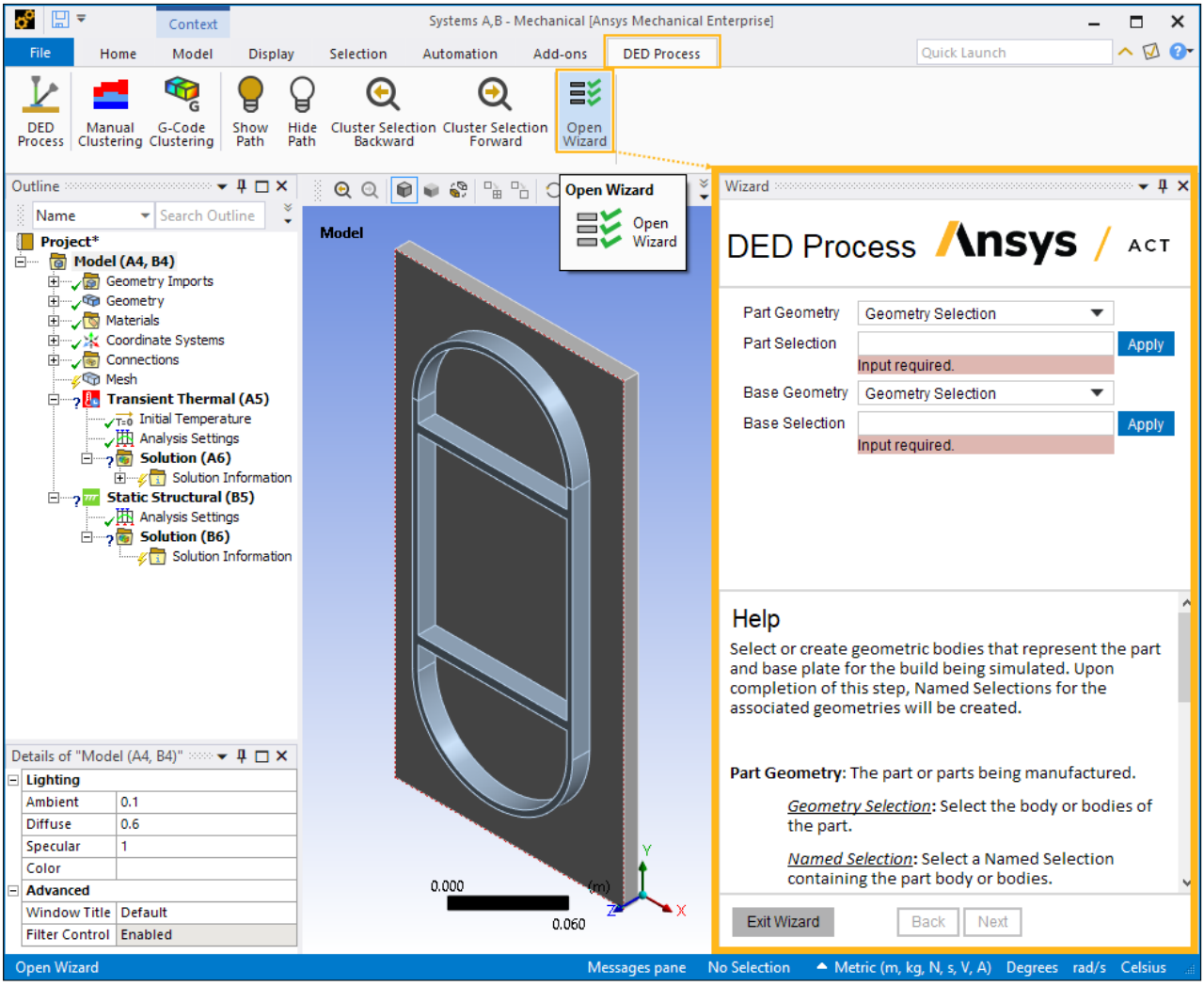
Chapter 4: Using the DED Process Wizard

Simplify and focus your workflow using the DED Process Wizard. Step through the wizard pages that are organized into workflow tasks. Actions take place when you click Next at the bottom of each page. As you progress through the steps, the wizard automatically adds objects into the project tree. Exit the wizard at any time and continue setting up your simulation manually in the project tree.

Open the Wizard

First make sure the DED Process Add-on is loaded (p. 3).

To start the wizard, select the **DED Process** tab and then select the **Open Wizard** button.



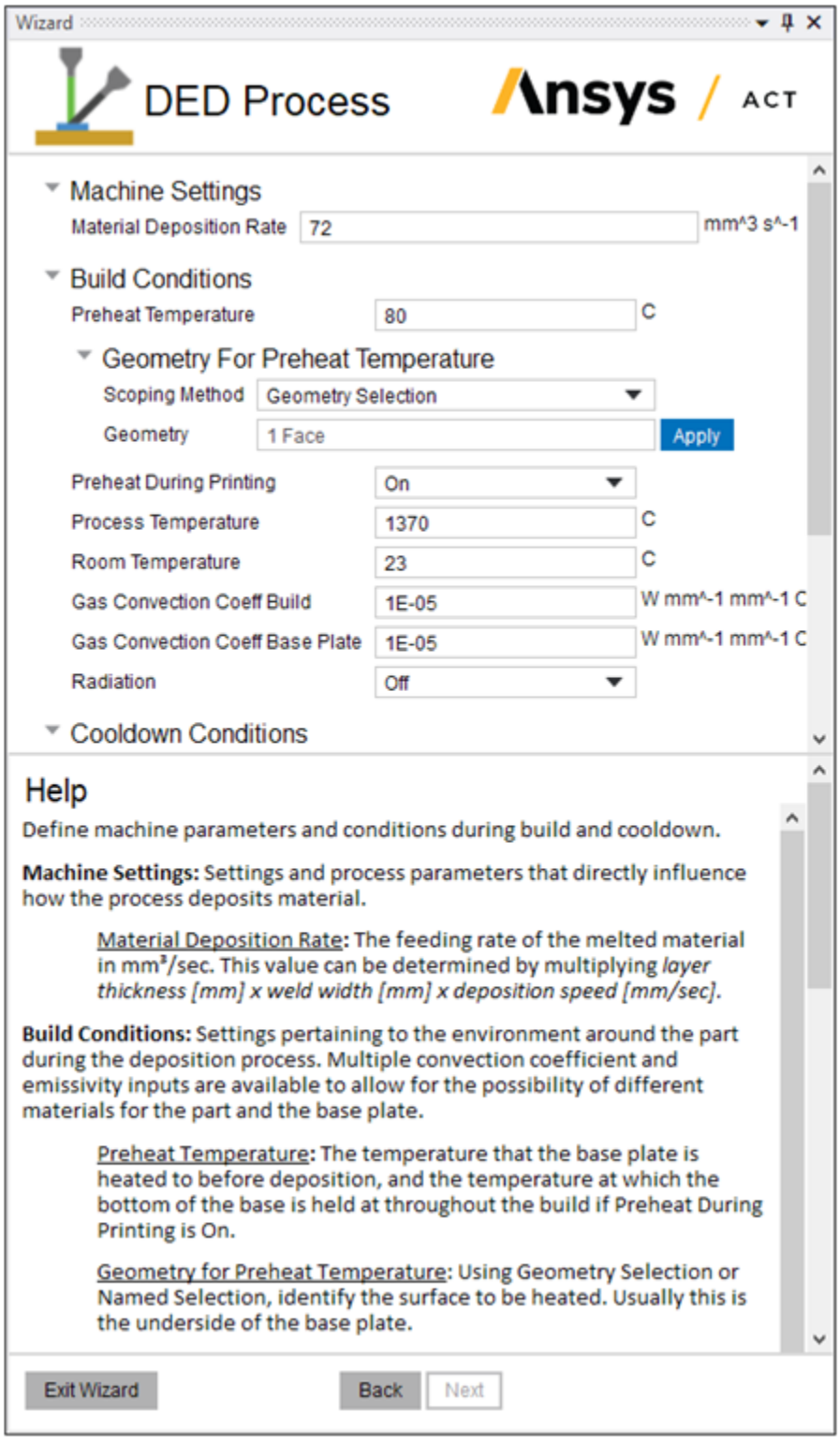
The DED Process Wizard guides you in a step-by-step manner through the following steps:

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

1. Identify geometries
2. Set up for element clustering
3. Generate mesh and create contact connections
4. Assign materials
5. Define machine parameters and thermal boundary conditions
6. Define structural boundary conditions and base removal

Refer to the in-wizard help at the bottom of each page. An example page, for defining machine parameters and thermal boundary conditions, is shown here:

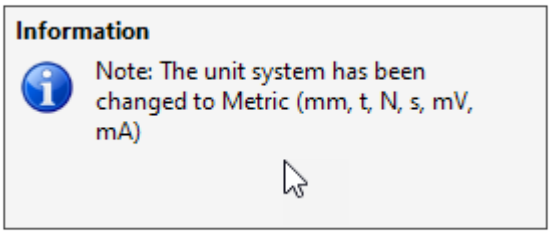
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Assumption and Restrictions When Using the Wizard

- When you launch the wizard, the unit system is checked and, if necessary, automatically switched to the Metric system (mm, t, N, s, mV, mA) because the DED Process Add-on requires that units be in millimeters. The following message appears if the unit system is changed:

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



- Upon completion of the material assignment step of the wizard, the Reference Temperature and Reference Temperature Value for the build body are automatically set to "By Body" and to the build material's melting temperature respectively, or 1000°C if the melting temperature is not defined for the assigned material. The Reference Temperature for the base is set to "By Environment."

- The options to read Feed Rate, Preheat Temperature, and Dwell Time from a G-Code file are not available in the wizard. Specify them on the G-Code Clustering object (p. 116) in the project tree (outside of the wizard). See Element Clustering: G-Code Approach (p. 62).

Close the Wizard

1. Upon completing the last page, click **Finish** to complete the wizard.

2. Click the X in the upper, right corner to close the wizard window.

Review Items in Project Tree

Continue the simulation in the Mechanical application interface at this point. It is a good idea to review the objects in the project tree that were created by the DED Process Wizard. Select the objects in the tree from top to bottom and examine the details view for each object to see how the wizard options are implemented in the project tree. New objects have been added, such as the AM Process for DED (p. 109) object and its child objects (Build Settings (p. 111), Manual Clustering (p. 121) or G-Code Clustering (p. 116)). Take note of the named selections for the build body, the base body, build and base body element faces and nodes for the removal step. A connection object, mesh objects, boundary condition objects, result objects, and others have been added or changed automatically by the wizard. You can make changes or add items as needed.

Perform Element Clustering

The element clustering step must be performed after you finish the wizard. Refer to one of the following sections, depending on which method you selected to define the build order in the wizard:

- If you specified a manual approach to define the building order, refer to Element Clustering: Manual Approach (p. 60).

- If you specified a G-Code file to define the building order, you must now define the G-Code clustering options and perform element clustering, as described in Element Clustering: G-Code Approach (p. 62).

After performing element clustering, the simulation is ready to solve.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Chapter 5: Workflow Through the Project Tree

We strongly recommend you use the DED Process Wizard (p. 23) to set up your simulation because it simplifies your workflow considerably. The wizard populates objects, options, Named Selections, and defaults for you in the Mechanical application's project tree. This chapter describes the steps required if you prefer to use your own workflow to set up a DED simulation in the project tree.

A DED process simulation involves most of the general steps found in any Ansys analysis with some additional steps and considerations as described in the following steps:

5.1. Create the Analysis System

5.2. Attach Geometry and Launch the Mechanical Application

5.3. Insert AM Process for DED and Identify Geometries

5.4. Assign Materials

5.5. Apply Mesh Controls and Generate Mesh

5.6. Create Named Selections for Contact and Clustering

5.7. Define Build Settings

5.8. Perform Element Clustering

5.9. Define Connections

5.10. Define DED Process Steps

5.11. Establish Thermal Analysis Settings

5.12. Apply Thermal Boundary Conditions

5.13. Solve the Transient Thermal Analysis

5.14. Establish Structural Analysis Settings

5.15. Apply Structural Boundary Conditions

5.16. Solve the Static Structural Analysis

5.17. Review Results

5.1. Create the Analysis System

A DED process simulation requires a transient thermal analysis followed by a static structural analysis. For simplification, it is assumed the physics are uncoupled in that data flows one-way from the thermal analysis to the structural. A customized, linked system is available for DED simulations.

1. Open Ansys Workbench.

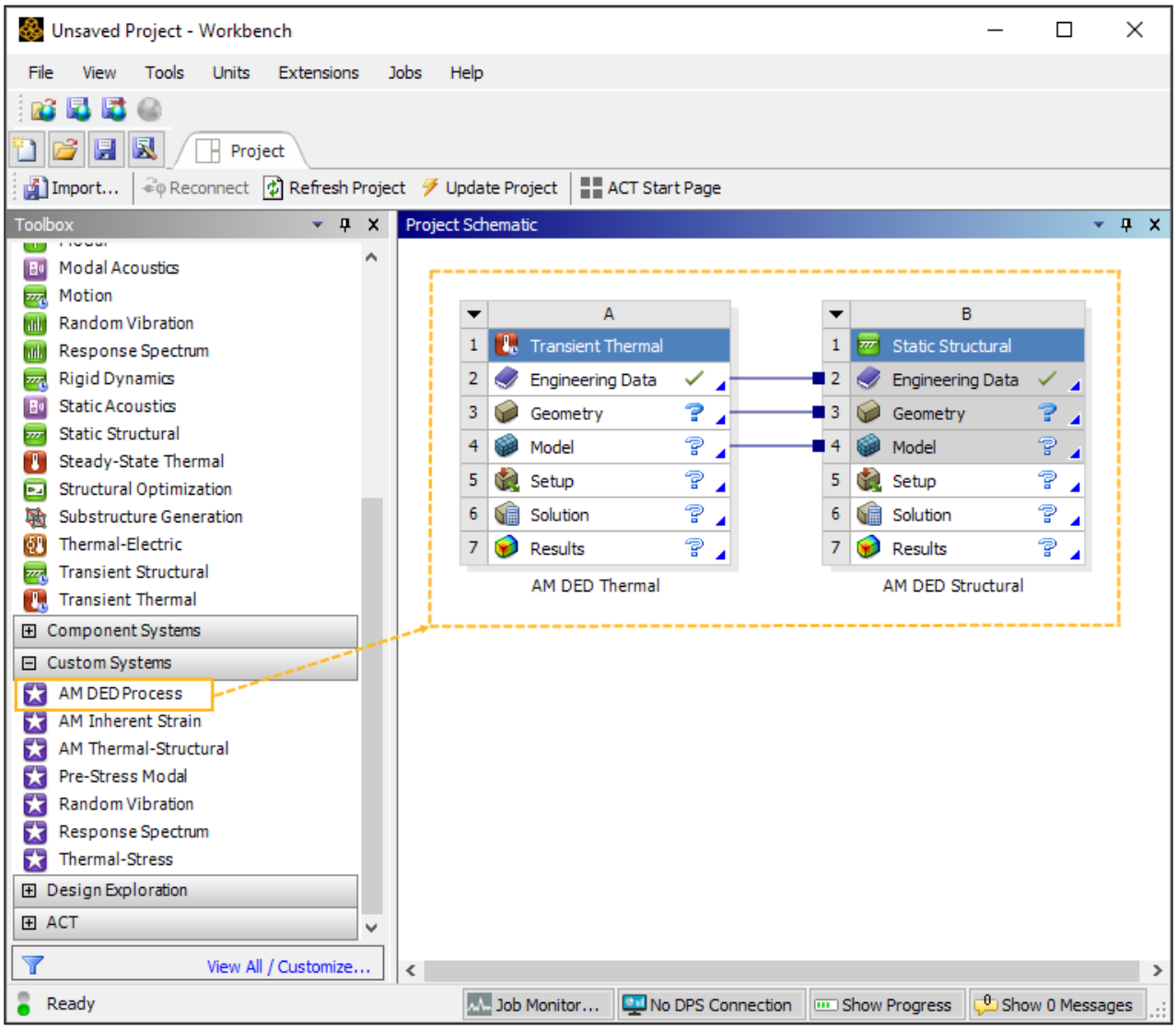
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

2. In Workbench, scroll down to custom systems, located in the toolbox on the left side of the window, and load the AM DED Process by double-clicking it.

Note:

If you use a Mechanical Model from Workbench's Component Systems, you must manually create the AM DED Process system. This is due to the fact that AM Custom Systems are pre-populated with AM sample materials in Engineering Data, and linking to the Mechanical Model requires the absence of materials in Engineering Data.

If you open the Mechanical application independent of Workbench, you must manually insert the transient thermal and static structural systems.



5.2. Attach Geometry and Launch the Mechanical Application

Typically, you'll have these geometric bodies in the simulation model:

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

- **Part** – This is the part you are manufacturing. It should be a closed volume (that is, watertight), and should be oriented with the global X, Y, or Z axis as the print direction. The part can be made of multiple bodies but they must have shared topology between them. See Shared Topology in the Discovery Documentation.

  For the part geometry, commonly imported file types include the Discovery application's `.dsco` files (non-faceted) and stereolithography `.stl` files (faceted). For other options, see Attach Geometry/Mesh in the *Mechanical User's Guide*.

- **Base Plate** – This is the platform on which the part is to be printed. It is included in the simulation because it acts as a heat sink. It can be planar or curved.

Note:

If you are modeling a curved base plate where either the part or the baseplate will have a Cartesian mesh, we recommend you also set shared topology between the part and the base plate to force a better connection than the default contact connection.

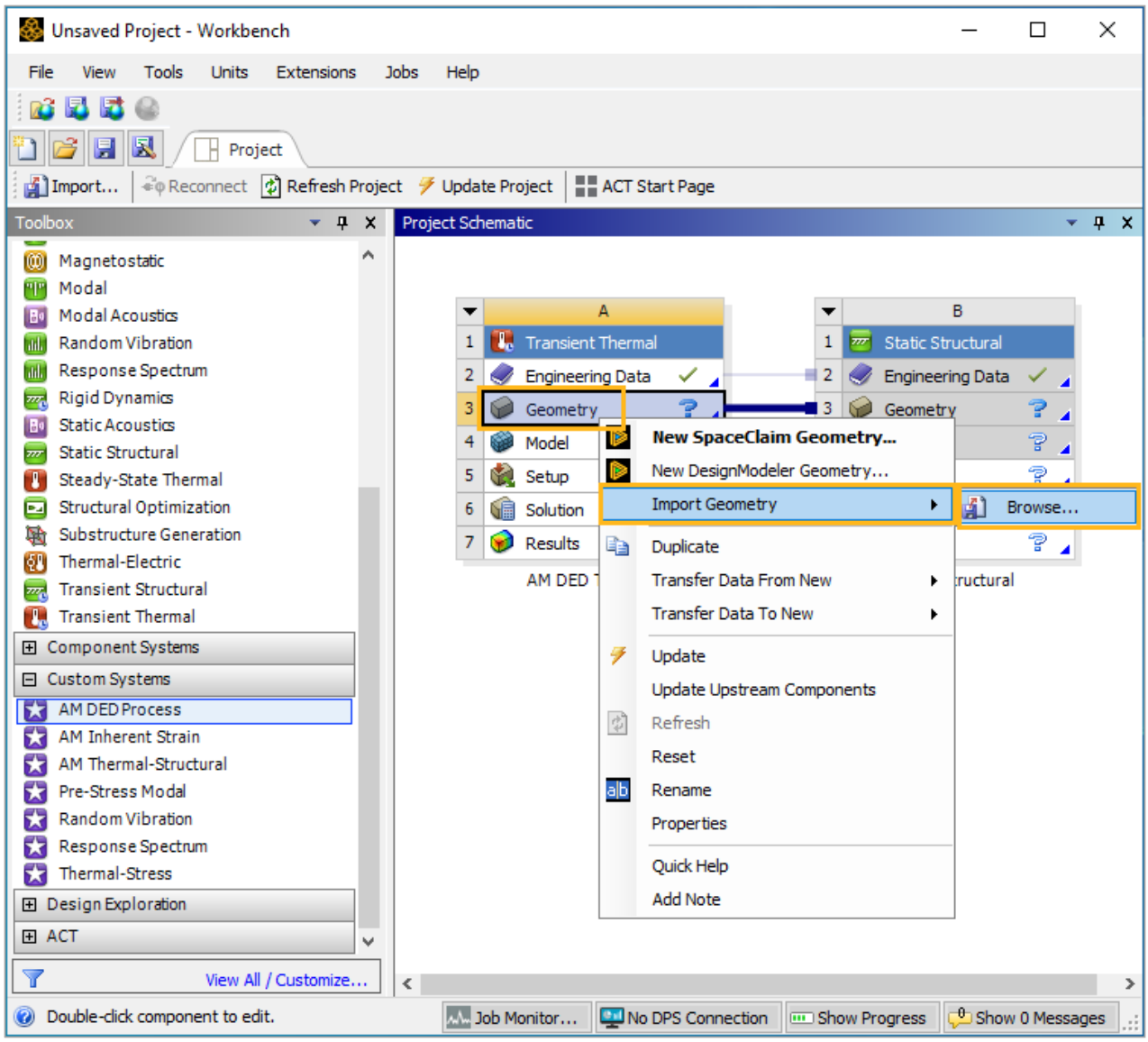
You set shared topology in the CAD/design application. See Apply Mesh Controls and Generate Mesh (p. 38) for options to change Projection Factor if you have unshared topology.

Note that supports are not used in a typical DED process because the multiple axis turntables rotate the build platform to facilitate material deposition. This is unlike in a powder bed fusion process where supports are used to anchor the part to the base plate and supports must therefore be included in the simulation.

Procedural Steps

1. Right-click the **Geometry** cell of the Geometry block, and select **Import**. Browse for your file and import it. (Double-clicking the Geometry cell opens the Ansys Discovery application.)

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



2. After importing geometry, double-click the **Model** cell (or right-click, and select **Edit**) of the Transient Thermal analysis block to launch the Mechanical application. It may take a few minutes for Mechanical to open and your geometry to appear in the geometry window.

3. Once your geometry is loaded in the Mechanical application, review how it is presented in the project tree.

4. Once the Mechanical application opens, it is a good time to adjust the number of processors (cores) you are using on your computer. Depending on the complexity of your model, DED process simulations may be computer intensive. If you have an Ansys HPC license, access the option in the Solve group on the **Home** tab and change the **Cores** to something appropriate for your simulation.

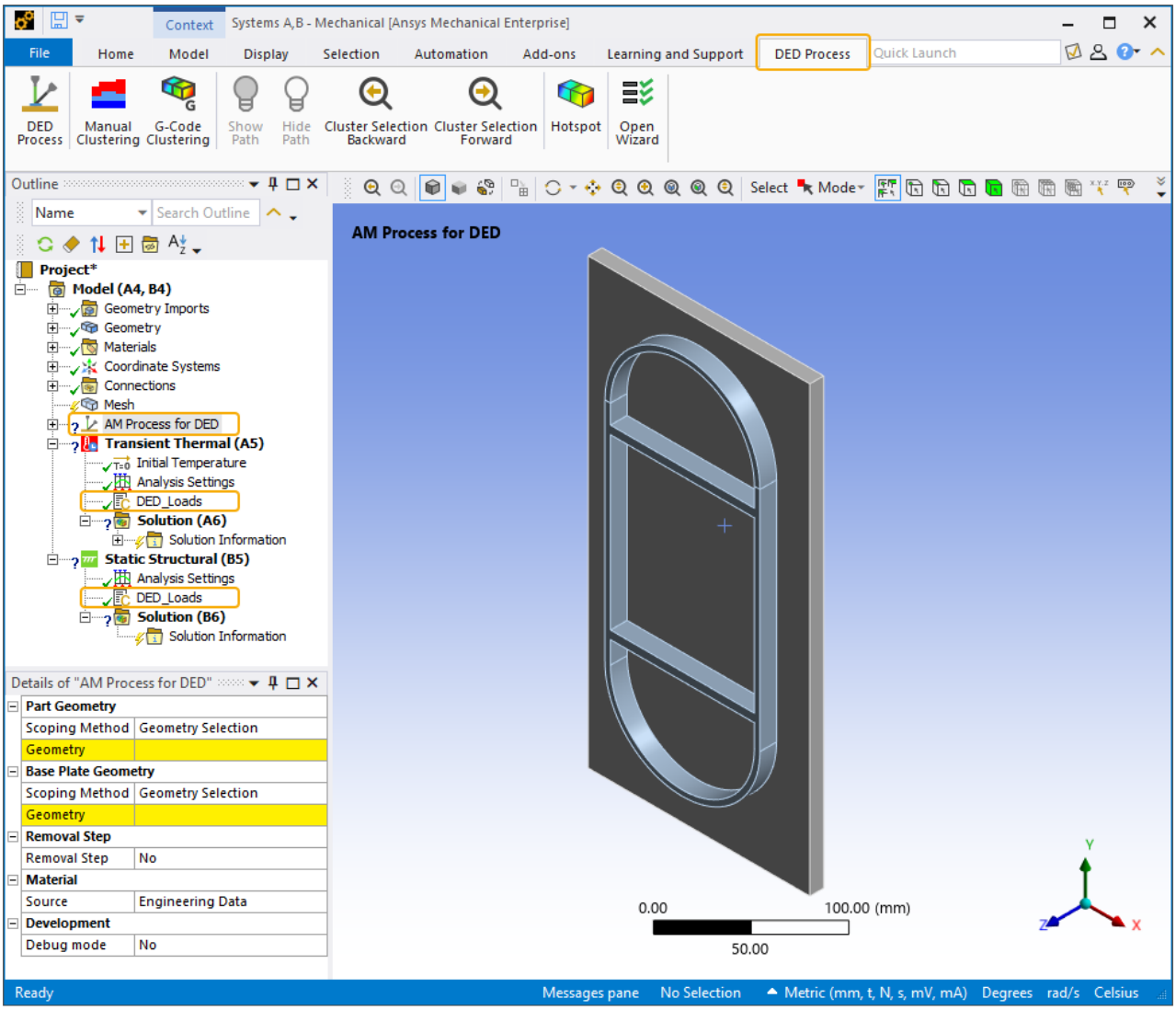
5.3. Insert AM Process for DED and Identify Geometries

To establish the options and assumptions appropriate for a DED process simulation, you need to insert the **AM Process for DED (p. 109)** object into the project tree. Here you will identify which bodies are associated with the build and the base plate.

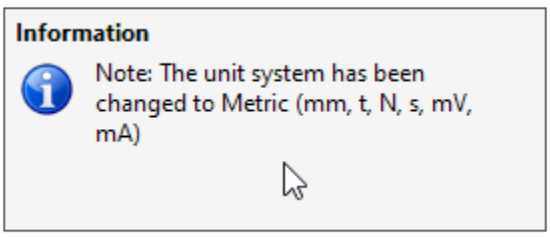
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Procedural Steps

1. Select the **DED Process** tab on the ribbon and then click the **DED Process** option from the DED Process tool bar. Notice that the AM Process for DED (p. 109) object is added to the project tree, as well as DED_Loads objects under each system (transient thermal and static structural). Keep these objects in the tree as they are required for solution.



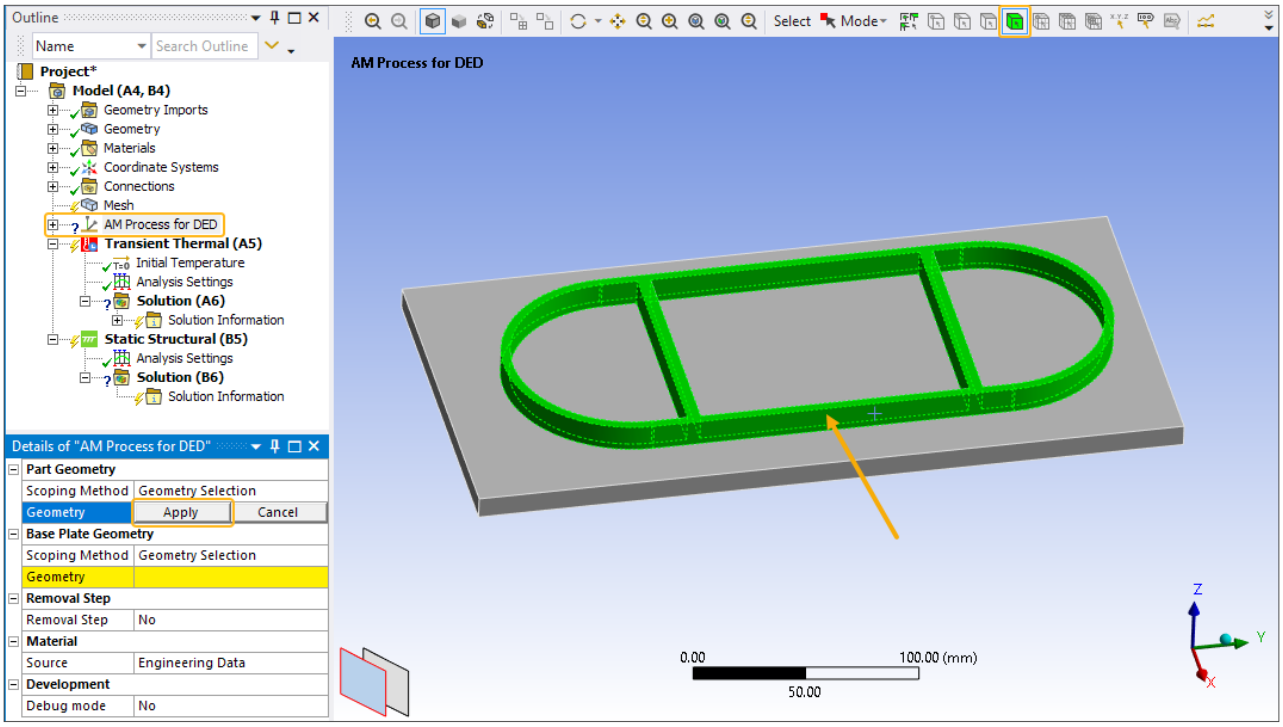
Also, the unit system is checked automatically and, if necessary, changed to the Metric system (mm, t, N, s, mV, mA) since the DED Process Add-on requires that units be in millimeters. The following message appears if the unit system is changed:



2. Next you need to identify which bodies are which in the geometry you imported. Select the **AM Process for DED** object and then:

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

a. Select the body or bodies representing the part that make up the **Build Geometry** and click **Apply** in the Details view under Build Geometry. To select a body, be sure that your *mode of selection* is on body  (rather than on face, edge, or vertex). Use the Ctrl key while clicking the left mouse button to select multiple entities (or click, hold, and drag). Or use named selections as a basis for selection. When selected, the build geometry is shown in green. Clicking Apply displays the build geometry in blue.



b. Select the body that is the **Base Plate Geometry** and click **Apply** in the Details view. (If your imported geometry does not have a base plate, construct one in the next step.)

3. To create the base plate if not imported with the CAD geometry, select the **Model** object and then:

   a. Right-click the **Model** object or in the geometry window and select **Insert** > **Construction Geometry**.

   b. Right-click the new **Construction Geometry** object (under Model) and choose **Insert** > **Solid**.

   c. In the Details view, fields appear for you to enter coordinates with respect to the center of the top of the base. Enter values for **X1**, **X2**, **Y1**, **Y2**, **Z1** and **Z2**. An outline is provided to preview the dimensions. Right-click **Solid** and select **Add to Geometry**. For more information, see Construction Geometry in the *Mechanical User's Guide*.

   d. To identify this newly created body as the Base Plate Geometry, highlight the **DED Process** object, select the new body and click **Apply** in the Details view for Base Plate Geometry.

5.4. Assign Materials

Assign a material to each geometry body. The build and the base plate can be a different material.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

The reference temperature is the temperature at which thermal strains do not exist in a material. We recommend you set the reference temperature value to be the melting temperature of your chosen material. In the simulation of the DED process, our assumption is that each finite element cluster is added (with the element birth and death technique) at or above the melting temperature for the material and is initially strain-free. As the build cools, thermal strains develop. The static structural analysis will use the temperature results of the transient thermal analysis to compute the displacements, stresses, strains, and forces due to these induced thermal strains.

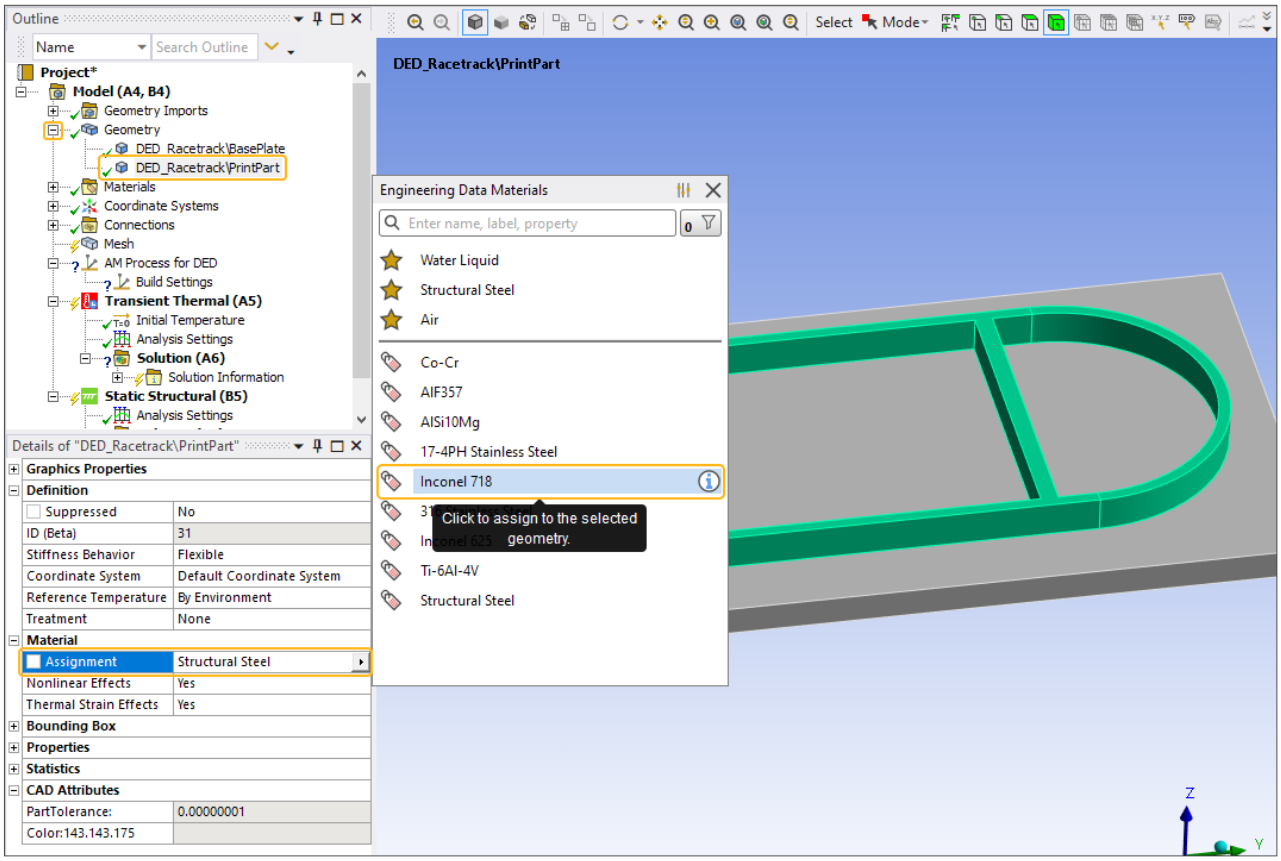
As its name implies, the reference temperature is used as a reference value to determine the state of the material - whether it is melted or not. It is also used as a criterion in the simulation to determine whether elements should be alive (activated) or dead (deactivated). The *remelting* phenomenon can be captured in the simulation by distinguishing between process temperature and reference temperature (melting temperature). Because the process temperature can be higher than the reference temperature, newly activated elements in the build (prior to the current cluster) might still be at a temperature higher than the reference temperature (melting temperature). Consequently, these elements will be deactivated to emulate remelting and prevent the accumulation of strain. This distinction becomes particularly crucial in the case of power-based temperature solutions (Heating Method = Power Based) (p. 57), as the maximum temperature can be significantly higher than the material's melting temperature.

Procedural Steps

When you select a geometry entity in the project tree, the Details view lists all the settings associated with that body.

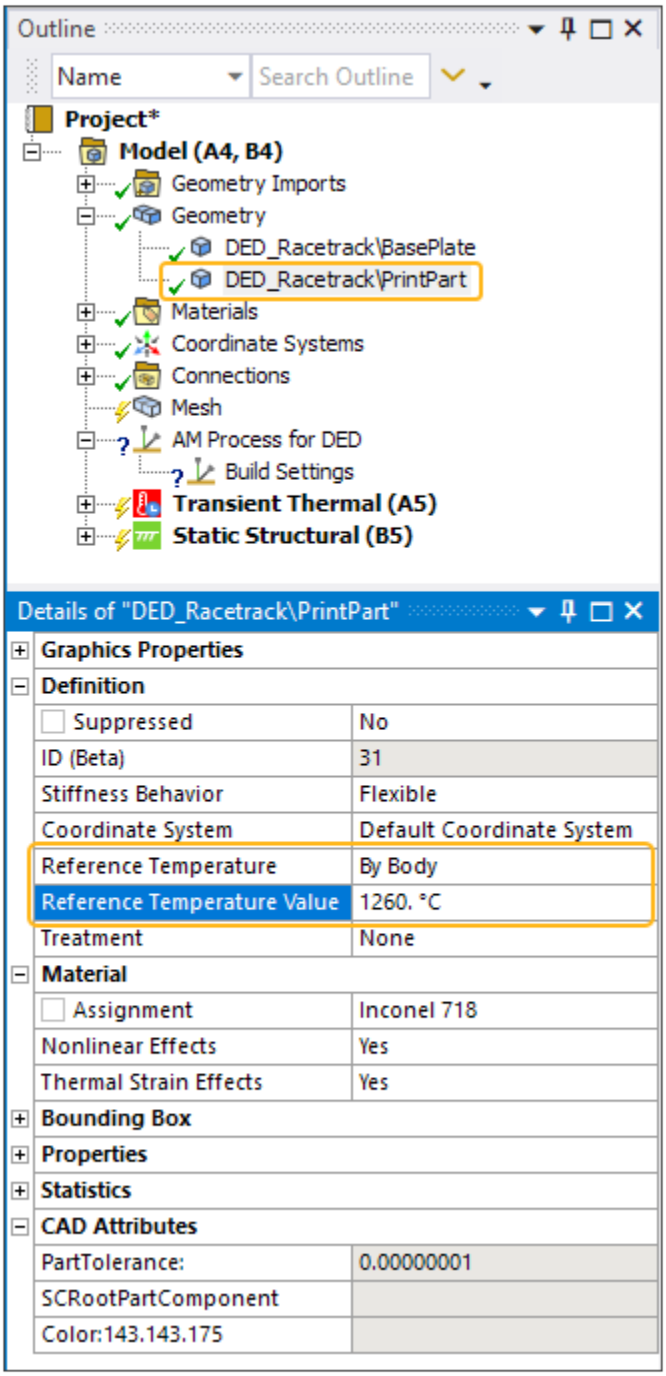
1. In the project tree under Model, expand the **Geometry** object to see its child objects below it. Select the geometric entity that is the *Print /part geometry* and, in Details view, under Material, change the **Assignment** to be an additive manufacturing material.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



2. For the print/part geometry, set the **Reference Temperature** to be **By Body**, and enter a **Reference Temperature Value** to be the *melting temperature*. This step is very important to ensure that the proper birth and death behavior occurs in the simulation. See Methodology and Abstractions (p. 4).

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Note:

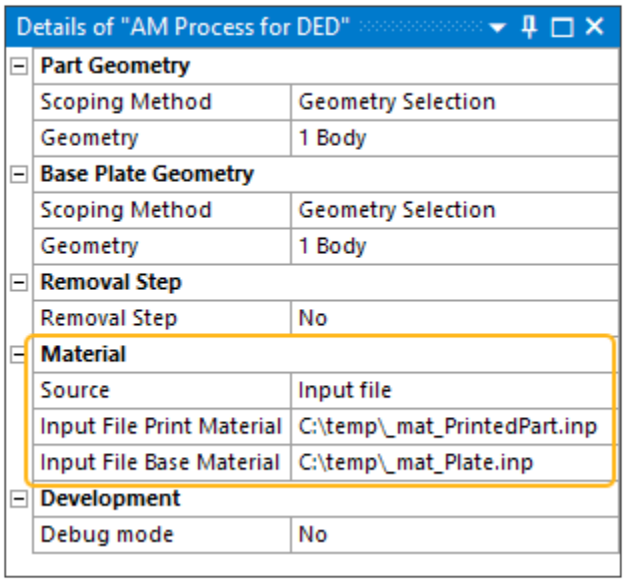
If the Reference Temperature for the print/part geometry is set to By Environment, or set to By Body but the Reference Temperature Value is left as the default of 22°, you may see odd behavior in the structural results, such as missing elements.

3. Select the entity that is the *base plate body* and change its material **Assignment**, as desired. Leave the Reference Temperature option to be By Environment (default).

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Custom Material Files

Alternatively, advanced users may want to define text files containing material data for the build part and base plate, useful for sensitivity analysis of material properties, perhaps. (See example files _mat_Plate.inp and _mat_PrintedPart.inp that show examples of the required definitions.) Define the custom material files on the Details view of the AM Process for DED (p. 109) object.



Example files are shown here for reference.

`_mat_Plate.inp`:

```
/nopr
!+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
!
! Bodenplatte AL-2 ( Isotropic Hardening ) DHV
!
!+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
!
!   Reference Temperature [C]
!
MP,REFT,  matpoi, 20,
!
!   Thermal Expansion Coefficient [mm/mm C]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,ALPX,  matpoi,  1, 2.3860E-05, 2.6843E-05
!
!-----------------------------------------------------------------------------------------------------------
!
!   Thermal Conductivity [mW/mm C]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,KXX ,  matpoi,   1,  122.50,  147.00

!-----------------------------------------------------------------------------------------------------------
!
!   Thermal Heat Capacity [mW s/ton C]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,C   ,  matpoi,   1, 9.00000E+08, 10.1000E+08
!
```


Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

```
!-----------------------------------------------------------------------------------------------------
!
!   Mass Density [t/mm*mm*mm]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,DENS,  matpoi,   1, 2.66000E-09, 2.53000E-09
!
!-----------------------------------------------------------------------------------------------------
!
!   Young's Modulus [MPa]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,EX  ,  matpoi,   1,  70400.0,  56320.0
!
!-----------------------------------------------------------------------------------------------------
!
!   Poisson's Ratio [-]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,PRXY,  matpoi,   1, 0.30, 0.30
!
!-----------------------------------------------------------------------------------------------------
!
!   Plastic Stress Strain Curves [C] ; [-] ; [MPa]
!
TB,PLASTIC,matpoi,2,2,MISO
TBTEMP, 20.0
TBPT,,0.0  ,  123.64
TBPT,,0.1325,  246.40
TBTEMP,250.0
TBPT,,0.0  ,  100.0
TBPT,,0.2875,  160.0
!
!+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
```

_mat_PrintedPart.inp:

```
/nopr
/com,
/com,!!!
```

```
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,KXX ,  matpoi,   1,  122.50,  147.00

!-----------------------------------------------------------------------------------------------------
!
!   Thermal Heat Capacity [mW s/ton C]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,C   ,  matpoi,   1, 9.00000E+08, 10.1000E+08
!
!-----------------------------------------------------------------------------------------------------
!
!   Mass Density [t/mm*mm*mm]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,DENS,  matpoi,   1, 2.66000E-09, 2.53000E-09
!
!-----------------------------------------------------------------------------------------------------
!
!   Young's Modulus [MPa]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,EX  ,  matpoi,   1,  70400.0,  56320.0
!
!-----------------------------------------------------------------------------------------------------
!
!   Poisson's Ratio [-]
!
MPTEMP
MPTEMP,    1,   20.0,  250.0
MPDATA,PRXY,  matpoi,   1, 0.30, 0.30
!
!-----------------------------------------------------------------------------------------------------
!
!   Plastic Stress Strain Curves [C] ; [-] ; [MPa]
!
TB,PLASTIC,matpoi,2,2,MISO
TBTEMP, 20.0
TBPT,,0.0  ,  123.64
TBPT,,0.1325,  246.40
TBTEMP,250.0
TBPT,,0.0  ,  100.0
TBPT,,0.2875,  160.0
!
!+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
```

5.5. Apply Mesh Controls and Generate Mesh

For the part, we recommend one finite element layer per real deposition layer. A DED process simulation does not need a strictly layered mesh with identical layer height for every layer as does a Powder Bed Fusion simulation, but the DED weld track must be represented in the mesh.

A slightly coarser mesh is acceptable for the base plate because it is simply serving as a heat sink and a fixed support in the simulation.

Mesh types to consider include Cartesian (i.e., hex), tetrahedrons, and sweep mesh. We recommend a Cartesian rather than a tetrahedrons mesh if you will be using manual clustering because, for some geometries, some tetrahedrons may be excluded from element clusters. We also recommend using linear elements rather than quadratic.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Procedural Steps

The following steps describe the Cartesian mesh technique:

1. Set mesh controls for the part. The size of the mesh is required to be the same throughout all part bodies.

   a. Right-click **Mesh** and select **Insert** > **Method**. An Automatic Method object appears under the Mesh object.

   b. Select the part body. In the details pane of the Automatic Method object, click **Apply** next to Geometry.

   c. Select the dropdown in the **Method** option and choose **Cartesian**. Notice the Automatic Method object has been renamed to Body Fitted Cartesian.

   d. In the details pane of the Body Fitted Cartesian object, under Definition, set the **Element Size** to the desired value. (Click in the right column on the word Default to activate the text field.)

   e. Under Advanced, set the **Projection Factor** to a value between 0 and 1. The Projection Factor defines how well the mesh fits to the geometry. A value of 0 results in cubic elements with a rough fit to the geometry. Increasing the Projection Factor changes the shape of the elements to better fit the geometry and may yield better results in some cases, but may also result in an overlapping or failed mesh. Our recommendation is to use 0, or close to 0, initially and then iterate from there to see the effects of changing Projection Factor.

      Too large of a Projection Factor may result in:

      - Highly distorted mesh resulting in cluster layers that are unacceptably uneven: While a small amount of irregularity in the mesh layers is acceptable, if you increase the Projection Factor too much, at some point element clustering is adversely affected, or clusters may even be impossible to generate.

      - Overlapping mesh at the contact region: An overlapping mesh between the part and the base is not desirable. The required contact connection (p. 91) may still be generated but during the solution, the heat does not get properly conducted to the base because of the small negative gap.

   You can apply a Coordinate System for the mesh. Typically, the default global coordinate system is appropriate but you may want to set it to one located on the top surface of the base plate.

   For additional information about Cartesian meshes, see Cartesian Method Control in the *Meshing User's Guide*.

2. Set mesh controls for the base plate. Highlight the **Mesh** object and right-click **Insert** > **Sizing**. A Sizing object is created under the Mesh object and becomes active.

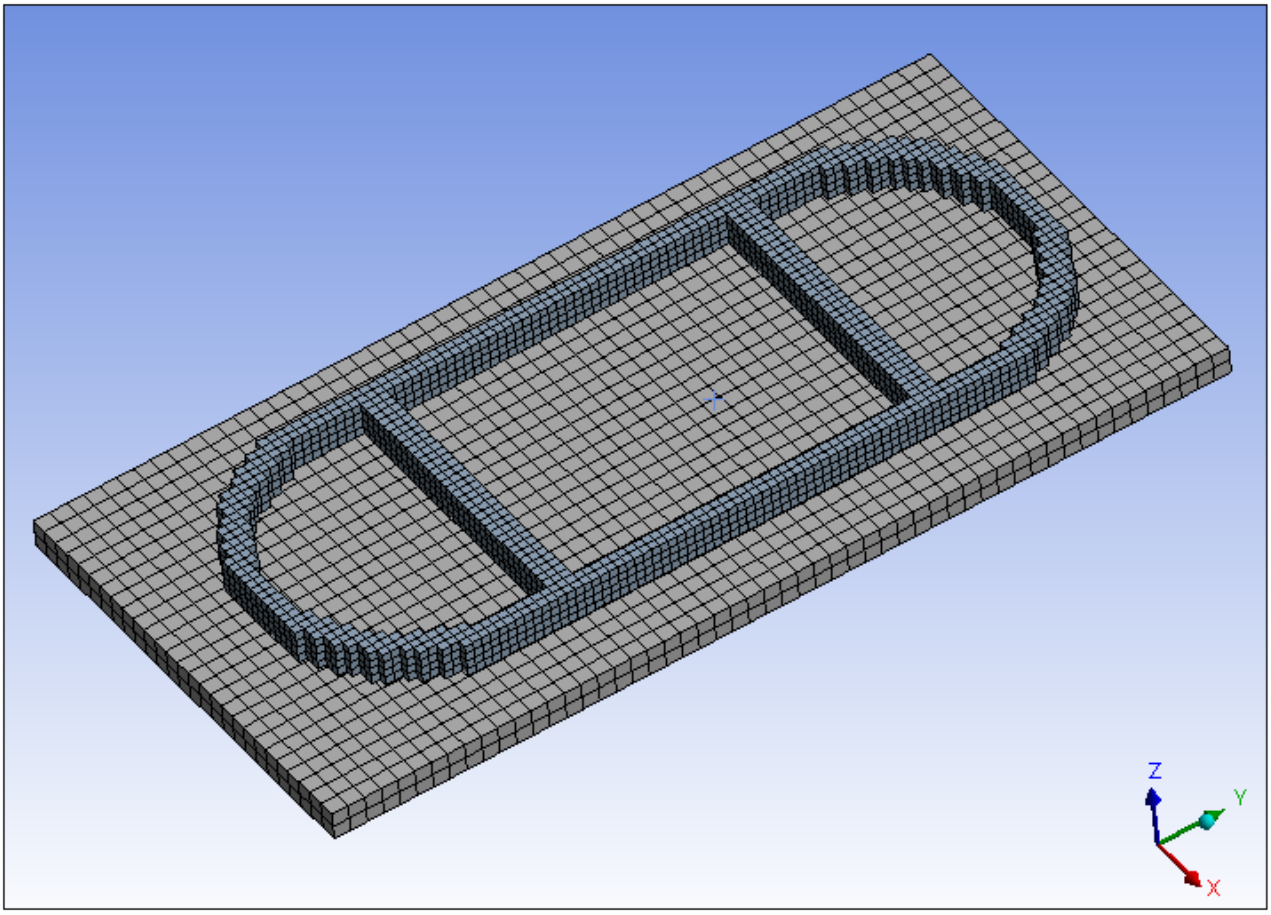
   a. Select the base plate body and hit **Apply** in the Details view.

   b. Under Definition, change the **Element Size** to the desired value. It can be larger than the element size for the part.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

3. Click the **Mesh** object and, in Details, change the Element Order to **Linear**. This will set a global control for all bodies to be meshed.

4. Finally, highlight the **Mesh** object and right-click **Generate Mesh**. If you don't like the generated mesh you can easily go back and change element sizes, then right-click Mesh and select Update.



5.6. Create Named Selections for Contact and Clustering

Now that meshing is complete, it's time to define the Named Selections required for the simulation. The Named Selections feature allows you to create groupings of similar geometry or meshing entities for convenience to be used elsewhere in the simulation. Review the table below to determine which Named Selections are required based on your DED model setup. The Named Selections must be named exactly as they appear in the table. One of the reasons we recommend using the DED Process Wizard is that it determines which of these Named Selections are needed and creates them automatically.

Named Selections Required for DED Simulation		
Name of Named Selection	**Definition**	**Required for**
print_part	The geometry body representing the part	Creating other Named Selections
base_plate	The geometry body representing the base	Creating other Named Selections

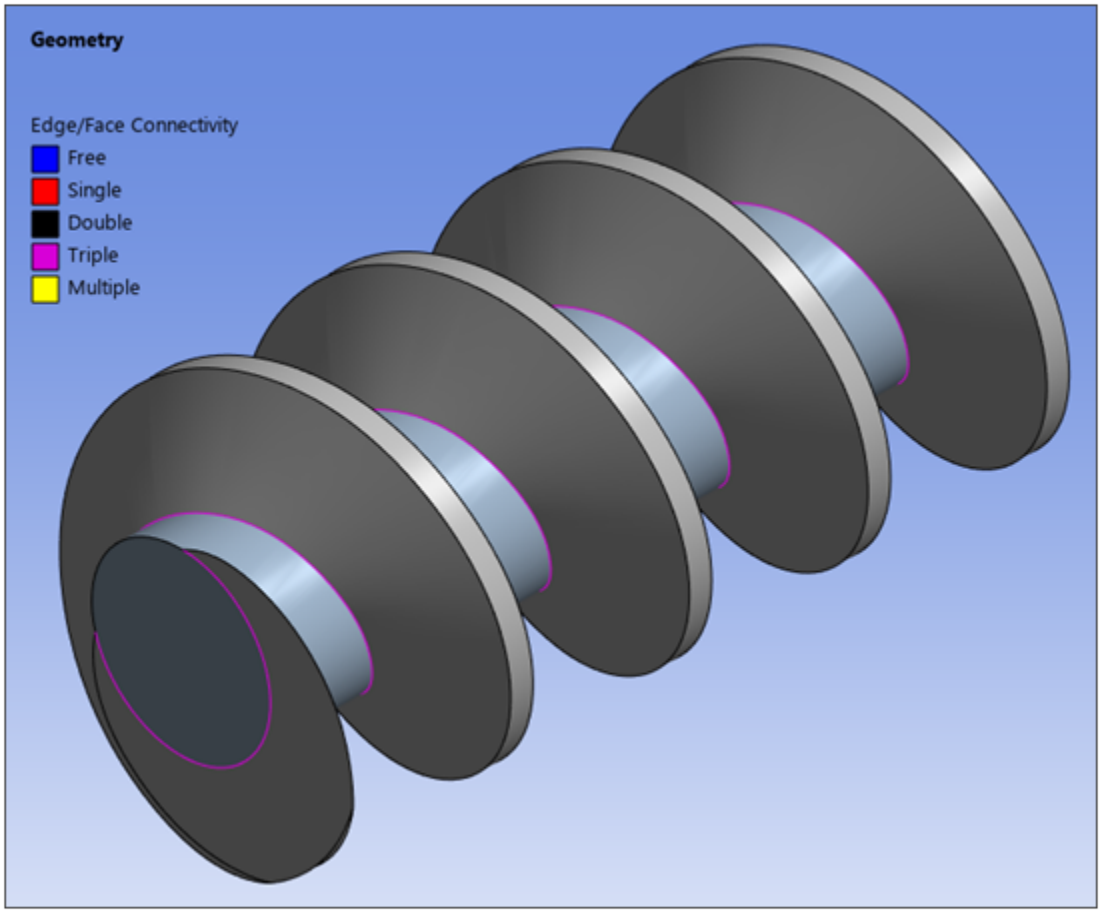
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Named Selections Required for DED Simulation		
Name of Named Selection	**Definition**	**Required for**
part_contact_face	The faces of the part that will be in contact with the base	Contact generation for the following scenarios: • Shared topology, or • Unshared topology with a curved base
base_contact_face	The faces of the base that will be in contact with the part	Contact generation for the following scenarios: • Shared topology, or • Unshared topology with a curved base
Print part lower element faces	The *element faces* of the part(s) that will be in contact with the base	Contact generation with unshared topology for both planar and curved bases
Base plate upper element faces	The *element faces* of the base that will be in contact with the part	Contact generation with unshared topology for both planar and curved bases
Part Elements	All elements in the part body	Generating clusters for multi-axis simulation
Part Nodes	All nodes in the part body	Generating clusters for multi-axis simulation
Part Contact Elements	Elements on the surfaces of the part that will be in contact with the base	Cluster connectivity check for multi-axis simulation. The way to create this is different depending on shared or unshared topology.

Example: Named Selection for Curved Base Multi-axis Model with Shared Topology

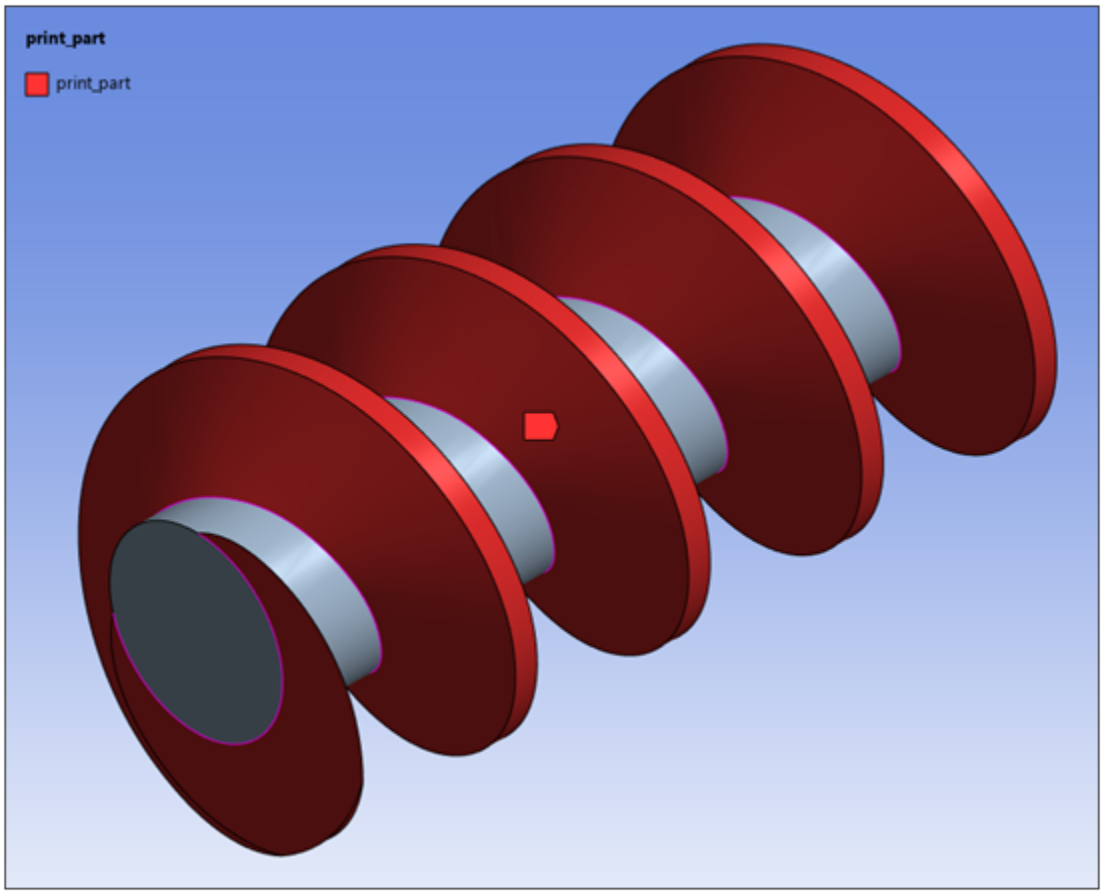
The following example shows the Named Selections defined for a screw-shaped model that has shared topology between part and base bodies. The first image shows the shared edges between the part and base. A pink edge indicates it is shared by three faces (labeled Triple). The three faces are two faces belonging to the part and one face belonging to the base.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



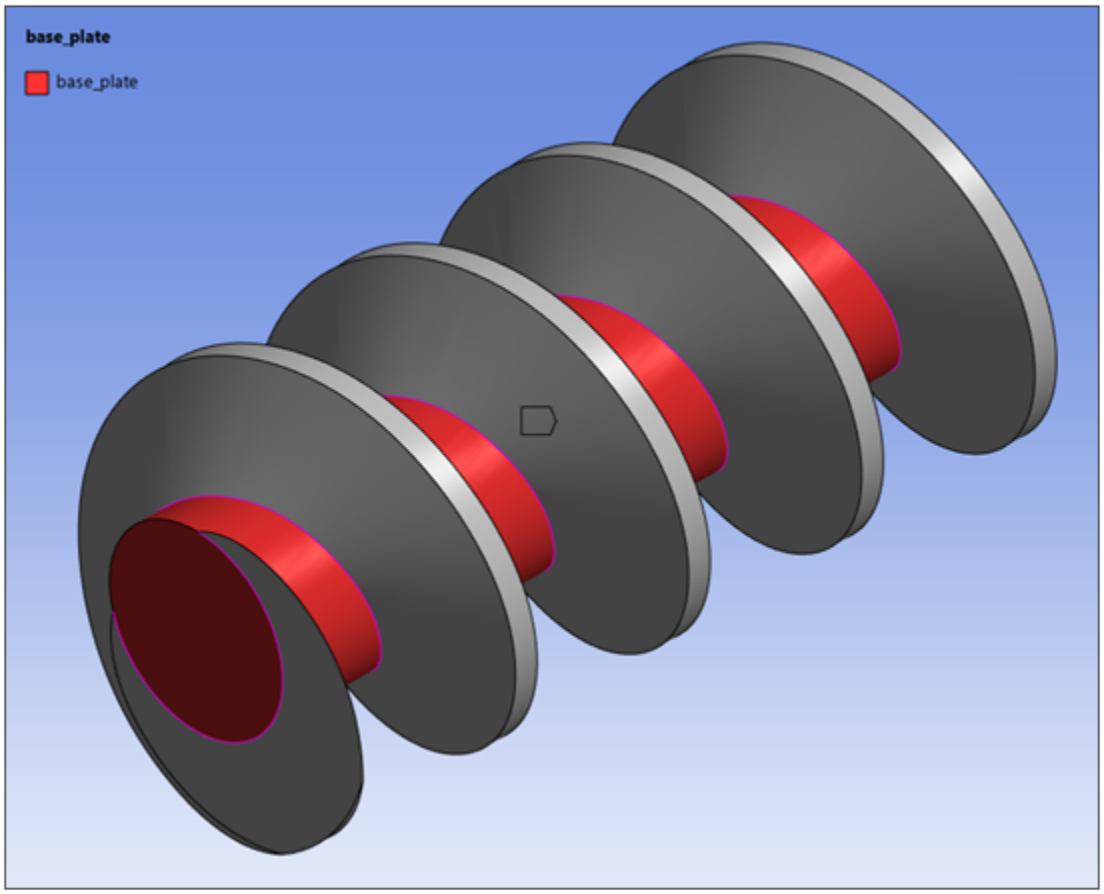
print_part: This Named Selection is easily created using geometry selection.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



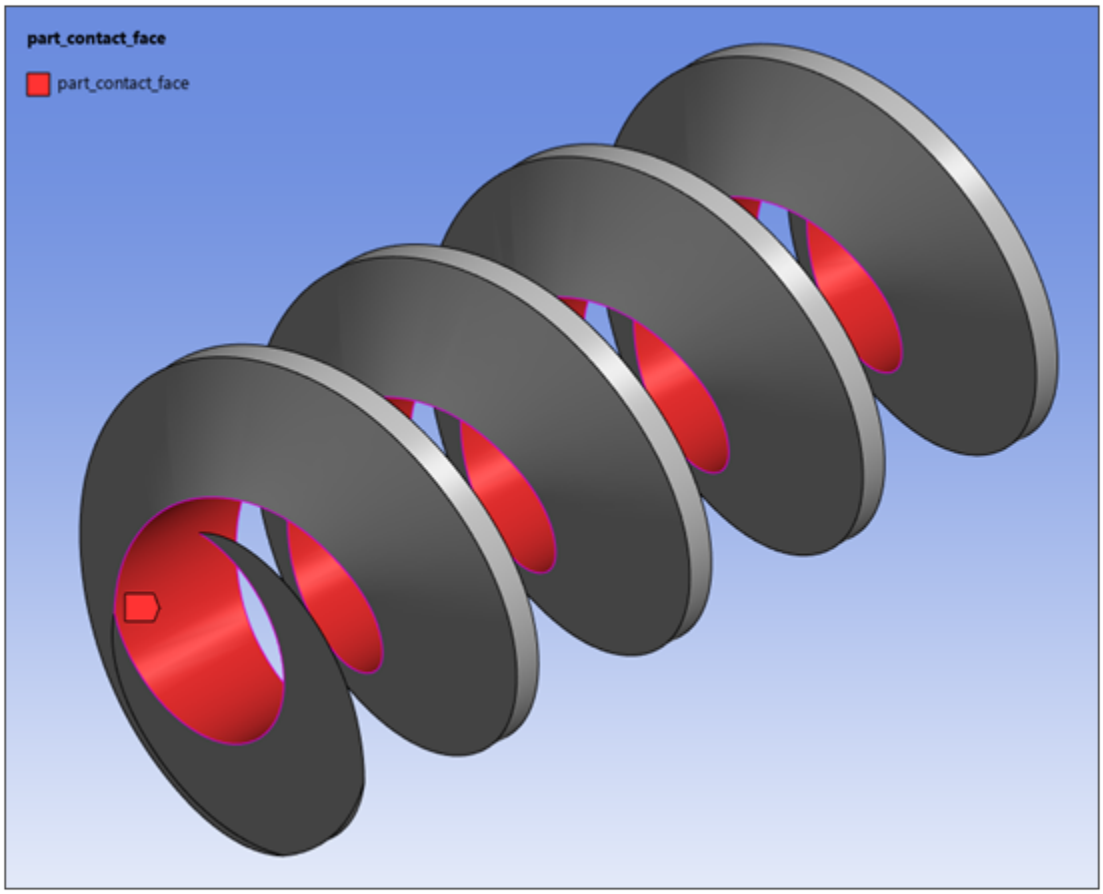
base_plate: This Named Selection is easily created using geometry selection.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



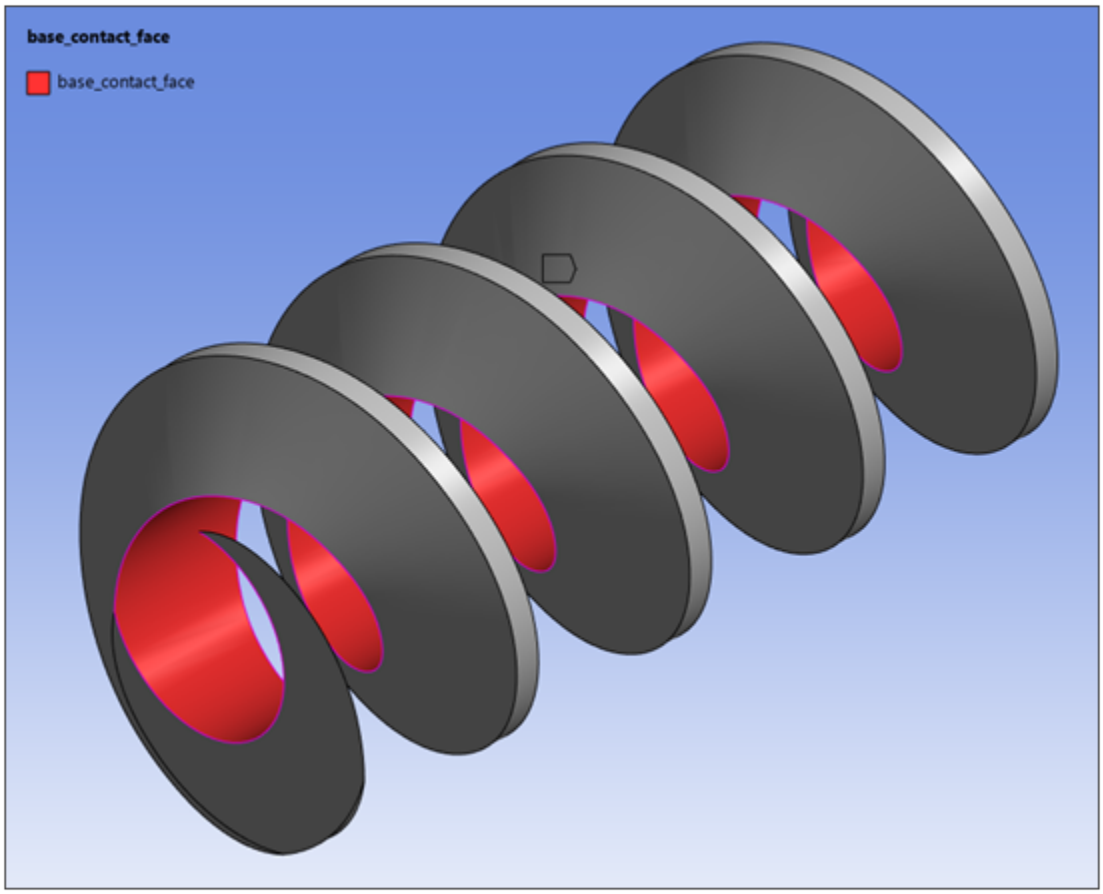
part_contact_face: This Named Selection is created using geometry selection with the base body hidden.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information
of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



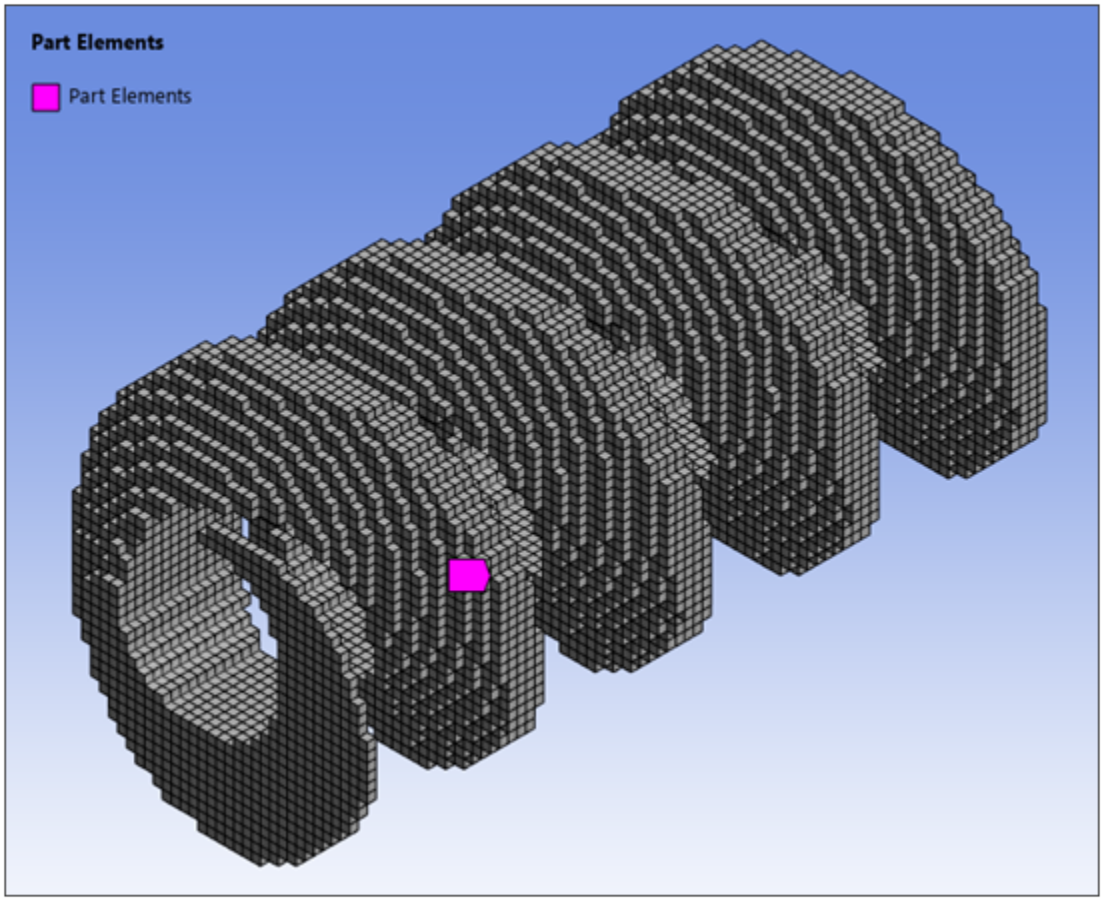
base_contact_face: This Named Selection is created using geometry selection with the base body hidden.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Part Elements: This Named Selection is required because it is a multi-axis simulation. We create the Named Selection by using a worksheet. Notice that the elements are voxels (Cartesian mesh with Projection Factor = 0).

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Worksheet

Part Elements

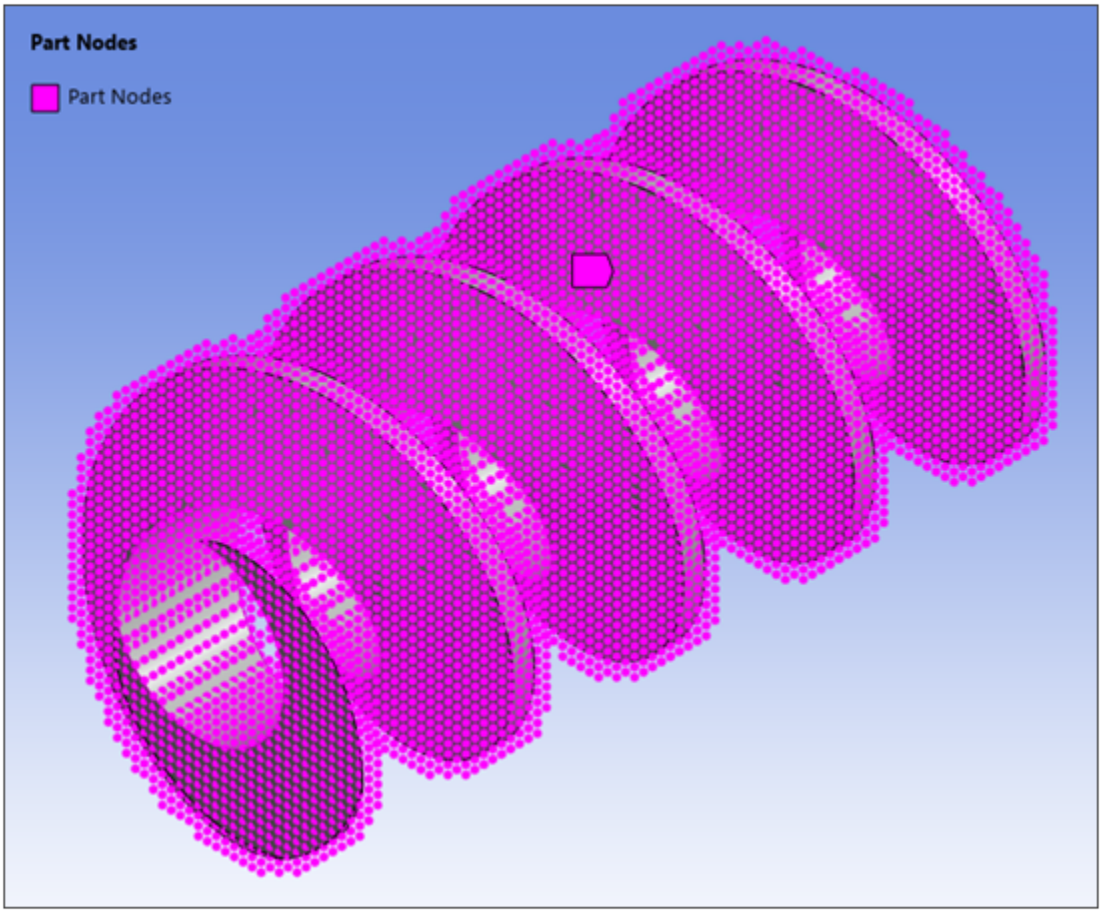
Generate

Note: Internal comparisons of values that have units are done in the CAD Unit System. See help for more information.
Current CAD Unit System: Metric (m, kg, N, s, V, A)

	Action	Entity Type	Criterion	Operator	Units	Value	Lower Bound	Upper Bound	Coordinate ...
☑	Add	Body	Named Sele...	Equal	N/A	print_part	N/A	N/A	N/A
☑	Convert To	Mesh Element	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Part Nodes: This Named Selection is required because it is a multi-axis simulation. We create the Named Selection by using a worksheet.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Worksheet

Part Nodes

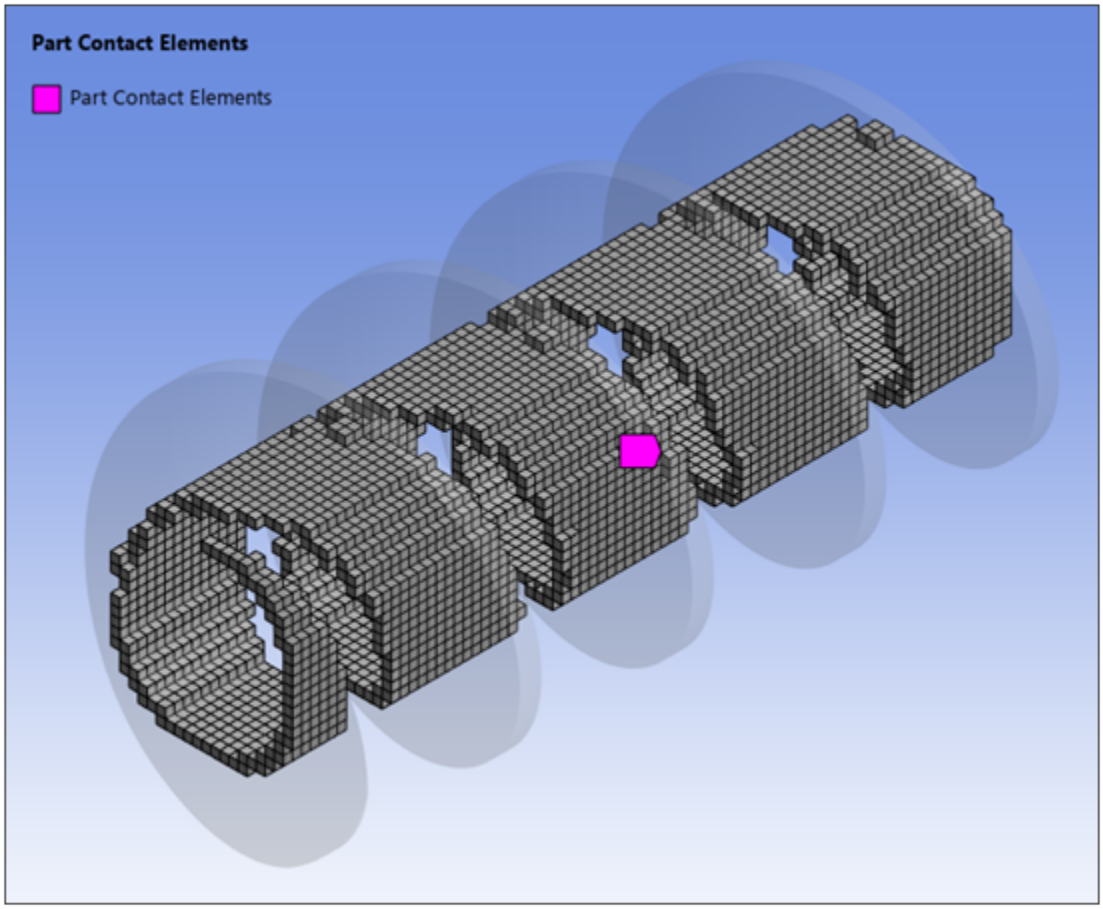
Generate

Note: Internal comparisons of values that have units are done in the CAD Unit System. See help for more information.
Current CAD Unit System: Metric (m, kg, N, s, V, A)

	Action	Entity Type	Criterion	Operator	Units	Value	Lower Bound	Upper Bound	Coordinate ...
☑	Add	Body	Named Sele...	Equal	N/A	print_part	N/A	N/A	N/A
☑	Convert To	Mesh Element	N/A	N/A	N/A	N/A	N/A	N/A	N/A
☑	Convert To	Mesh Node	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Part Contact Elements: This Named Selection is required because it is a multi-axis simulation. We create the Named Selection by using a worksheet.

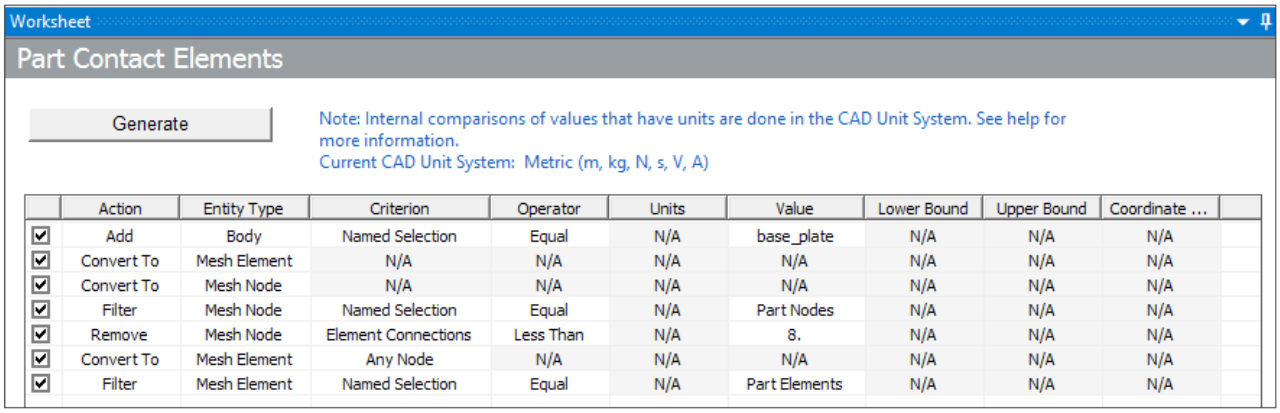
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



There are several actions required to isolate the part's elements that will be in contact:

1. Add the base plate body.
2. Convert the body selection to its elements.
3. Convert the elements selection to their nodes. Now we have the nodes of the base. Because of shared topology, some of these nodes are shared with the part.
4. Filter the nodes to just those included in the part. Remember that these nodes are connected to both base and part elements so now we must narrow our selection to just the elements of the part.
5. Remove any nodes that are connected with less than 8 elements. Since multi-axis simulations are limited to Cartesian mesh type, 8 is the maximum number of elements a node can connect with. This helps us to identify the innermost layer of *part* elements in contact with the base and sets us up to eliminate the base elements.
6. Convert the nodes selection to their elements.
7. Filter the elements to just those included in the part. Now we have the part's elements that will be in contact.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.


Worksheet

Part Contact Elements

Generate

Note: Internal comparisons of values that have units are done in the CAD Unit System. See help for more information.
Current CAD Unit System: Metric (m, kg, N, s, V, A)

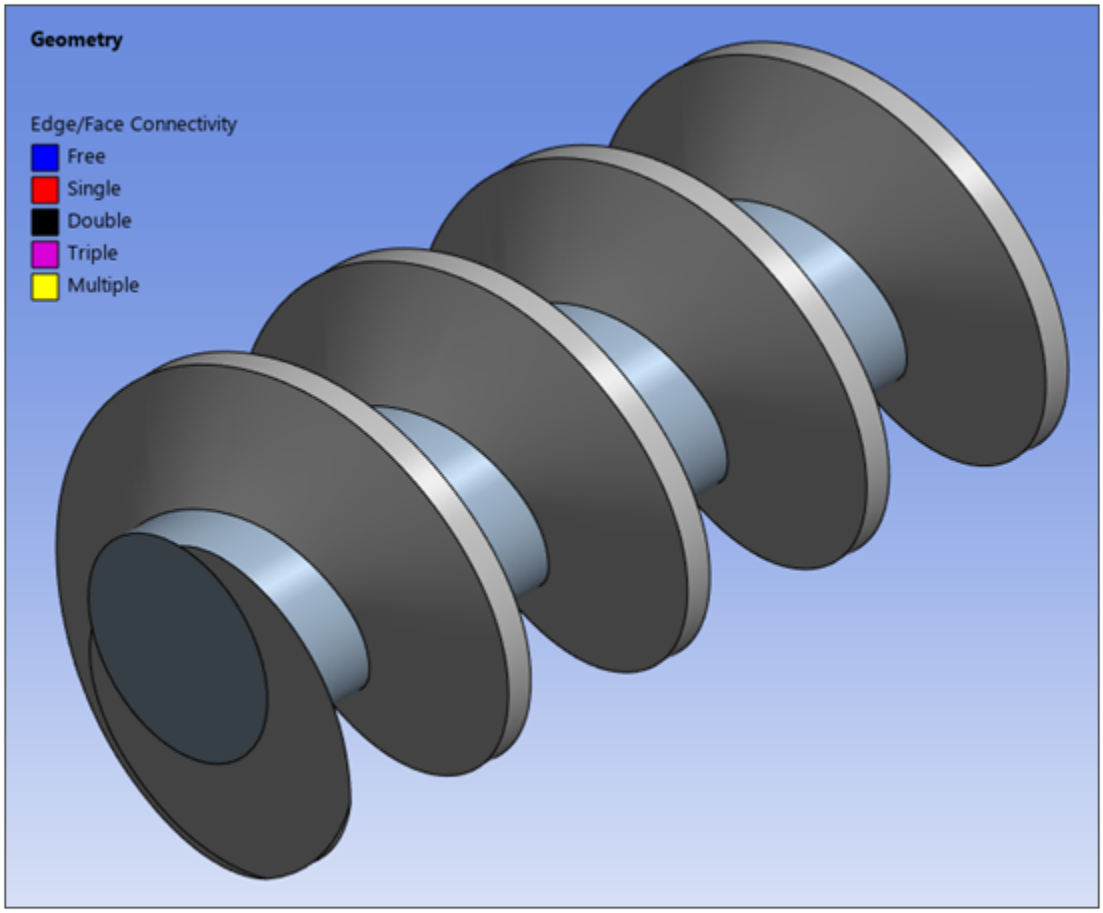
	Action	Entity Type	Criterion	Operator	Units	Value	Lower Bound	Upper Bound	Coordinate ...
☑	Add	Body	Named Selection	Equal	N/A	base_plate	N/A	N/A	N/A
☑	Convert To	Mesh Element	N/A	N/A	N/A	N/A	N/A	N/A	N/A
☑	Convert To	Mesh Node	N/A	N/A	N/A	N/A	N/A	N/A	N/A
☑	Filter	Mesh Node	Named Selection	Equal	N/A	Part Nodes	N/A	N/A	N/A
☑	Remove	Mesh Node	Element Connections	Less Than	N/A	8.	N/A	N/A	N/A
☑	Convert To	Mesh Element	Any Node	N/A	N/A	N/A	N/A	N/A	N/A
☑	Filter	Mesh Element	Named Selection	Equal	N/A	Part Elements	N/A	N/A	N/A

For this example, the Named Selections of **Print part lower element faces** and **Base plate upper element faces** are not needed because the model has shared topology.

Example: Named Selection for Curved Base Multi-axis Model with Unshared Topology

The following example shows the Named Selections defined for the same screw-shaped model but this time with unshared topology between part and base bodies. The first image shows there are no shared edges between the part and base. A black edge indicates it is shared by two faces (labeled Double). The two faces are the two belonging to the part.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



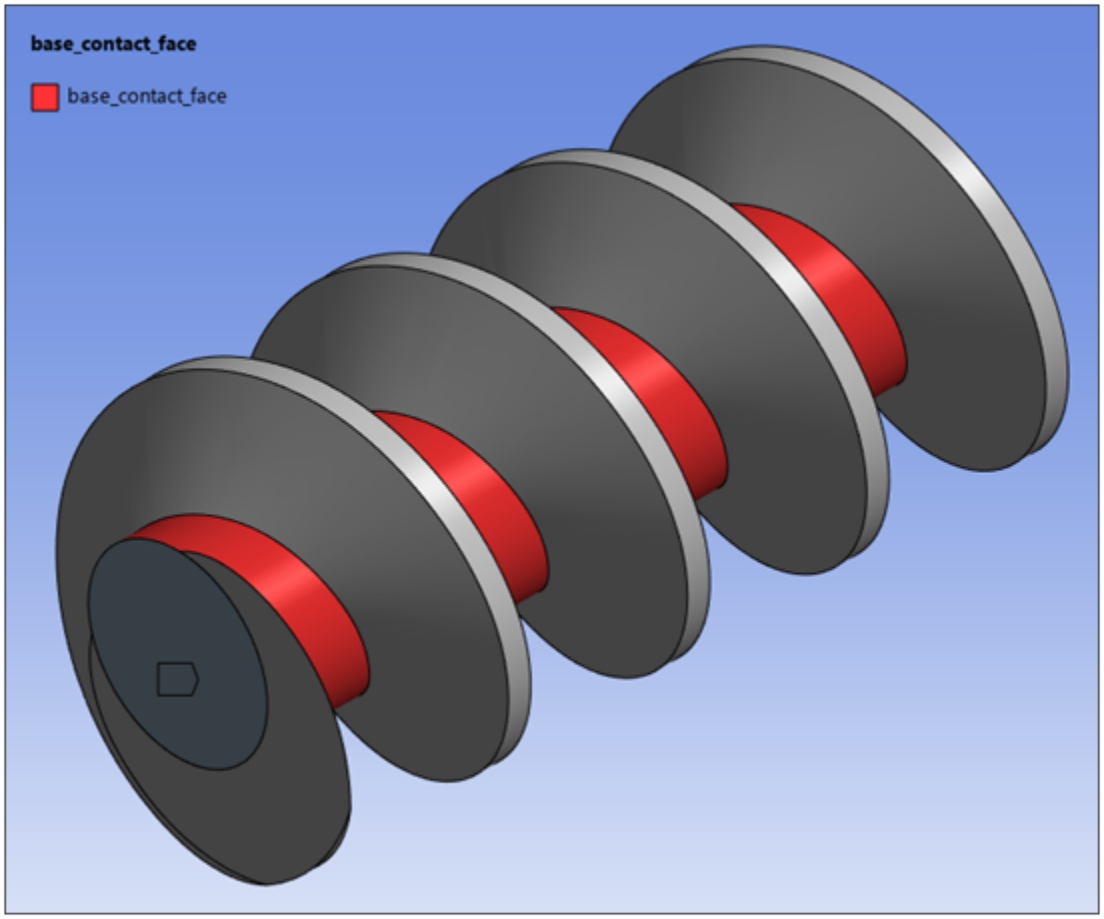
print_part: Same as shared topology example above.

base_plate: Same as shared topology example above.

part_contact_face: Same as shared topology example above.

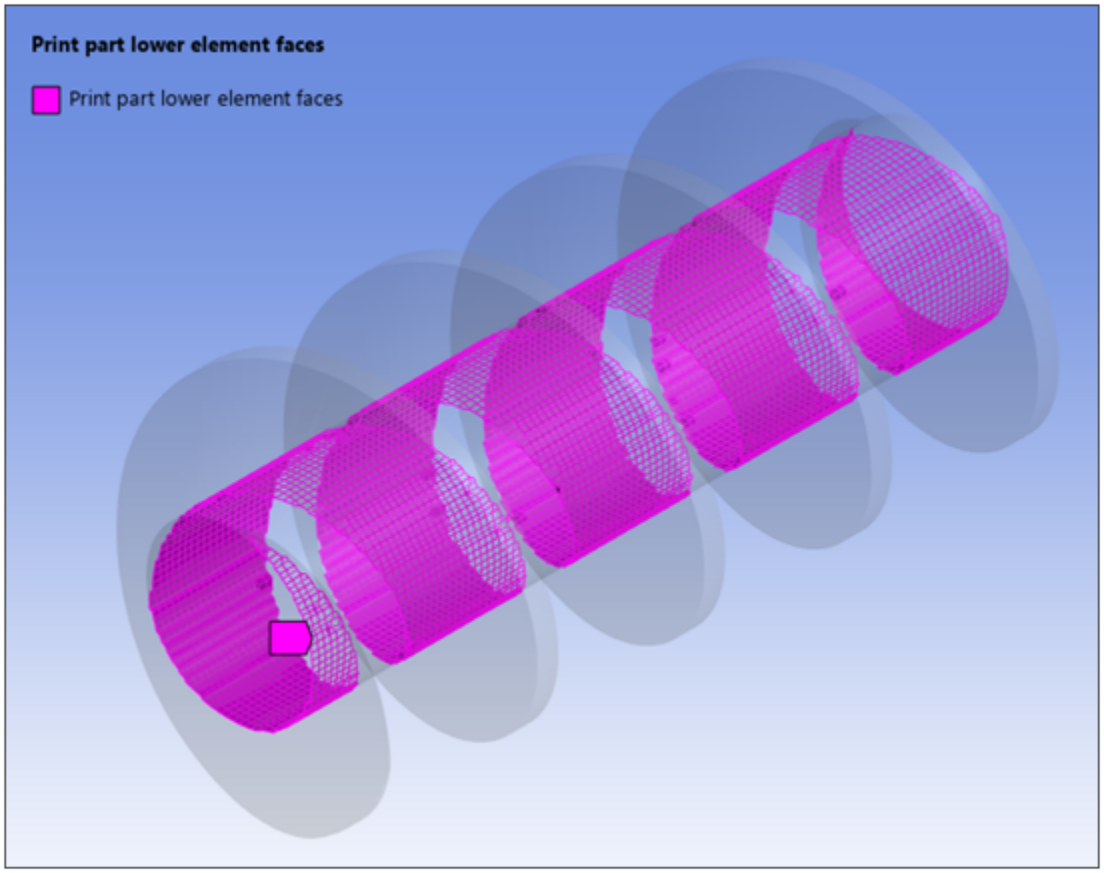
base_contact_face: This Named Selection is created using geometry selection.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Print part lower element faces: This Named Selection is created using a worksheet.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Worksheet

Print part lower element faces

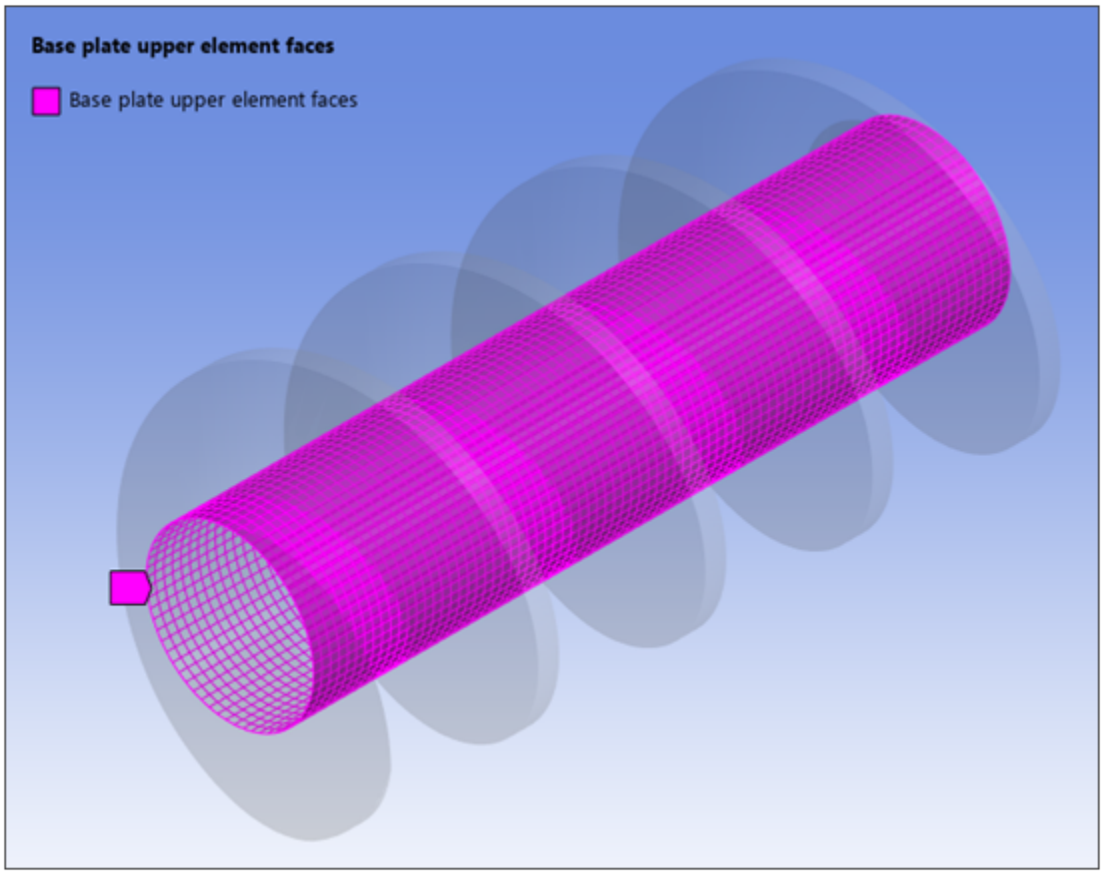
Generate

Note: Internal comparisons of values that have units are done in the CAD Unit System. See help for more information.
Current CAD Unit System: Metric (m, kg, N, s, V, A)

	Action	Entity Type	Criterion	Operator	Units	Value	Lower Bound	Upper Bound	Coor
☑	Add	Face	Named Sele...	Equal	N/A	part_contact_face	N/A	N/A	
☑	Convert To	Element Face	N/A	N/A	N/A	N/A	N/A	N/A	

Base plate upper element faces: This Named Selection is created using a worksheet.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Worksheet

Base plate upper element faces

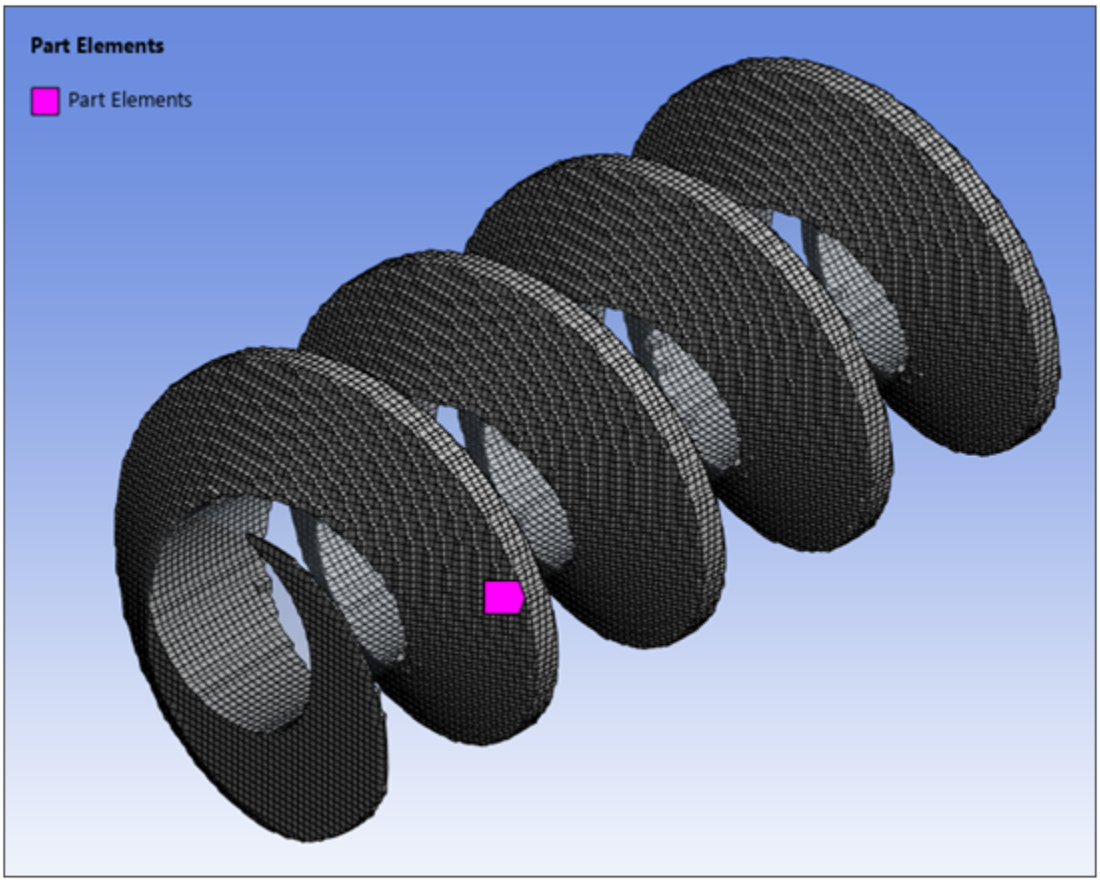
Generate

Note: Internal comparisons of values that have units are done in the CAD Unit System. See help for more information.
Current CAD Unit System: Metric (m, kg, N, s, V, A)

	Action	Entity Type	Criterion	Operator	Units	Value	Lower Bound	Upper Bound	Coor
☑	Add	Face	Named Sele...	Equal	N/A	base_contact_face	N/A	N/A	
☑	Convert To	Element Face	N/A	N/A	N/A	N/A	N/A	N/A	

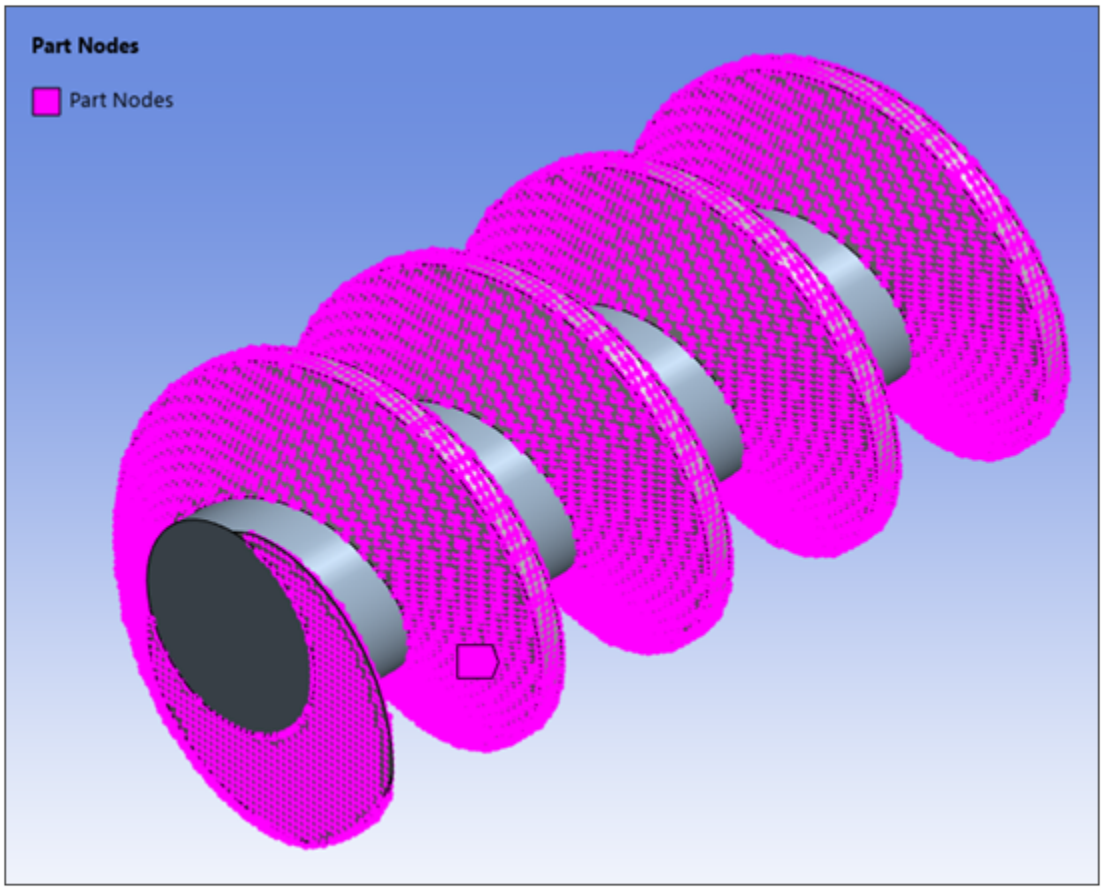
Part Elements: This Named Selection is created using a worksheet, the same way as in the shared topology example above. Notice that the elements are well-fitted hex elements (Cartesian mesh with Projection Factor = 0.8). Because of unshared topology, we used a Projection Factor (p. 38) other than zero to improve contact quality between the build and the base.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



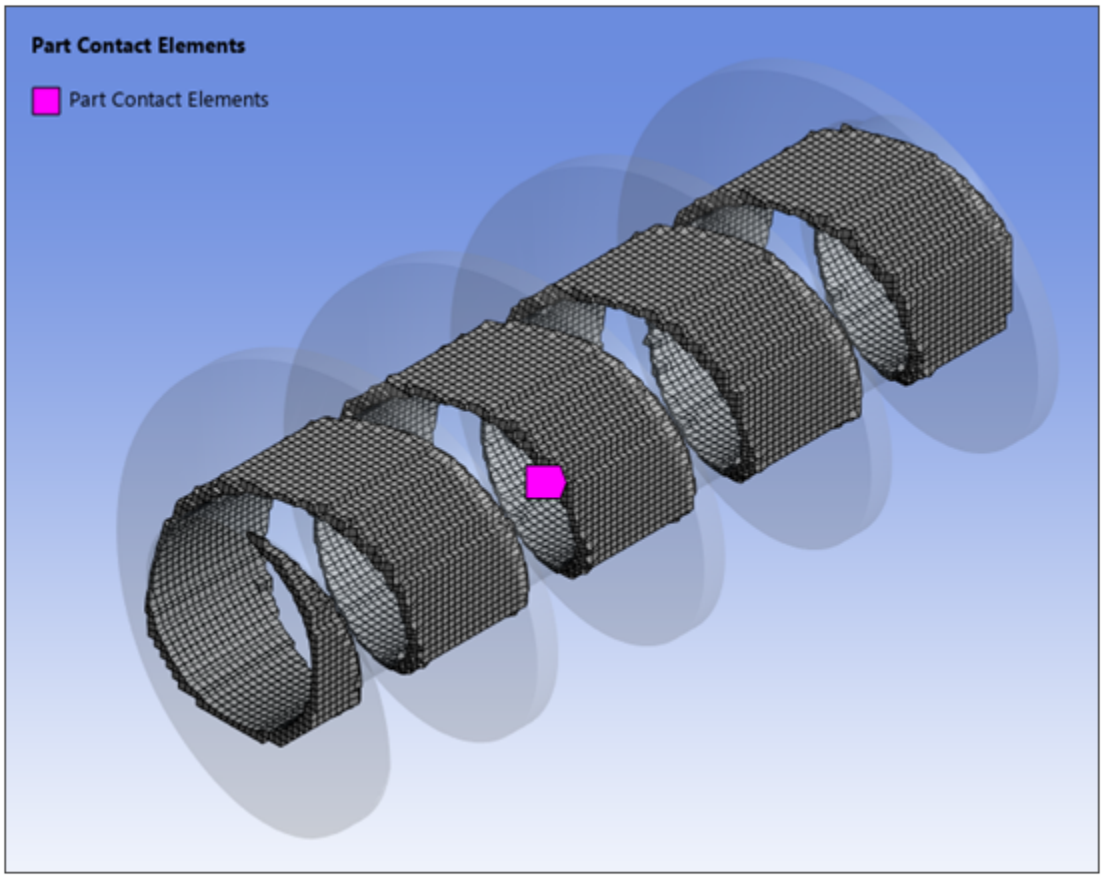
Part Nodes: This Named Selection is created using a worksheet, the same way as in the shared topology example above.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Part Contact Elements: This Named Selection is created using a worksheet, with *different* actions than in the shared topology example above. In this case it is relatively easy to isolate the part's elements that will be in contact with the base by using the already defined Print part lower element faces Named Selection.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Worksheet

Part Contact Elements

Generate

Note: Internal comparisons of values that have units are done in the CAD Unit System. See help for more information.
Current CAD Unit System: Metric (m, kg, N, s, V, A)

	Action	Entity Type	Criterion	Operator	Units	Value	Lower Bound	Upper Bound	Coordinate ...
☑	Add	Element Face	Named Sele...	Equal	N/A	Print part lower element faces	N/A	N/A	N/A
☑	Convert To	Mesh Node	N/A	N/A	N/A	N/A	N/A	N/A	N/A
☑	Convert To	Mesh Element	Any Node	N/A	N/A	N/A	N/A	N/A	N/A

5.7. Define Build Settings

In this step, specify the settings and conditions related to the machine and the process, grouped into the following three categories:

Machine Settings refer to process parameters that vary for each DED machine as well as for the material used in the deposition process.

- Material Deposition Rate: The feeding rate of the melted material, in mm^3/sec. This value can be determined by multiplying *layer thickness* (mm) x *weld width* (mm) x *deposition speed* (mm/sec).

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Build Conditions are the settings pertaining to the environment in the build chamber around the part as it is being printed, including the preheat temperature. Multiple convection coefficient and emissivity inputs are available to allow for the possibility of different materials and/or conditions for the part and the base plate.

- Preheat Temperature: The starting temperature of the base plate.
- Scoping Method: Choose to select geometry directly in the geometry window or use a named selection to identify the bottom of the base plate that will be preheated.
- Preheat Geometry: The bottom face of the base plate.
- Preheat During Printing: If set to "Off (default)," the preheat temperature is applied in only the first load step. Option "On" maintains the preheat temperature throughout the complete build process.
- Heating Method: Controls how new clusters are heated in the transient thermal analysis. Choose Temperature Based (default) or Power Based.
  - Temperature Based: Heating is based on the process temperature.
    - → Process Temperature: The temperature of the feed material as it is exiting the nozzle—usually melting temperature or above. (When using the DED Process wizard, this value automatically defaults to the build material's melting temperature.)
  - Power Based: Heating is based on laser power and material absorptivity.
    - → Power: The power of the laser in Watts.
    - → Absorptivity: The average fraction of energy that is absorbed by the deposited material and contributes to the heating process. Value must be between 0 and 1. Defaults to 0.4, or 40%.
- Room Temperature: Temperature of the gas in the build chamber during the build.
- Gas Convection Coefficient: Convection coefficients from the part and base plate surfaces to the surrounding gas in the chamber during printing. Convection is applied to the element faces of every element cluster, which can be especially relevant in the case of a bulky structure.
- Radiation: Set to Off (default) or On to include the effect of radiation in the simulation. Define emissivity if Radiation = On. Turning on radiation effects may provide higher accuracy but at significant computational cost.
- Thermal Calibration: Off (default)/On - If set to Off, default heating and cooling behavior is used for cluster deposition. Setting the option to On allows you to change the Cluster Cooling Time Ratio, essentially using it as a calibration factor.

  Cluster Cooling Time Ratio: A value between 0 and 1 (default). The total deposition time, t, for each cluster is determined by the volume of the cluster and the deposition rate. By default, a very tiny fraction of this total time (1e-6 seconds) is used to raise the temperature to the Process Temperature, and the remainder is used for cooling. Use the Cluster Cooling Time Ratio to adjust the heating and cooling distribution. Lowering this value from the default of 1 has the effect of lowering cooling time, thereby increasing heating time. For example, a value of 0.8 means 20% of time, t, is used for heating and 80% for cooling (during printing). The lower the number, the higher the temperature spikes that will exist in the thermal history across the part.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Cooldown Conditions are the settings pertaining to the environment in the build chamber around the part in the cooldown step after the last layer is printed.

- Room Temperature: Temperature of the gas in the build chamber during cooldown.
- Gas Convection Coefficient: Convection coefficient from the part to the surrounding gas in the chamber during cooldown.
- Time: The duration of the cooldown process.

Tip:

To neglect convection for testing purposes, set the Gas Convection Coefficient for both the build and cooldown conditions to very small values.

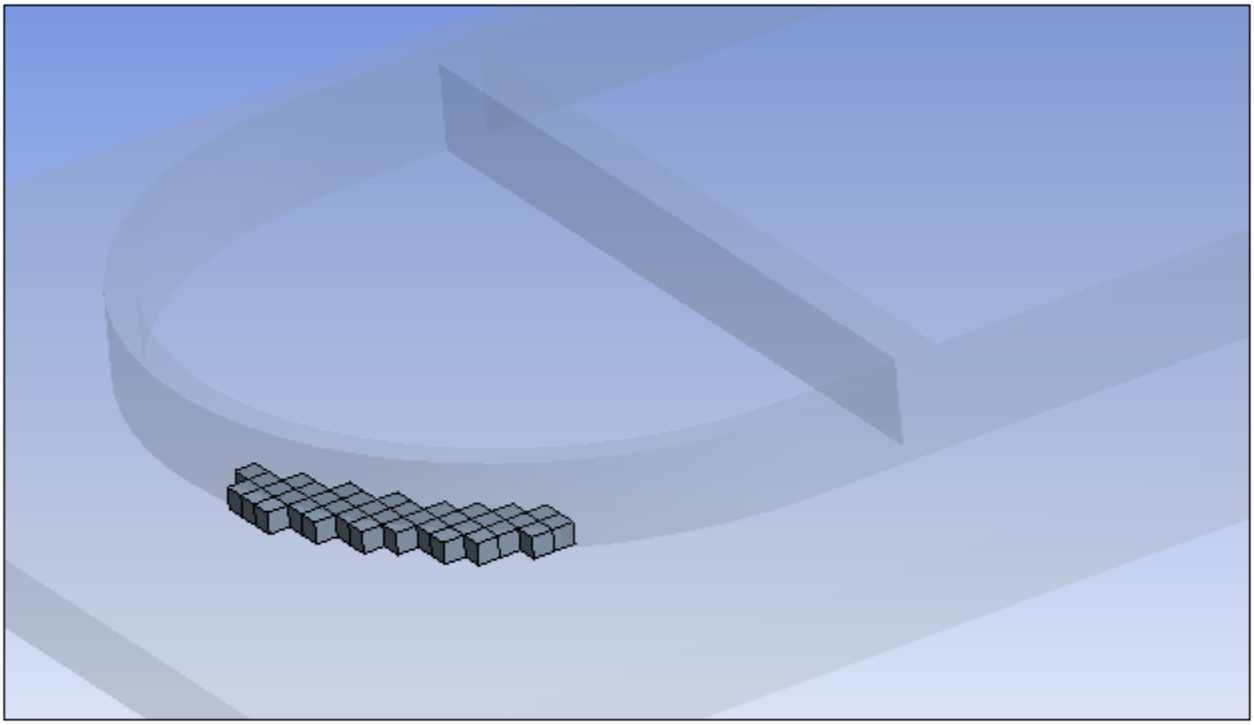
Cooldown Between Layers are options to adjust the dwell time, which is the pause, in seconds, between *layers* as measured after the last cluster in a layer is deposited, not including laser travel time. Two methods are available, Time Based and Temperature Based.

- Time Based is a method allowing direct input of dwell time, in seconds or milliseconds, from the G-Code file which is then interpreted through the element clustering process (p. 59). Dwell time for individual *clusters* is modifiable in the cluster settings table (p. 84). You can use the convenient "List only last cluster in each layer" filter to manually adjust dwell time between layers.
- Temperature Based uses a Target Temperature together with time step and convergence options to automatically adjust dwell time. See Cooldown Between Layers options in the Build Settings object (p. 111). With this method, the ability to manually adjust dwell time for individual *clusters* is disabled.

5.8. Perform Element Clustering

Clustering is used to split weld lines into smaller pieces of mesh clusters that are exposed to one temperature in a time step. In the racetrack example that has four layers, one cluster in one layer is shown here.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Cluster volume, in mm^3, is used to control the size of the clusters, and hence, has a direct influence on simulation time. This value determines how many elements are activated per load step, the time for this load step is then determined by volume/deposition rate. A smaller cluster volume tends to provide a more accurate result. Based on the overall dimension of the build geometry, this value should be determined by balancing the computational cost and desired accuracy.

Set cluster volume based on the total volume of your part. If, for instance, your build part has a total volume of 10,000 mm^3 and you define a cluster volume of 100 mm^3, the clustering process will generate approximately 100 element clusters.

Two approaches are available to define element clusters, either manually or using G-Code. For both of these approaches, a license check for the Additive Suite license is done when generating element clusters. Upon completion of cluster generation, you can use the Cluster Settings table to adjust machine parameters for individual element clusters, as needed.

See the following topics:

5.8.1. Element Clustering: Manual Approach

5.8.2. Element Clustering: G-Code Approach

5.8.3. Editing Individual Clusters for More Precise Control of Process Settings

5.8.1. Element Clustering: Manual Approach

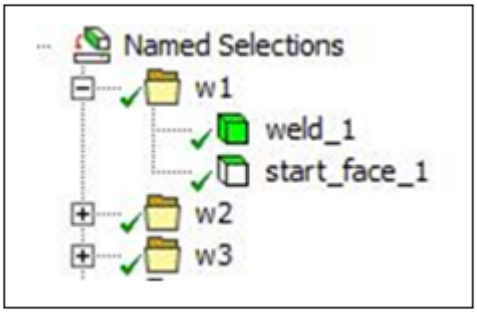
In this approach to element clustering, you manually define named selections in a very specific pattern and with a specific naming convention. Every weld track must be defined as a *body* named selection with the name "weld_X" where X is the number of the weld track describing the weld track order. Specifically, the name for the first weld track named selection must be weld_1, the name of the second weld track named selection must be weld_2, and so on.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Additionally, every weld track needs to have a *face* named selection as a start face called “start_face_X” where X is the number of the weld track describing the weld track order.

To summarize, for every body named selection weld_X there must be a face named selection start_face_X.

For the sake of clarity, weld_X and start_face_X can be, but do not have to be, grouped together.

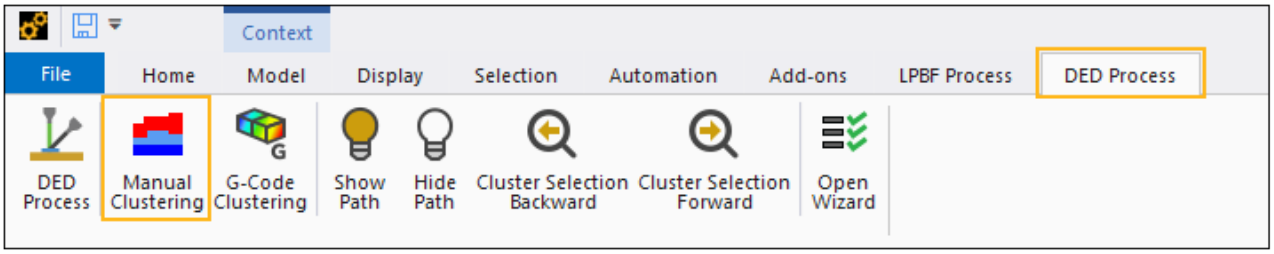


You must set a Cluster Volume per simulation step (default: 20 mm³) to determine the element cluster size.

Procedural Steps

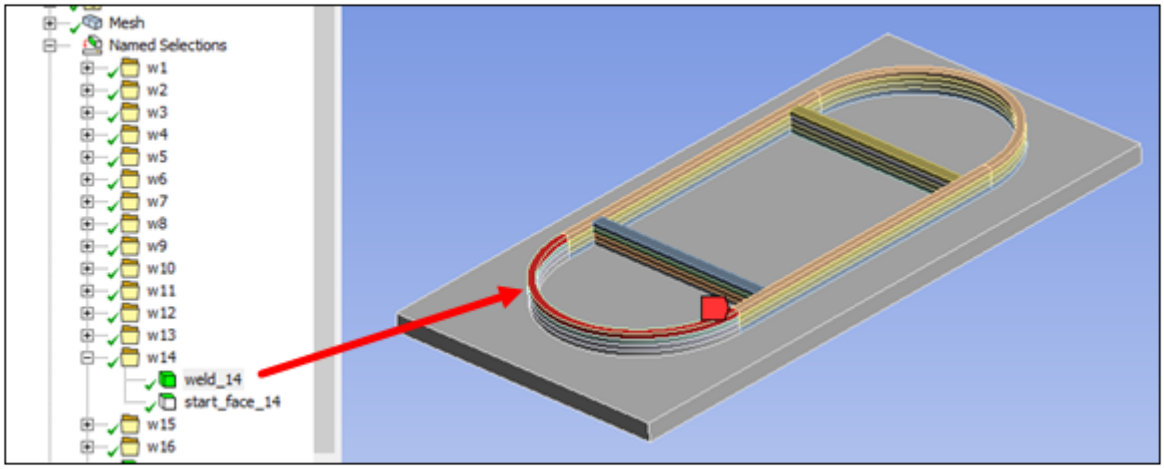
To create element clusters using the manual approach:

1. Select the **Manual Clustering** button in the **DED Process** ribbon. A **Manual Clustering** object is inserted below the **AM Process for DED** object in the project tree.



2. In the Details view of the Manual Clustering object, define a **Cluster Volume** per simulation step.
3. Select a **Print Direction** (default: Z direction)
4. Right-click Manual Clustering and click **Generate**.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



At the beginning of every weld track a new cluster will be created. In most cases, one or more element clusters with a predefined element cluster size do not fit perfectly into the weld track. That is why at the end of a weld track there may be small clusters with a much smaller cluster size than defined.

5. Use the buttons **Cluster Selection Forward** and **Cluster Selection Backward** in the **AM Process for DED** toolbar in the ribbon to display the clustering sequence in the geometry window.

6. The Cluster Settings object is automatically inserted under the AM Process for DED (p. 109) object in the project tree together with the Manual Clustering object. Click **Cluster Settings** to use a table to adjust machine parameters for individual element clusters as needed. For details, see Using the Cluster Settings Table (p. 84).

5.8.2. Element Clustering: G-Code Approach

G-Code is the most commonly used programming language for computer numerical controlled (CNC) machines. It is used to control cutting tools within a machine tool (like a lathe or mill) as well as non-cutting tools used for additive methods like DED. Within a DED simulation, we use the G-Code commands as a basis for creating clusters of elements that represent the building sequence of the part. Since G-Code can contain extensions and variations specific to machine tool manufacturers and control manufacturers, we focus on basic commands for linear movement (G00 and G01) and circular movement (CIP), and assume a planar tool path in the X-Y, X-Z, or Y-Z plane, normal to print direction. See Supported G-Code Commands (p. 103).

The general steps to create element clusters based on a G-Code file are:

1. Add G-Code Clustering object and import G-Code file. This step is done automatically if you use the DED Process Wizard to set up the simulation.

2. Visualize the G-Code path.

3. Define additional clustering and G-Code options.

4. Translate or rotate the G-Code in relation to the build part, as needed.

5. Perform element clustering.

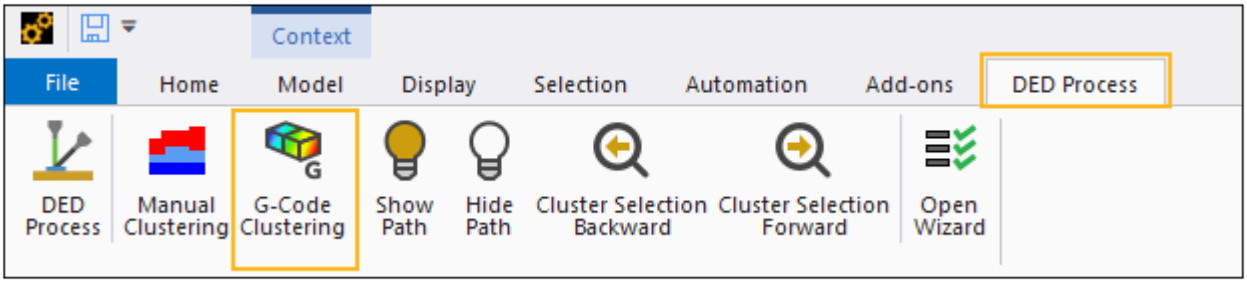
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

6. Review the cluster progression after element clustering is completed.

Procedural Steps

1. **Add G-Code Clustering object and import G-Code file**

   a. Select the **G-Code Clustering** button in the DED Process ribbon. A G-Code Clustering object is inserted below the AM Process for DED (p. 109) object in the project tree.



   b. In the Details view of the G-Code Clustering object, click **G-Code File** to browse to the G-Code file. The G-Code file must be in ASCII format. Select the G-Code file and then click **Open** in the file selection dialog.

2. **Visualize the G-Code path**

Important:

Don't skip this step. It is your responsibility to visually verify that the G-Code path is correct for the model and is correctly aligned (p. 67) with the part. If the simulation is run with an inappropriate G-Code path for the model, the solution will be inaccurate.

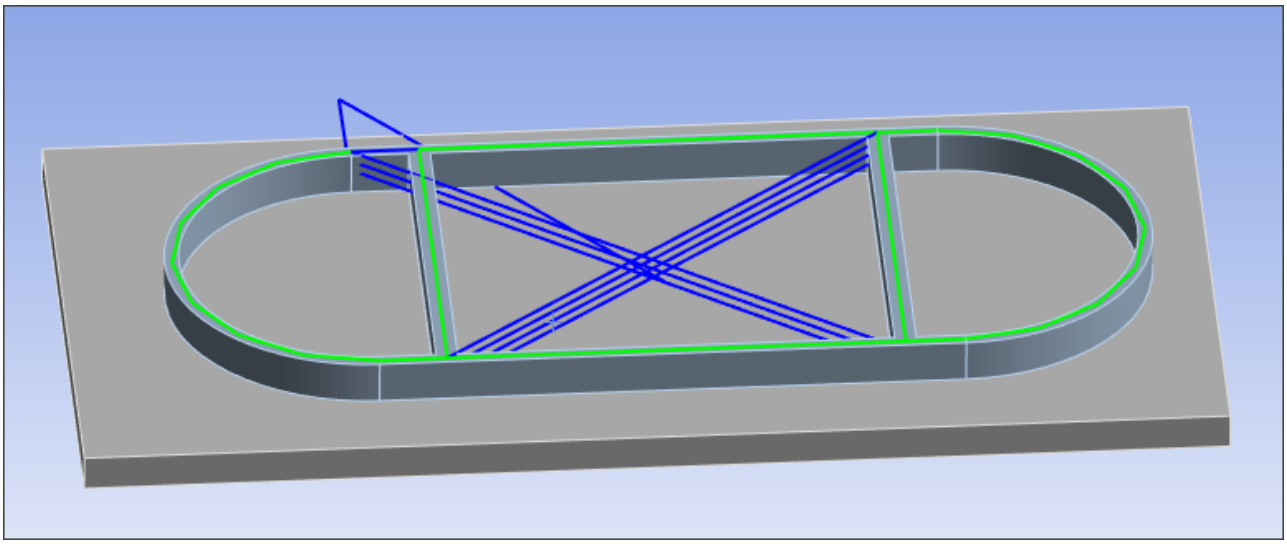
You should visualize the G-Code path to check the correct positioning of the build part with respect to the G-Code positions. Note that the following G-Code Path Visualization settings in the G-Code Clustering object have no influence on element clustering itself.

   a. Click **Show Path** in the AM Process for DED toolbar in the ribbon. The G-Code path is displayed as blue and green lines in the geometry window. (Note: It may take some time to read the G-Code file and display the path.)

      - **Blue lines** show rapid movement commands in the G-Code. With this type of movement, no material is deposited, and these paths are also not used in the subsequent element clustering.

      - **Green lines** show printing commands in the G-Code. With this type of movement, material is deposited, and these paths are used in subsequent element clustering. In the visualization, the green line appears at the *top* of each deposition layer.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



b. Click **Hide Path** in the toolbar to clear the G-Code path display from the geometry window.

c. You can adjust the visualization display by changing the following settings in the Details view of the G-Code Clustering object, under **G-Code Path Visualization**:

- **Path Layers to Show**: Path layers from the G-Code file to display. Enter a range of values with a hyphen and multiple values separated by commas. For example, *1-10, 25-30* displays layers 1 through 10 and 25 through 30. The value 0 displays all layers and is the default setting.
- **Show Every n-th Layer**: Displays every *n*-th path layer of the designated Path Layers to Show. For example, the number 2 displays every second layer, 3 displays every third layer, and so on. The number 1 displays all layers designated in Path Layers to Show and is the default setting.
- **Animation Frames**: Animation frames in which the designated Path Layers to Show are rendered. Click the Show Path button  to render the animation. The value 1 displays all Path Layers to Show in a single animation frame, that is, all at once. The value 2 divides the Path Layers to Show across two animation frames, rendering half of the layers first and then the other half. The greater the amount of Animation Frames, the clearer the G-Code path progression will be. The default setting is 10. Values greater than the total number of layers in Path Layers to Show are rendered with Animation Frames equal to the total number of layers in Path Layers to Show. Because the animation rate is fast, this option is most useful when visualizing large models and is less useful when visualizing smaller models.
- **Line Segments For Arc**: The graphical display of an arc path is approximated by a number of lines in the geometry window. The number of lines per arc can be modified, a useful option for G-Code files with many arcs in order to speed up the graphical display. The default number of line segments is 5.
- **Show Positioning Lines**: On (default) displays the rapid movement commands in the G-Code. Off suppresses the display of rapid movement commands.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Cluster generation depends on the G-Code being properly aligned with the build part. After visualizing the G-Code path, if it does not align with the build part, consider using the G-Code translation or rotation options described in step 4 (p. 67).

3. **Define additional clustering and G-Code options**

   The interpretation of the G-Code file commands can be modified before element clustering starts. Adjust the following additional options in the Details view of the G-Code Clustering object, under **Clustering Options** and **G-Code Options**:

   a. Define the **Cluster Volume**, in mm^3.

Details of "G-Code Clustering"	
Import G-Code file	
G-Code File	DED_Racetrack_G-Code.txt
Clustering Options	
Cluster Volume	1000 mm^3
Print Direction	Z
G-Code Options	
Move Commands	G0, G00
Laser On Commands	G1, G01
Laser Off Commands	G0, G00
Extrusion Tag	Off
Combine Layers	Off
Ignore Layers	0
Definition	
Read Feed Rate	No
Read Preheat Temperature	No
Read Dwell Time	No
G-Code Transformation	
Translate X	0 mm
Translate Y	0 mm
Translate Z	0 mm
Rotate X	0 °
Rotate Y	0 °
Rotate Z	0 °
G-code Path Visualization	
Path Layers to Show	0
Show Every n-th Layer	1
Animation Frames	10
Line Segments for Arc	5
Show Positioning Lines	On
G-Code Statistics	
Layers	to be set
Lines	to be set

   b. Define **Print Direction**: Choose X, Y, Z, -X, -Y, or -Z for three-axis planar builds, or Multi-axis for tool paths that move in any direction (X, Y, and Z simultaneously). Defaults to Z direction.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Multi-axis refers to a DED process that uses tools that move in four or more directions, implying that the print direction might be greater than one or that it changes throughout the process. If Print Direction is Multi-axis, the following additional options are available:

- **Force Cluster at Laser Off**: Yes (default) or No - Controls whether or not a cluster is forced to generate when the laser is turned off even when the volume is less than the specified Cluster Volume. If No, clusters are generated based on volume only.

  Keep Force Cluster at Laser Off set to Yes for most high-fidelity simulations. This ensures the simulation timing matches the machine's actual "on/off" cycle, which is critical for capturing the correct thermal history and cooling periods between weld beads. Only set it to No if you are performing a very coarse, fast preview and aren't concerned with localized thermal gradients.

- **Connectivity Check**: Yes (default) or No - Checks for and corrects badly connected elements if they exist. A badly connected element is one that: a) is not face-connected with any other elements of the current cluster and b) is not face-connected to any activated elements and c) is not in contact with the base. The connectivity check occurs during clustering and may cause the operation to take longer. Consider setting it to No to save time when working on preliminary simulation setup. Set Connectivity Check back to Yes for your final (non preliminary) simulation.

Selecting the Right Print Direction

While it may be tempting to use multi-axis print direction for all simulations, the three-axis options (X, Y, Z, -X, -Y, or -Z) are preferred for standard planar builds. The primary difference lies in the search algorithm used to activate elements:

- **Three-axis Clustering:** The algorithm projects a search volume specifically "underneath" the G-code path along the defined print vector. This directional constraint ensures that elements are activated in a way that produces a perfectly flat top surface, matching the intended layer height.

- **Multi-axis Clustering:** The algorithm performs a more complex 3D radial search to find the nearest elements in any direction. While this is necessary for robotic or 5-axis tool paths, it is more computationally intensive. Furthermore, because it lacks a fixed directional "up," it cannot guarantee a perfectly planar top surface, which may lead to slight "stair-stepping" or irregular element activation in a standard layer-by-layer build.

Choosing the correct clustering logic ensures that the thermal mass is added to the simulation in a way that accurately reflects the physical deposition process. Use three-axis clustering whenever the build direction is fixed to maximize the geometric accuracy of the printed layers.

Feature	Three-axis (X, Y, Z)	Multi-axis
Search Logic	Unidirectional projection (along the vector).	3D Radial/Proximity search.
Solve Speed	May sometimes be faster; optimized for layer-by-layer.	May sometimes be slower; handles complex 3D calculations.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Top Surface	Maintains a clean, planar top surface.	May result in uneven/stair-stepped activation.
Best Use Case	Standard 3-axis machines, plates, and prismatic parts.	Robotic arms, 5-axis machines, and curved substrates.
G-code Compatibility	Requires G-code oriented to a single axis.	Necessary for G-code with changing vectors (for example, dynamic A B C values).

c. Set these additional options as needed:

- **Move Commands**: Defines the G-Code commands for pure movement. Multiple commands can be specified using comma separation, such as "G00, G01."
- **Laser On Commands**: Defines the G-Code command(s) for switching the laser to on (Laser On). All following move commands are used for element clustering until a Laser Off command is found. Multiple commands can be specified using comma separation, such as "M24, M174."
- **Laser Off Commands** Defines the G-Code command(s) for switching the laser to off (Laser Off). All following move commands are not used for element clustering until a Laser On command is found. Multiple commands can be specified using comma separation, such as "M28, M175."
- **Extrusion Tag**: G-Code commands may be extended by an extrusion tag "E." This can be used to additionally control if material is deposited or not. With the Extrusion Tag option set to On, only commands with the "E"-tag are considered for subsequent element clustering. Default setting is Off.
- **Ignore Layers**: With this option, specific layers can be excluded from the element clustering. Multiple layer numbers can be excluded using comma separated values. For example, "2,3" ignores the layers two and three from subsequent element clustering, so the clusters are divided into two instead of four layers. The default is 0 which includes all layers for element clustering (that is, 0 exclusions).

4. **Translate or rotate the G-Code in relation to the build part, as needed**

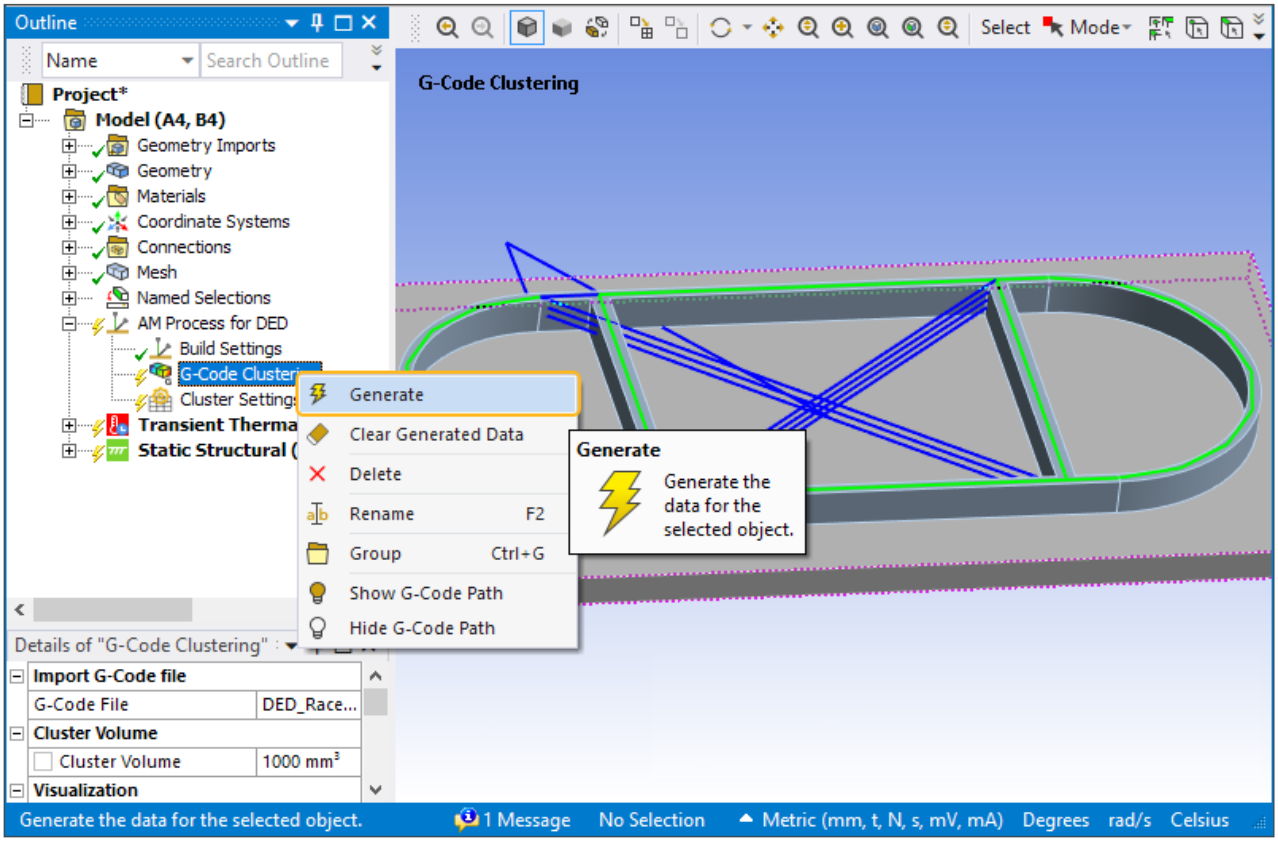
   Cluster generation depends on the G-Code being properly aligned with the build part. Use the fields under G-Code Transformation to position the G-Code properly with respect to the part. Entering a value in any of these fields transforms the G-Code path immediately. The example (p. 71) at the end of this section shows a case where rotation of the G-Code path is required.

5. **Perform element clustering**

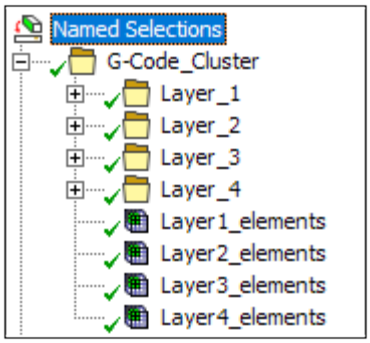
   The generation of element clusters requires the following steps:

   a. Right-click the **G-Code Clustering** object and select **Generate** to start the element clustering process. Depending on the number of commands in the G-Code file and the number of elements, this process may take some time.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

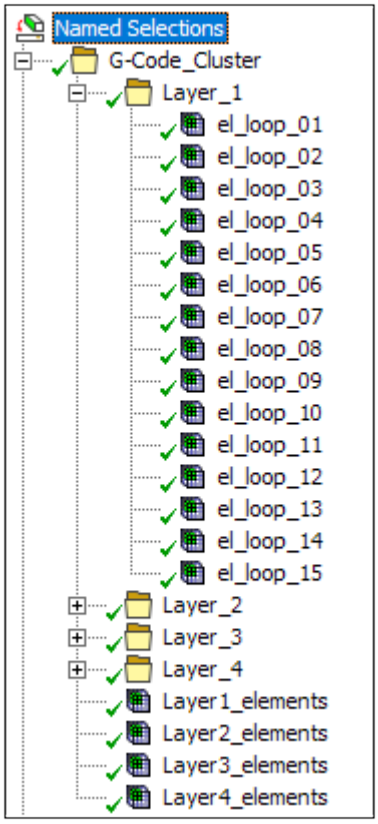


The following structure will be created under Named Selections in the project tree:



The folders "Layer_1," "Layer_2," and so on, contain named selections, one named selection for each element cluster. The cluster names "el_loop_001," "el_loop_002" and so on, must not be renamed.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



The named selections “Layer1_elements,” “Layer2_elements,” and so on, contain all elements within those detected layers. These named selections are created only for visualization purposes and have no influence on the simulation itself.

b. View the **G-Code Statistics** in the Details view of the **G-Code Clustering** object to see the total number of layers and lines read from the G-Code file, including rapid movements without material deposition.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Details of "G-Code Clustering"

Import G-Code file	
G-Code File	DED_Racetrack_G-Code.txt
Cluster Volume	
Cluster Volume	1000 mm^3
G-Code Options	
Print Direction	Z
Move Commands	G0, G00
Laser On Commands	G1, G01
Laser Off Commands	G0, G00
Extrusion Tag	Off
Combine Layers	Off
Ignore Layers	0
Definition	
Read Feed Rate	No
Read Preheat Temperature	No
Read Dwell Time	No
G-Code Transformation	
Translate X	0 mm
Translate Y	0 mm
Translate Z	0 mm
Rotate X	0 °
Rotate Y	0 °
Rotate Z	0 °
G-code Path Visualization	
Path Layers to Show	0
Show Every n-th Layer	1
Animation Frames	10
Line Segments for Arc	5
Show Positioning Lines	On
G-Code Statistics	
Layers	4
Lines	80

c. The Cluster Settings object is automatically inserted under the AM Process for DED (p. 109) object in the project tree together with the G-Code Clustering object. Click **Cluster Settings** to use a table to adjust machine parameters for individual element clusters as needed. See Using the Cluster Settings Table (p. 84) for details.

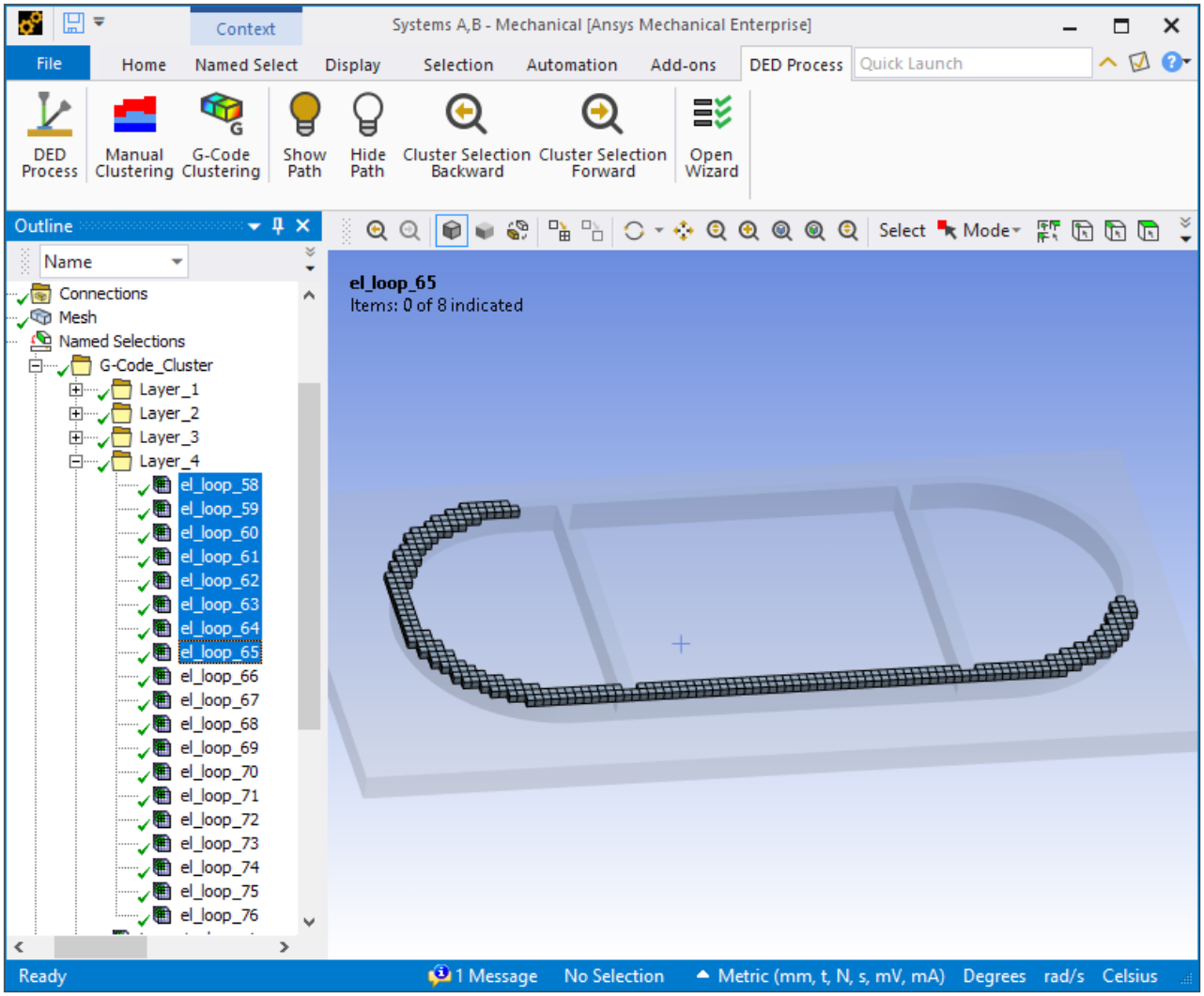
6. **Review the cluster progression after element clustering is completed**

Important:

Don't skip this step. It is very important to visually verify that the element clusters follow the G-Code path. The progression of the thermal solver follows the element clusters. If the cluster progression appears to jump around randomly back and forth to different portions of the model and does not follow the G-Code path, the results will be incorrect.

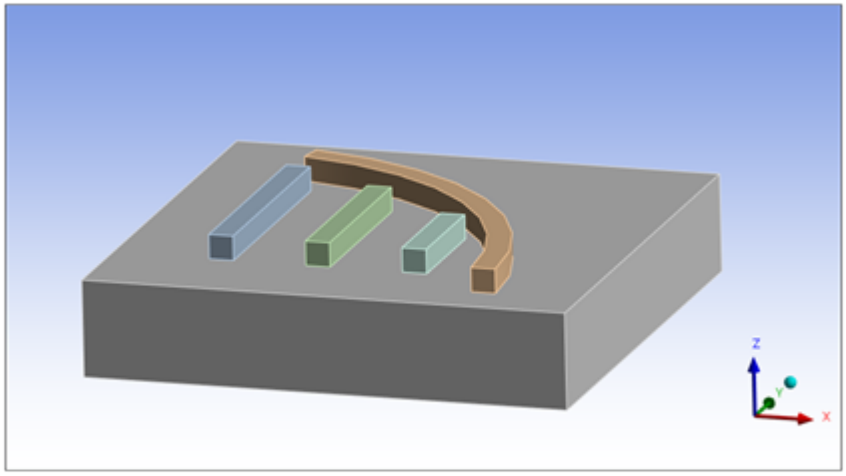
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Use the buttons **Cluster Selection Forward** and **Cluster Selection Backward** in the **AM Process for DED** toolbar in the ribbon to display the clustering sequence in the geometry window.



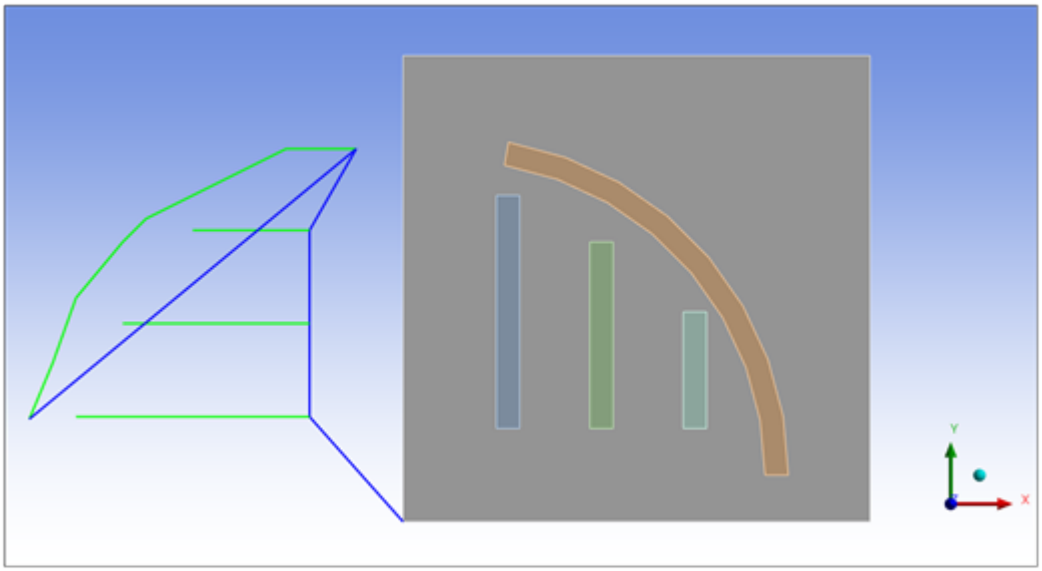
Example

In the following example, the build part consists of four bodies, as shown here:



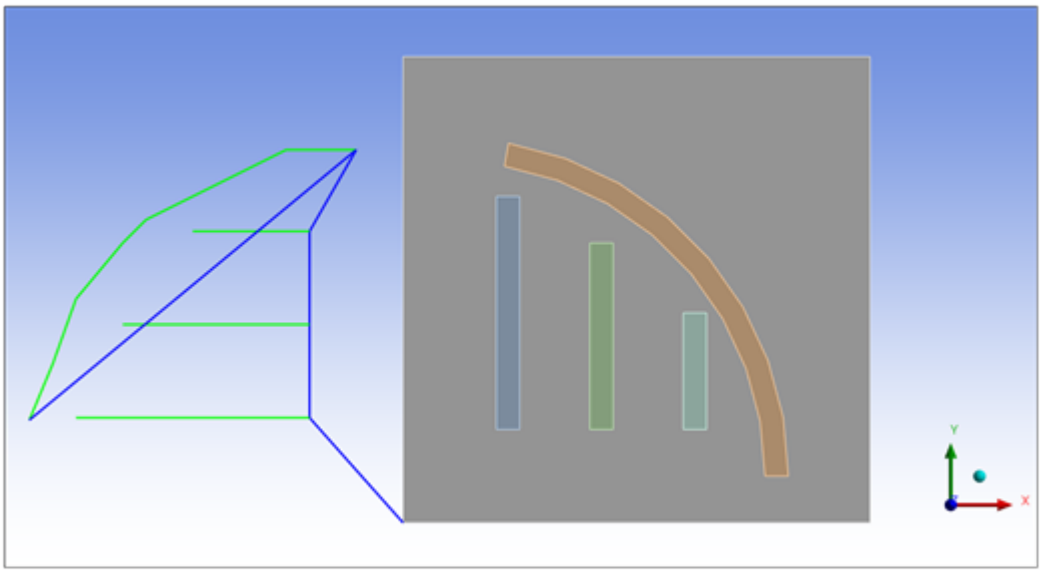
After importing the G-Code file and visualizing the G-Code path, we can see that it is not properly aligned on top of the parts.

Top-down view of G-Code path showing incorrect alignment with build parts



Indeed, after performing element clustering, we can review the cluster progression and see that the element clusters are adversely affected by the unaligned G-Code path and will result in an incorrect simulation.

Top-down view of G-Code path showing incorrect alignment with build parts



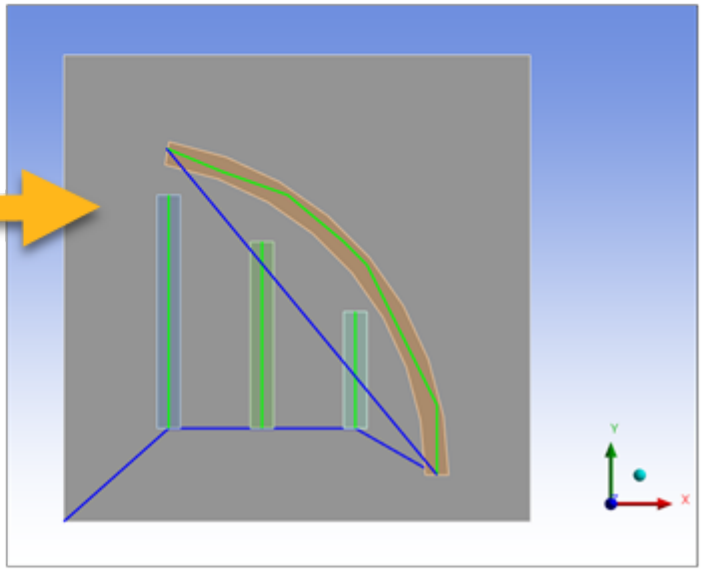
The solution to the problem is to rotate the G-Code path by 90° in the clockwise direction about the Z-axis. Note that rotation is allowed only in the printing direction. Since the Print Direction =Z in this example, we will use the Rotate Z field under the G-Code Transformation settings, as shown below. A top-down view of the model shows the corrected G-Code path after rotation.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Details of "G-Code Clustering"

Import G-Code file	
G-Code File	3linesAndArch.txt
Cluster Volume	
Cluster Volume	1.6E-07 m³
G-Code Options	
Print Direction	Z
Move Commands	G0, G00
Laser On Commands	G1,G01
Laser Off Commands	G0, G00
Extrusion Tag	Off
Combine Layers	Off
Ignore Layers	0
Definition	
Read Feed Rate	No
Read Preheat Temperature	No
Read Dwell Time	No
G-Code Transformation	
Translate X	0 m
Translate Y	0 m
Translate Z	0 m
Rotate X	0 °
Rotate Y	0 °
Rotate Z	-90 °
G-code Path Visualization	
Path Layers to Show	0
Show Every n-th Layer	1
Animation Frames	10
Line Segments for Arc	5
Show Positioning Lines	On
G-Code Statistics	
Layers	8
Lines	22

Top-down view of G-Code path showing corrected alignment with build parts



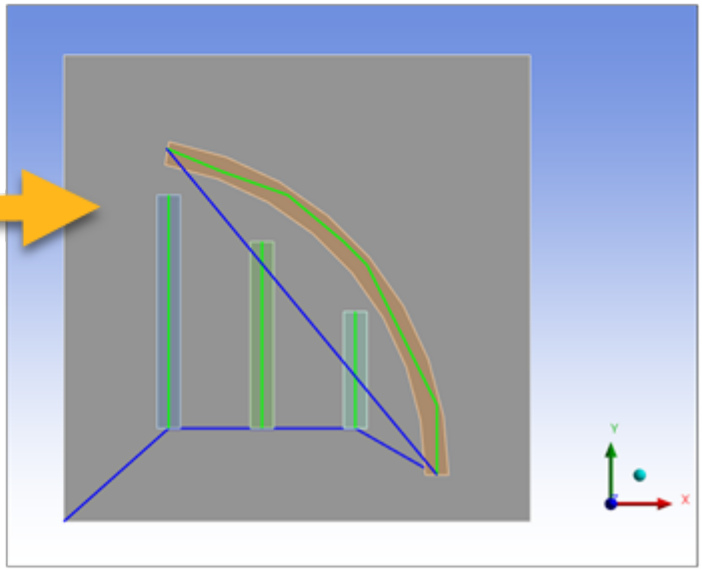
After element clustering, we can again review the cluster progression and verify it now follows the G-Code path.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Details of "G-Code Clustering"

Import G-Code file	
G-Code File	3linesAndArch.txt
Cluster Volume	
Cluster Volume	1.6E-07 m³
G-Code Options	
Print Direction	Z
Move Commands	G0, G00
Laser On Commands	G1,G01
Laser Off Commands	G0, G00
Extrusion Tag	Off
Combine Layers	Off
Ignore Layers	0
Definition	
Read Feed Rate	No
Read Preheat Temperature	No
Read Dwell Time	No
G-Code Transformation	
Translate X	0 m
Translate Y	0 m
Translate Z	0 m
Rotate X	0 °
Rotate Y	0 °
Rotate Z	-90 °
G-code Path Visualization	
Path Layers to Show	0
Show Every n-th Layer	1
Animation Frames	10
Line Segments for Arc	5
Show Positioning Lines	On
G-Code Statistics	
Layers	8
Lines	22

Top-down view of G-Code path showing corrected alignment with build parts



Known Issue where G-Code Clustering Does Not Follow the G-Code Path

For DED simulations using G-Code clustering with a *single print direction*, such as ±X, ±Y, or ±Z (that is, other than Multi-axis), depending on how your G-Code is written, the clustering may not reflect the real build sequence due to how the G-Code clustering algorithm is implemented within the DED Process Add-on. In certain cases, the DED simulation results may be invalid.

First, check to see if the G-Code clustering matches your actual build sequence by reviewing the cluster progression once clusters are generated. To do so, click the **Cluster Selection Forward** button in the AM Process for DED (p. 109) toolbar on the ribbon repeatedly to see the cluster progression in the geometry window. (Alternatively, choose the first Named Selection in Layer 1 under Named Selections and use the down arrow key to view the cluster progression through all layers.) If the cluster progression follows the G-Code path as expected, there is no need to be concerned, and you can proceed to solve the simulation. If the cluster progression does *not* follow the G-Code path, it is most

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

likely because the G-Code layers do not align evenly in the print direction everywhere on the model. One indication of this is that the cluster progression appears to jump around randomly back and forth to different portions of the model. This can happen in both multi-body parts and single-body parts with non-simple geometries, even if the G-Code file itself is valid. Common scenarios are shown below (p. 75).

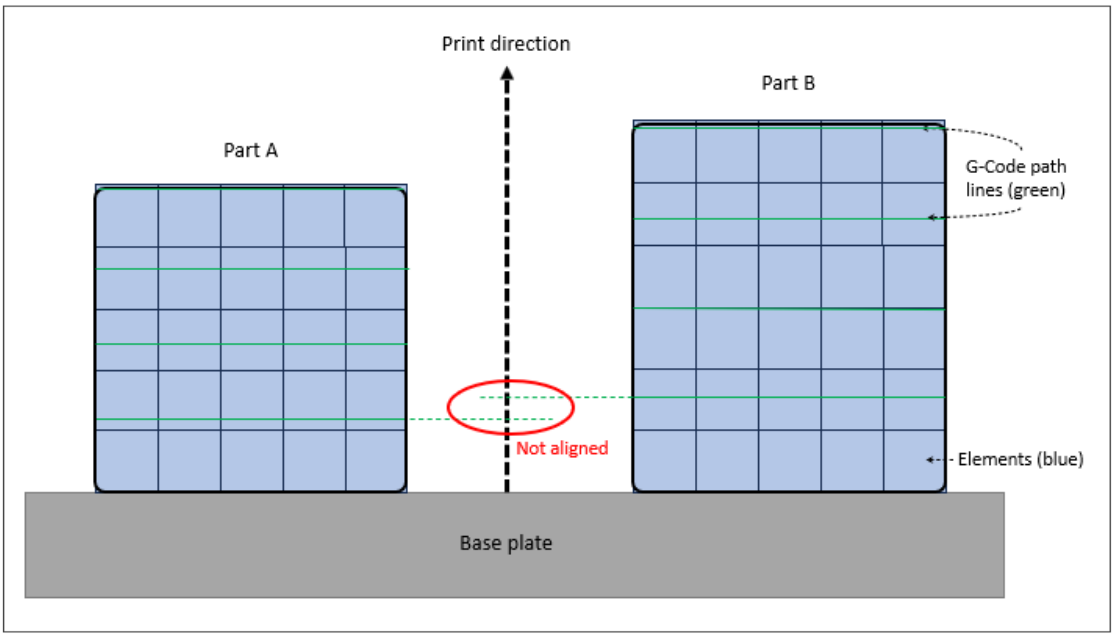
When G-Code clustering does not follow the G-Code path, the simple workaround is to change the Print Direction to Multi-axis, in which case a different algorithm is used resulting in a more rigorous clustering technique that most always solves the problem.

Common Scenarios where G-Code Clustering Does Not Follow the G-Code Path

While every geometry is unique, below are some build scenarios where you may experience G-Code clustering not following the G-Code path.

Different/multiple layer thicknesses resulting in unaligned G-Code layers

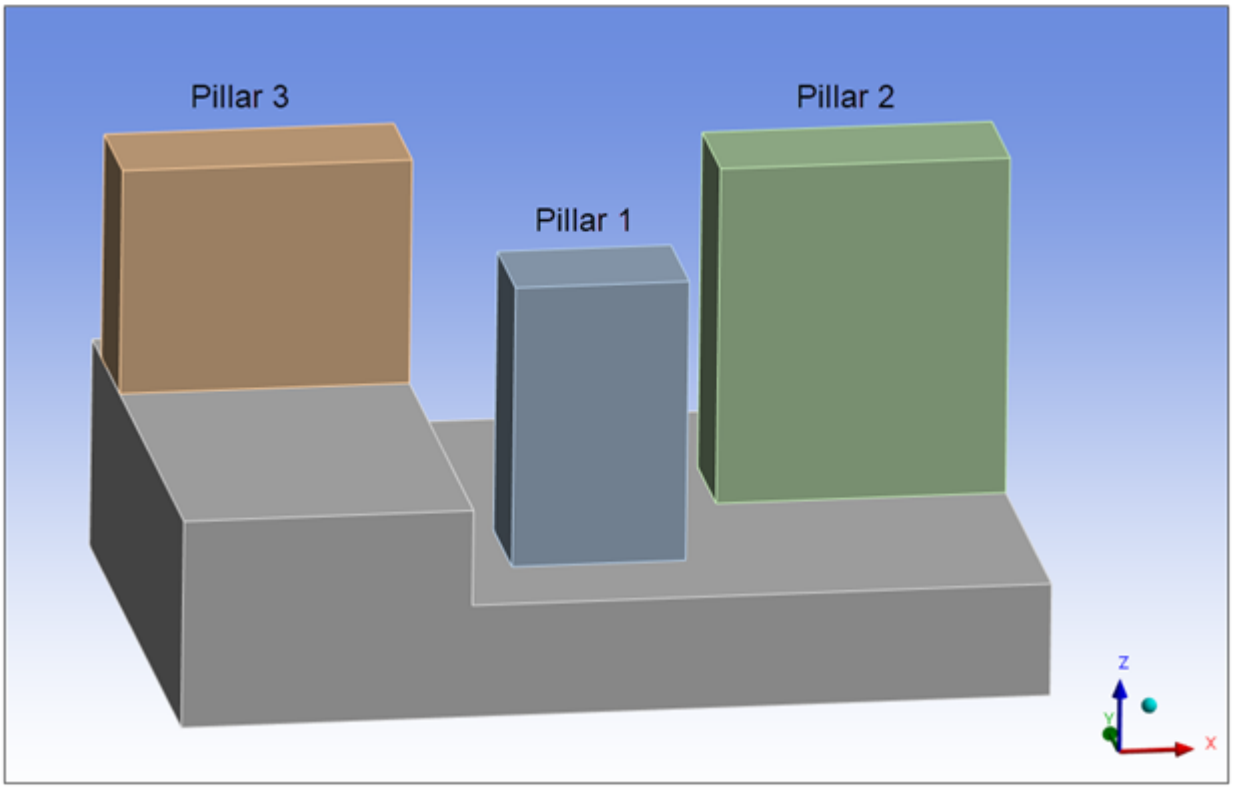
In this scenario, multiple parts on the base plate each have distinct deposition layer thicknesses that are not aligned in the print direction, as illustrated below. While this is a valid DED set-up, the current clustering implementation in the DED Process Add-on for a single print direction may result in unexpected behaviour because of the current limitation. Adjusting the G-Code layer thicknesses to be the same for both parts is an easy enough fix in this example if that is an acceptable build scenario. Or set Print Direction to Multi-axis as the workaround.



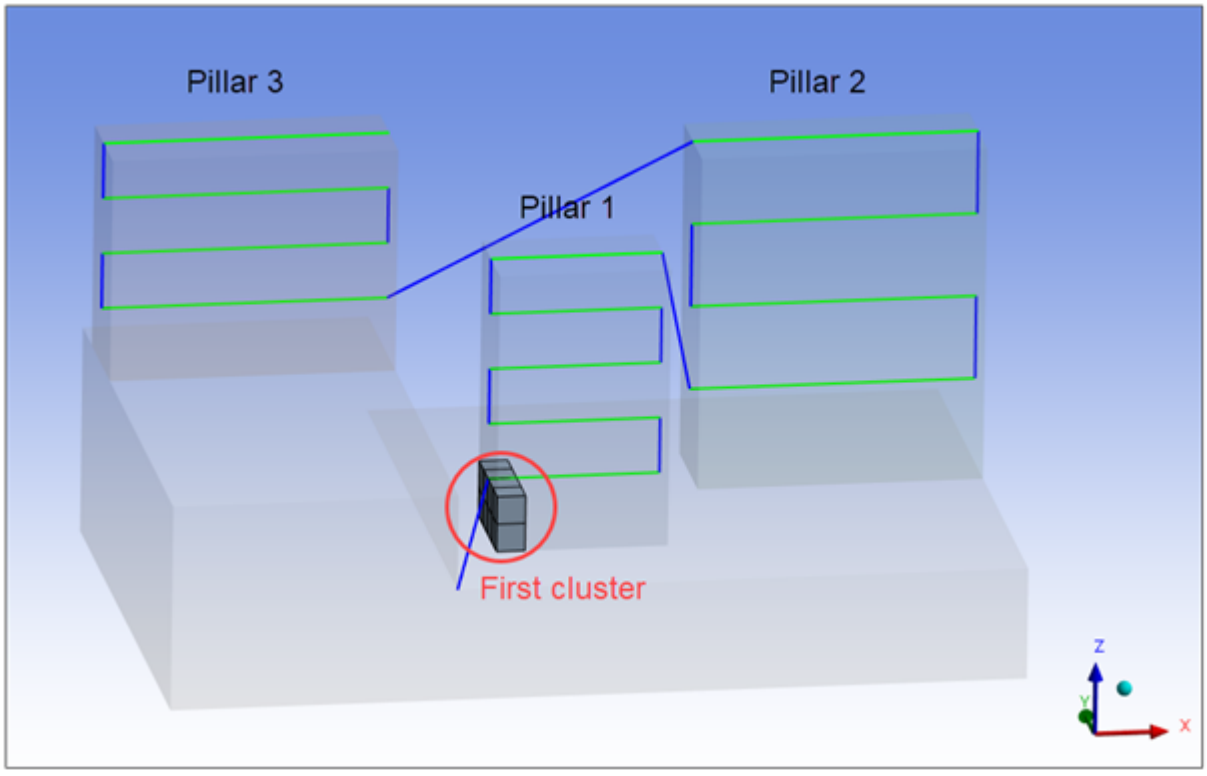
Non-planar base resulting in unaligned G-Code layers

The next example illustrates the scenario described above—a multi-body part with different layer thicknesses resulting in unaligned layers— as well as introduces another problem scenario caused by a non-planar base plate. Three pillars are to be printed one-by-one on a non-planar base. The intention is to build pillar 1 first, followed by pillar 2, and lastly, pillar 3.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

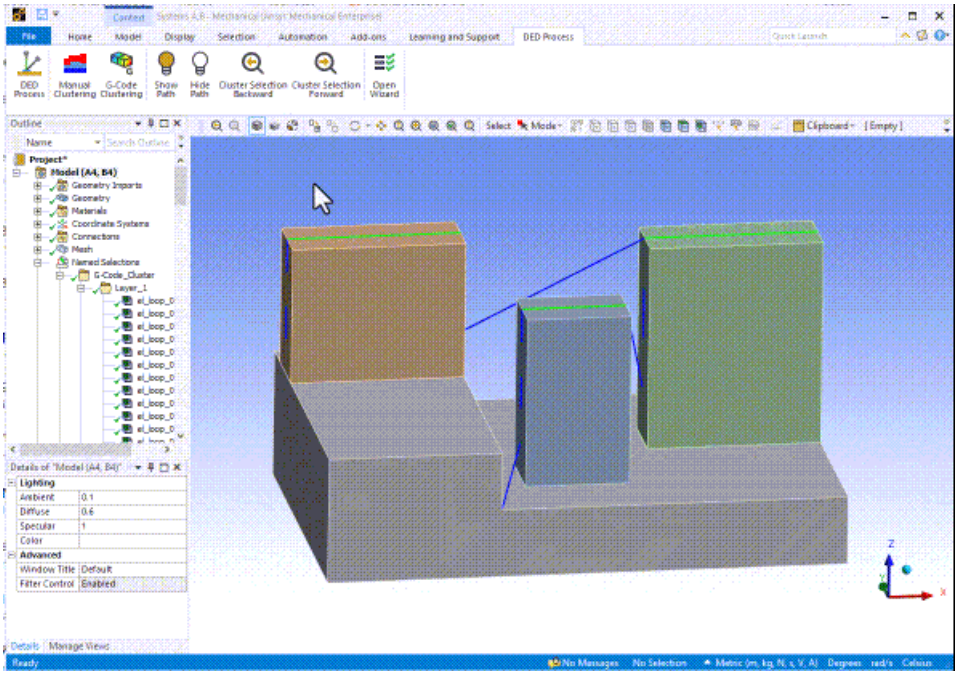


If we look at the G-Code path (highlight the first cluster under Named Selections and then click **Show Path** in the DED Process ribbon), we see a valid path starting at the lower, left corner of pillar 1, progressing upward back and forth until pillar 1 is completed, then moving to the bottom of pillar 2, moving upward back and forth until pillar 2 is completed, and finally moving to pillar 3 and moving upward back and forth until it is completed.

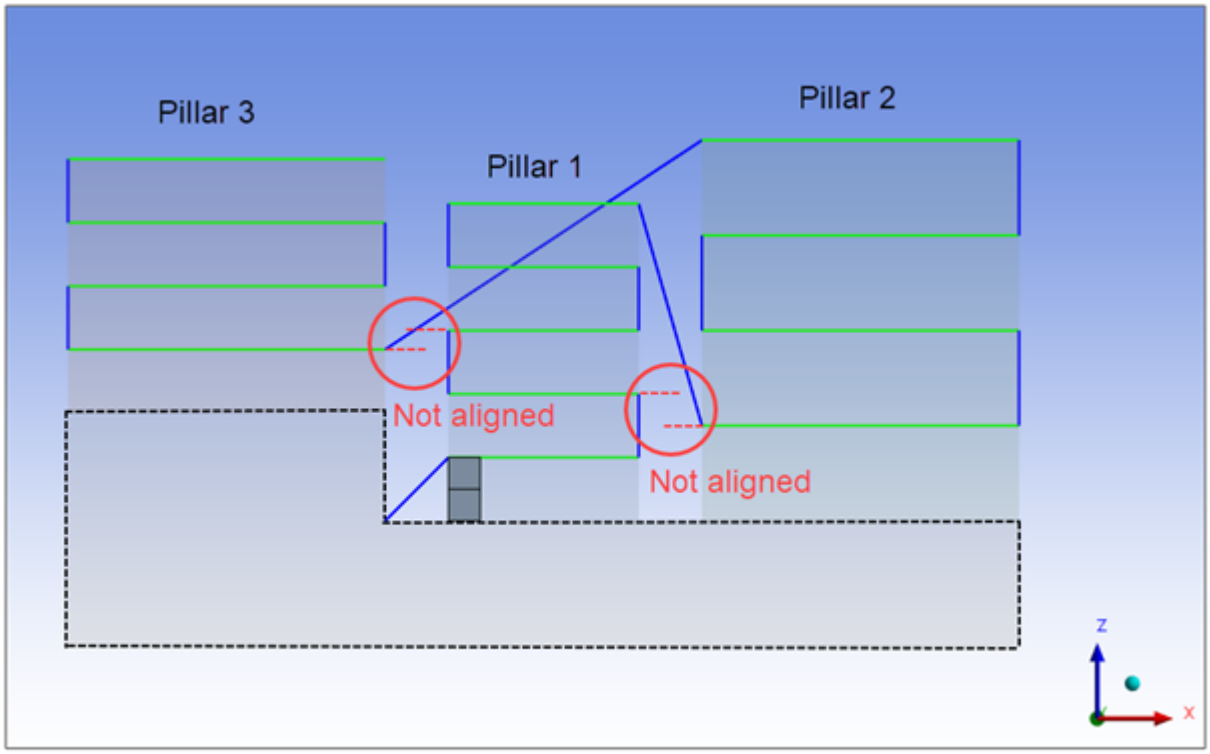


Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

But when we review the cluster progression, we see that the clustering does not follow the G-Code path, as shown in the following animation:



A front view of the model shows that the G-Code path lines for the pillars do not align with each other in the Z direction. For pillars 1 and 2, which are positioned at the same Z location on the base, their layer thicknesses are different. Pillars 1 and 3, while having the same layer thickness, start at different Z-locations on the base.



Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

There is no way to reconcile the unaligned layers problem due to the non-planar base plate without changing the G-Code and/or geometries. In this example, set Print Direction to Multi-axis as the workaround.

5.8.2.1. G-Code Clustering for Multi-axis Printing

Multi-axis DED refers to a Directed Energy Deposition process that uses tools that move in four or more directions, implying that the print direction might be greater than one or that it changes throughout the process.

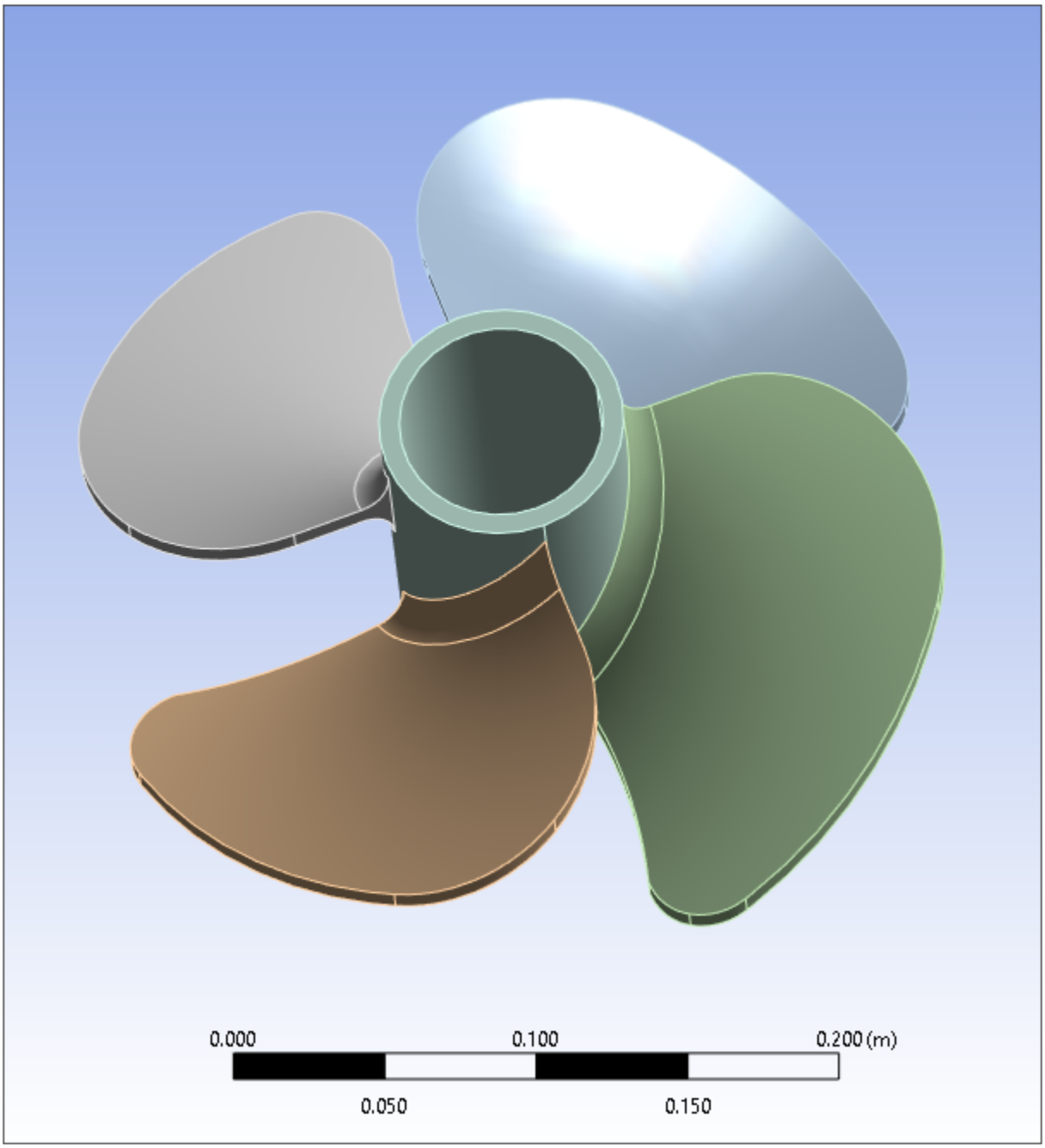
To simulate a multi-axis DED process:

- We recommend you set shared topology between the part bodies and the base if it is a curved base.
- Named Selections (p. 40) required for generating clusters and performing the connectivity check for multi-axis are:
  - Part Elements
  - Part Nodes
  - Part Contact Elements
- Use G-Code clustering and set the **Print Direction** to Multi-axis on the G-Code Clustering object (p. 116). Additional options include Force Cluster at Laser Off and Connectivity Check.
- For G-code path visualization purposes, one "layer" is defined by laser status changes. The lines between each pair of laser on and off commands are grouped into a layer.

5.8.2.1.1. Example

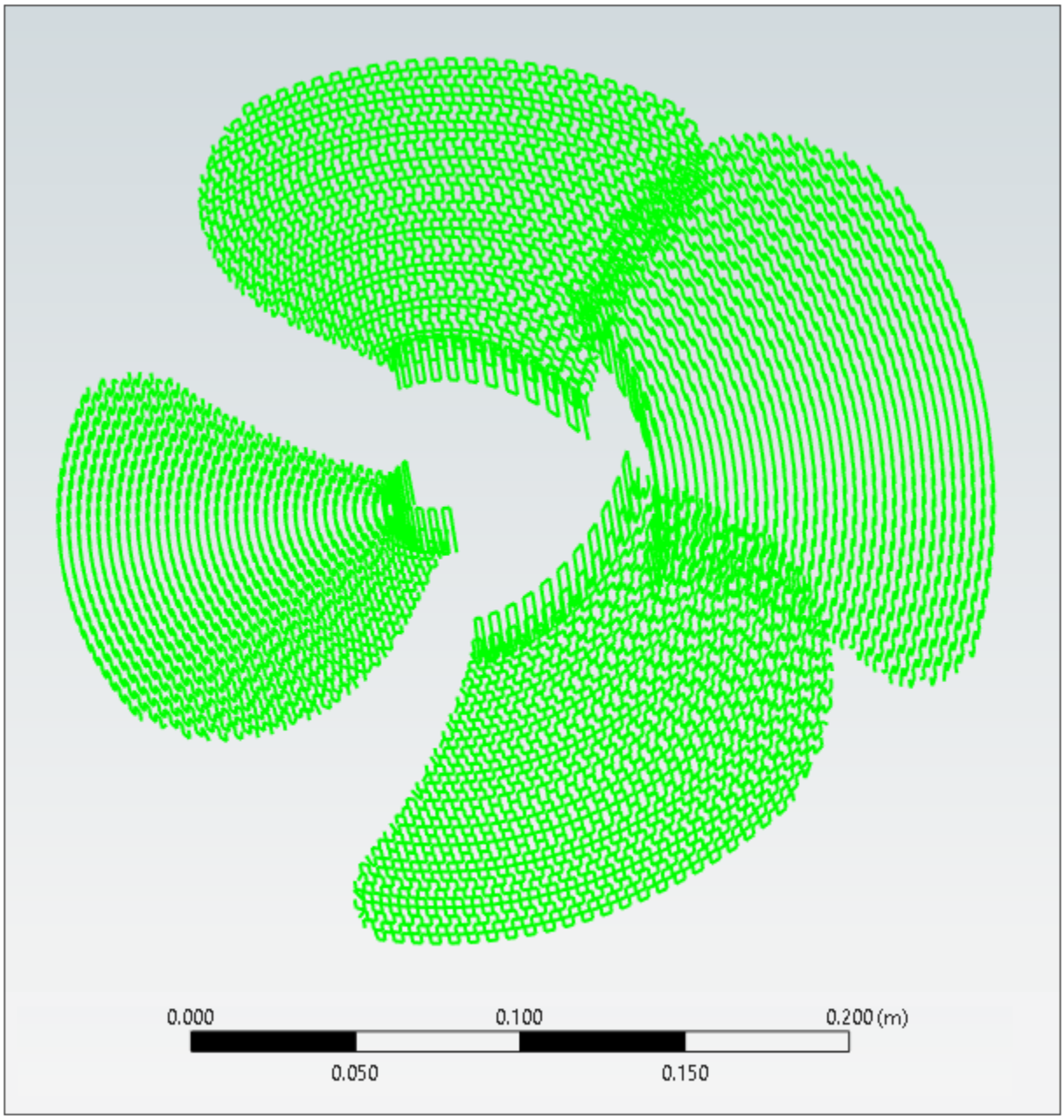
Consider the following example of a propeller.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



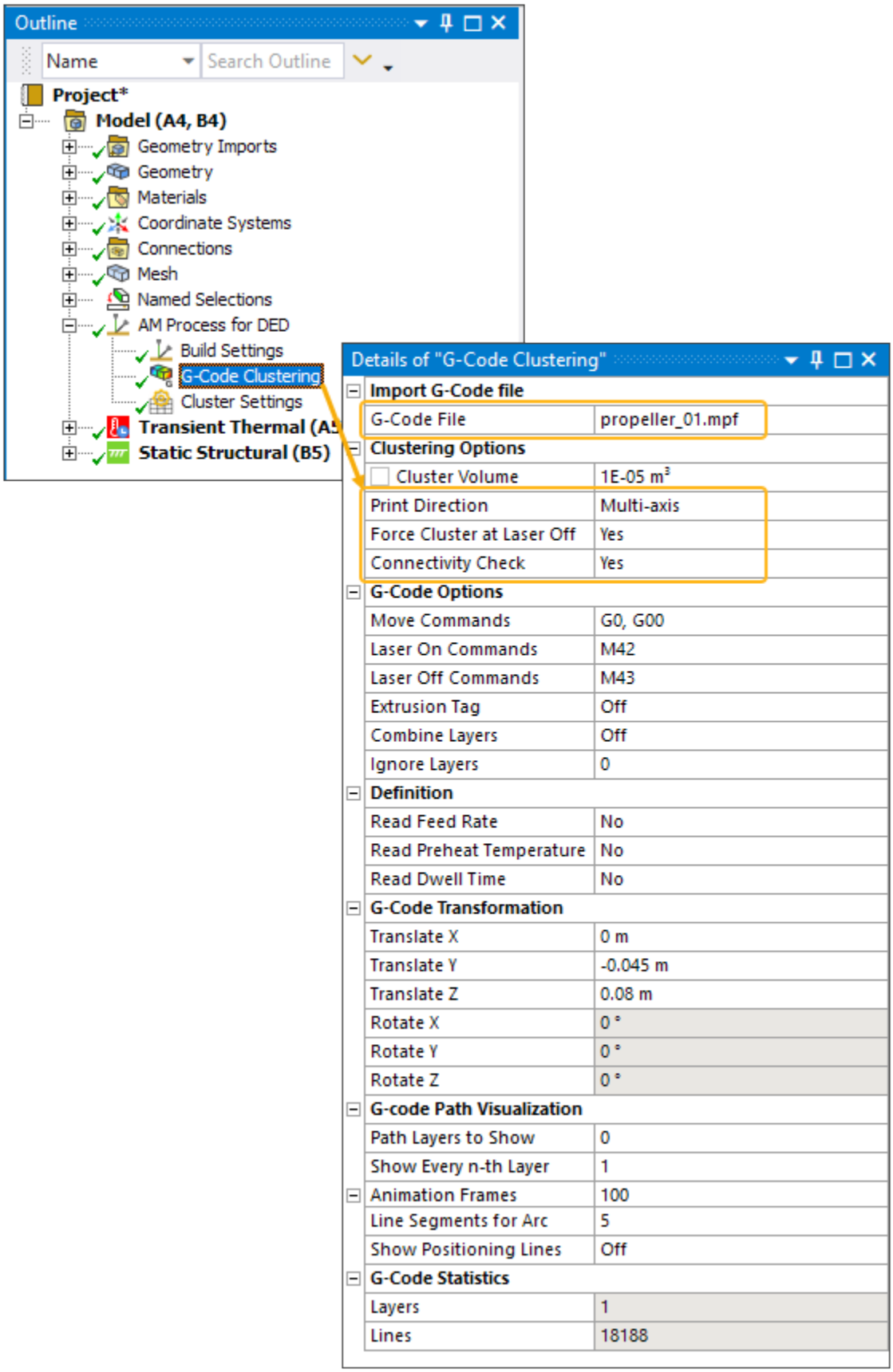
The G-Code file defines a print path starting from the hub and directed radially outward, as shown below. In the visualization, a green line appears at the top of each new deposition layer.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



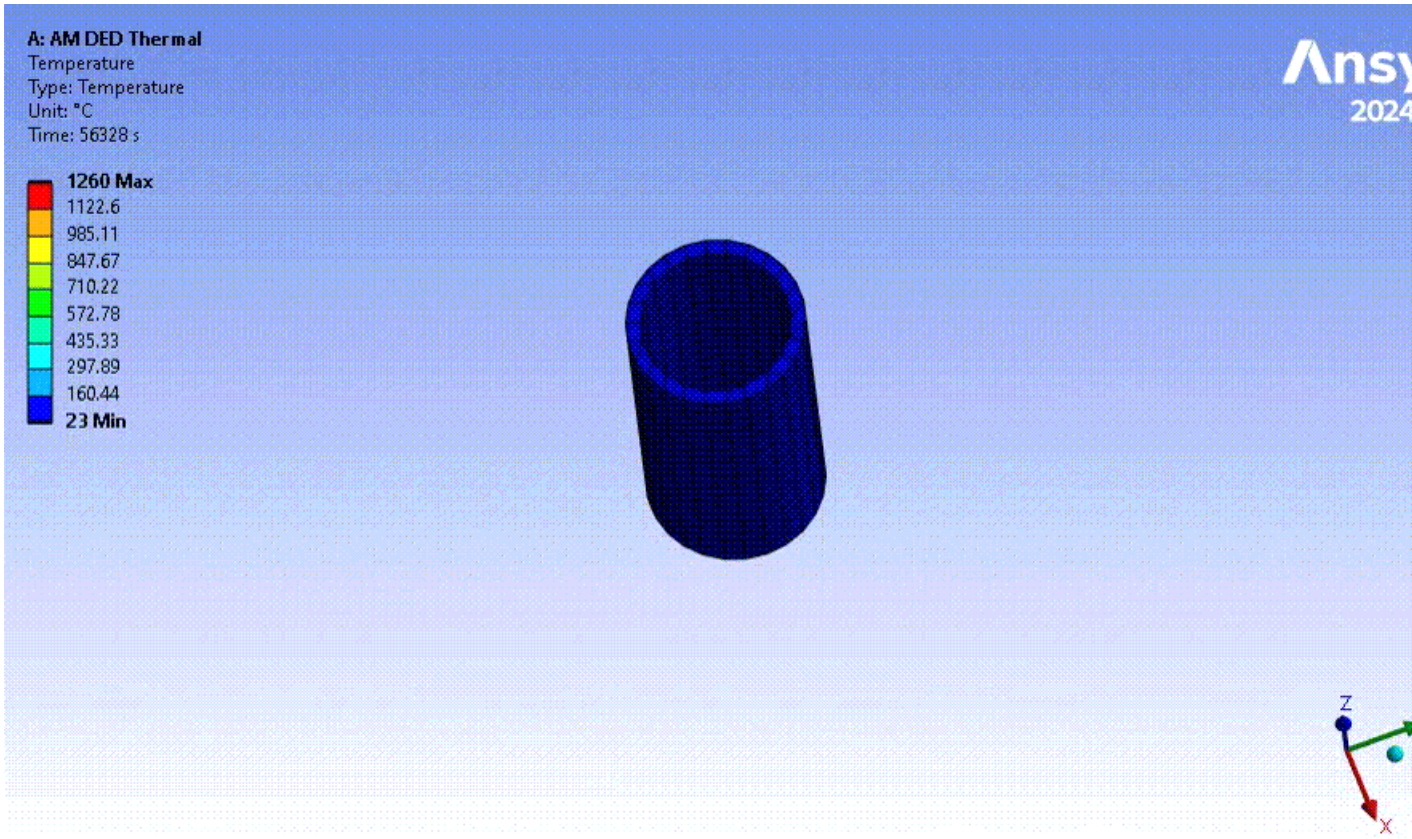
On the G-Code Clustering object, choose **Multi-axis** for Print Direction.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



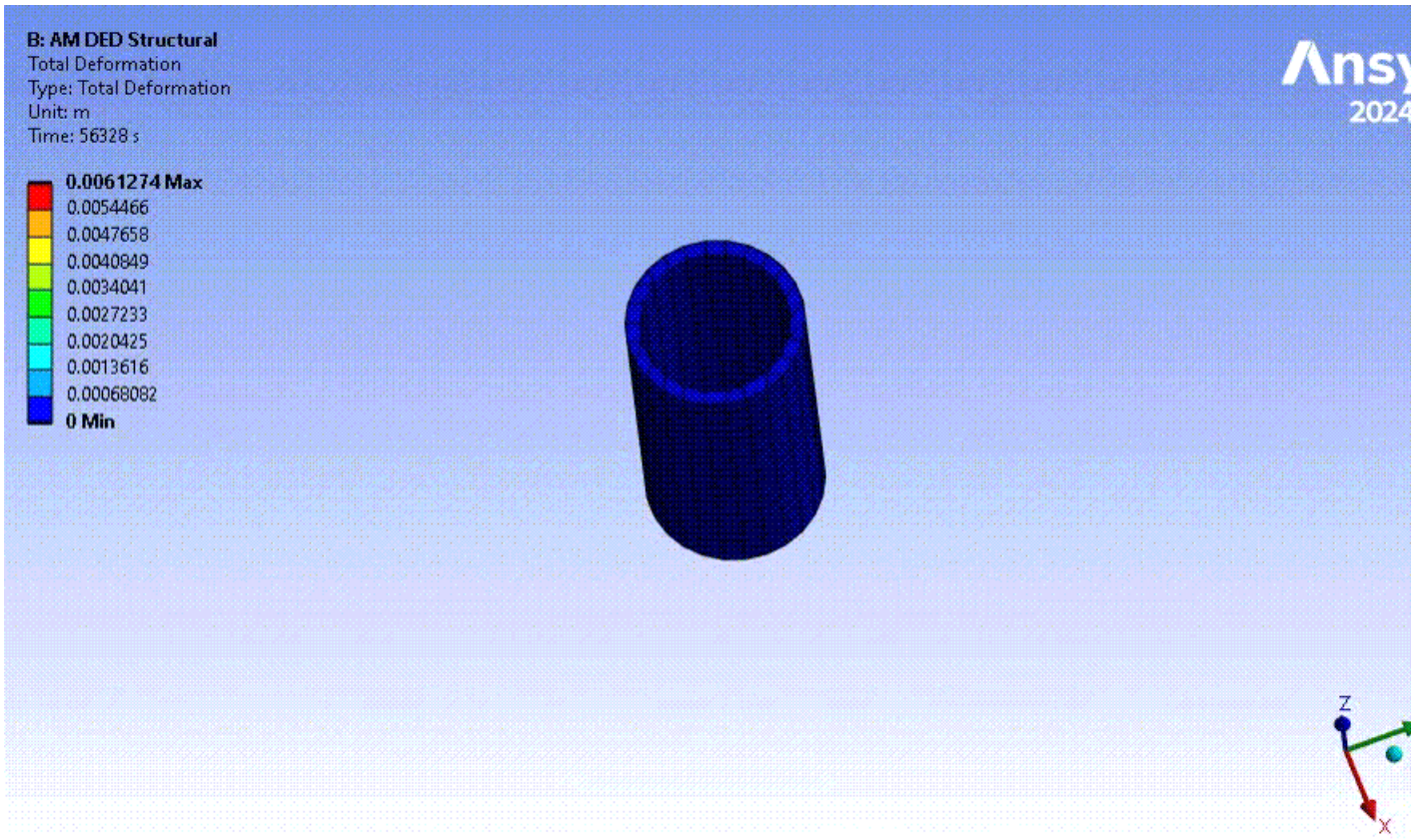
Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

The following animation shows the temperature results. *Refresh the page, as needed, to see the following animation. View online if you are reading the PDF version of the help.*



The following animation shows the structural displacement results. *Refresh the page, as needed, to see the following animation. View online if you are reading the PDF version of the help.*

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



5.8.3. Editing Individual Clusters for More Precise Control of Process Settings

Machine parameters are sometimes adjusted to optimize process settings and improve printing quality. You can exert more precise control over regions of the model, for example, to reduce distortion or local overheating by editing settings for individual clusters. Three methods are available to do this:

5.8.3.1. Using the Cluster Settings Table

5.8.3.2. Editing the Cluster Output File Directly

5.8.3.3. Controlling Heating and Cooling During the Build via Parameters (Command Snippets)

The following table shows which method is available for editing each process parameter. You can also add, remove, or move elements among clusters, which is listed as the last row of the table (Elements within the cluster).

Options for Editing Clusters			
Item to be Edited	**Cluster Settings Table**[a]	**Cluster Output File**[b]	**Parameters (Command Snippets)**[a]
Deposition Rate	✓	✓	
Heater/Preheat Temperature	✓	✓	
Dwell Time	✓	✓	
Process Temperature or Power		✓	✓
Convection coefficient for the part		✓	✓

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Options for Editing Clusters			
Item to be Edited	**Cluster Settings Table**[a]	**Cluster Output File**[b]	**Parameters (Command Snippets)**[a]
Convection coefficient for the base		✓	✓
Radiation emissivity for the part		✓	✓
Radiation emissivity for the base		✓	✓
Room temperature		✓	
Elements within the cluster		✓	

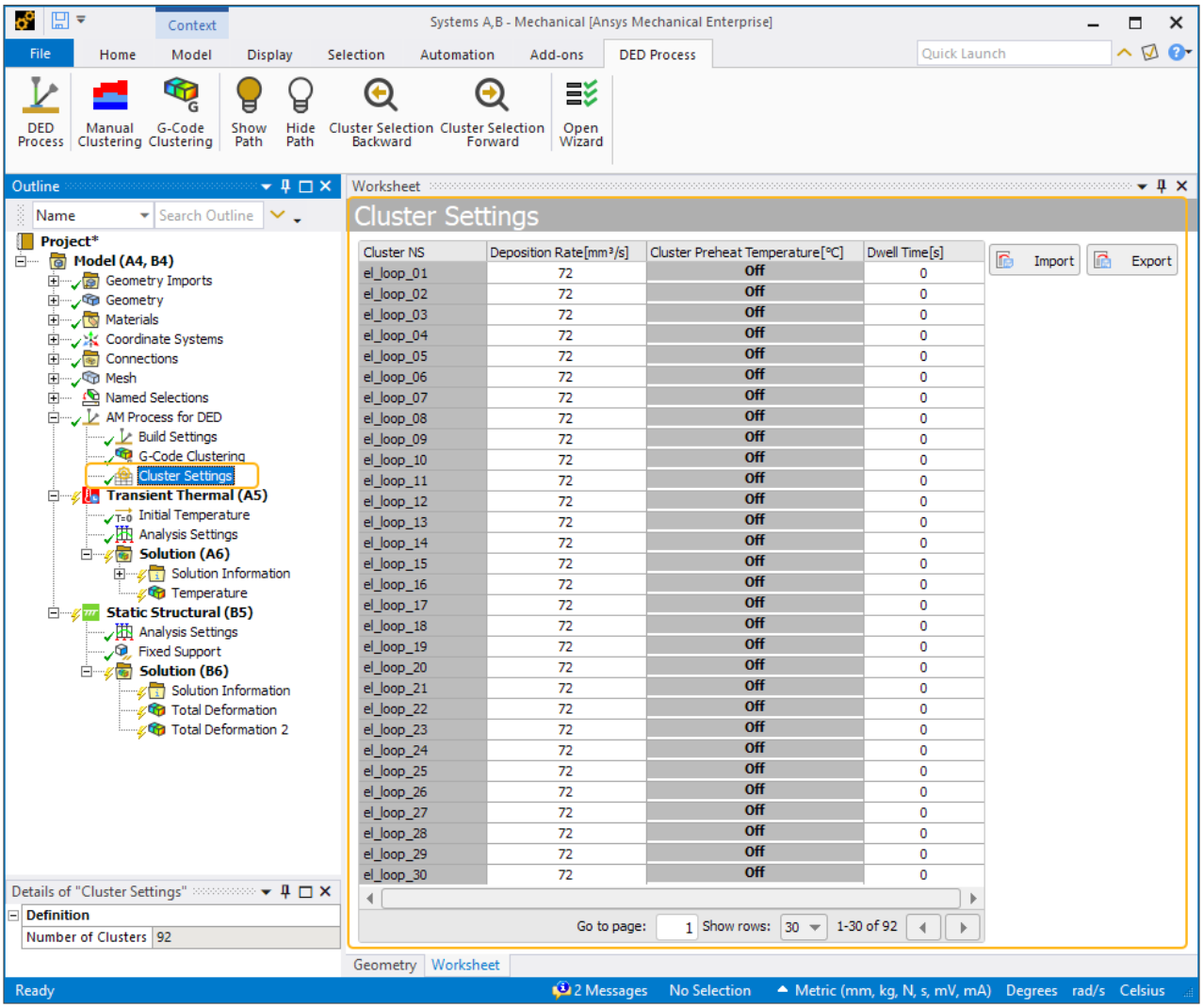
[a] Applicable for G-Code clustering and manual clustering

[b] Applicable for G-Code clustering only

5.8.3.1. Using the Cluster Settings Table

You can adjust some machine parameters for individual element clusters using the Cluster Settings table, accessible by selecting the Cluster Settings object (p. 122). The Cluster Settings object is automatically inserted under the AM Process for DED (p. 109) object in the project tree when either the Manual Clustering (p. 121) or G-Code Clustering (p. 116) object is inserted. (Manual Clustering and G-Code Clustering are inserted either by the DED Setup Wizard or by manual insertion.)

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



5.8.3.1.1. Type and Format of Table Data

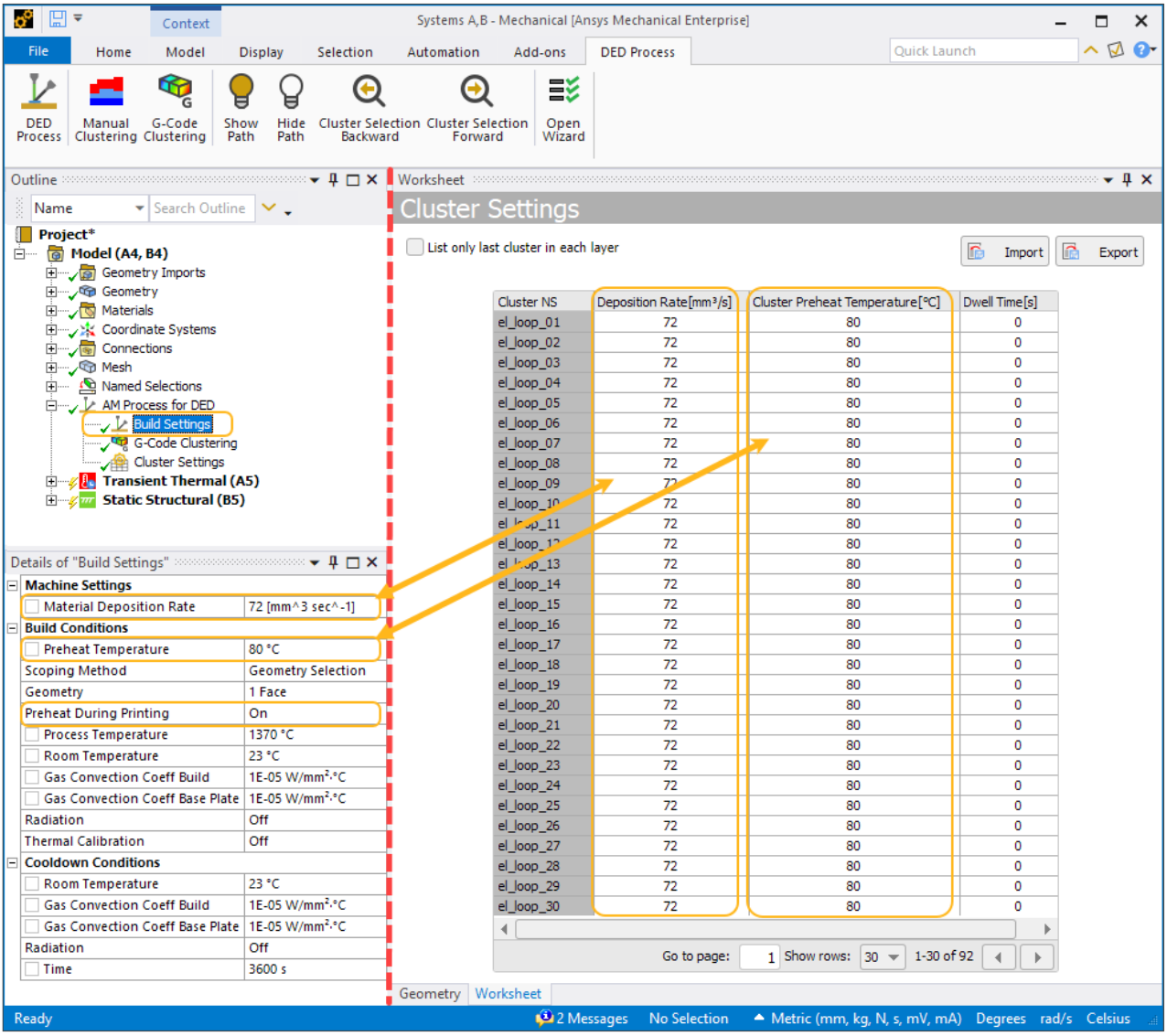
The table consists of the following columns of data:

- **Cluster NS**: Name of the named selection for the cluster.
- **Deposition Rate** (same as Material Deposition Rate in Build Settings): The feeding rate of the melted material, in mm^3/sec. This value can be determined by multiplying *layer thickness* (mm) x *weld width* (mm) x *deposition speed* (mm/sec).
- **Cluster Preheat Temperature**: The starting temperature, in °C, of the face scoped for preheating; usually this is the underside face of the base plate.
- **Dwell Time**: The pause, in seconds, between the completion of this *cluster* and the beginning of the next, not including laser travel time.

There is a relationship between the data in the table and the data in the Build Settings object (p. 111), as shown in the following image. In the image, the Cluster Settings table is juxtaposed next to the project tree for comparison purposes only. In the application, the table does not appear when Build Settings is selected.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Initially, the values in the table reflect what is entered in the Build Settings object.



Note that **Preheat During Printing** must be set to **On** for any temperature values to be included in the table. If Preheat During Printing is set to Off, the word Off appears in every row in the Cluster Preheat Temperature column, as shown in the image at the top of this section.

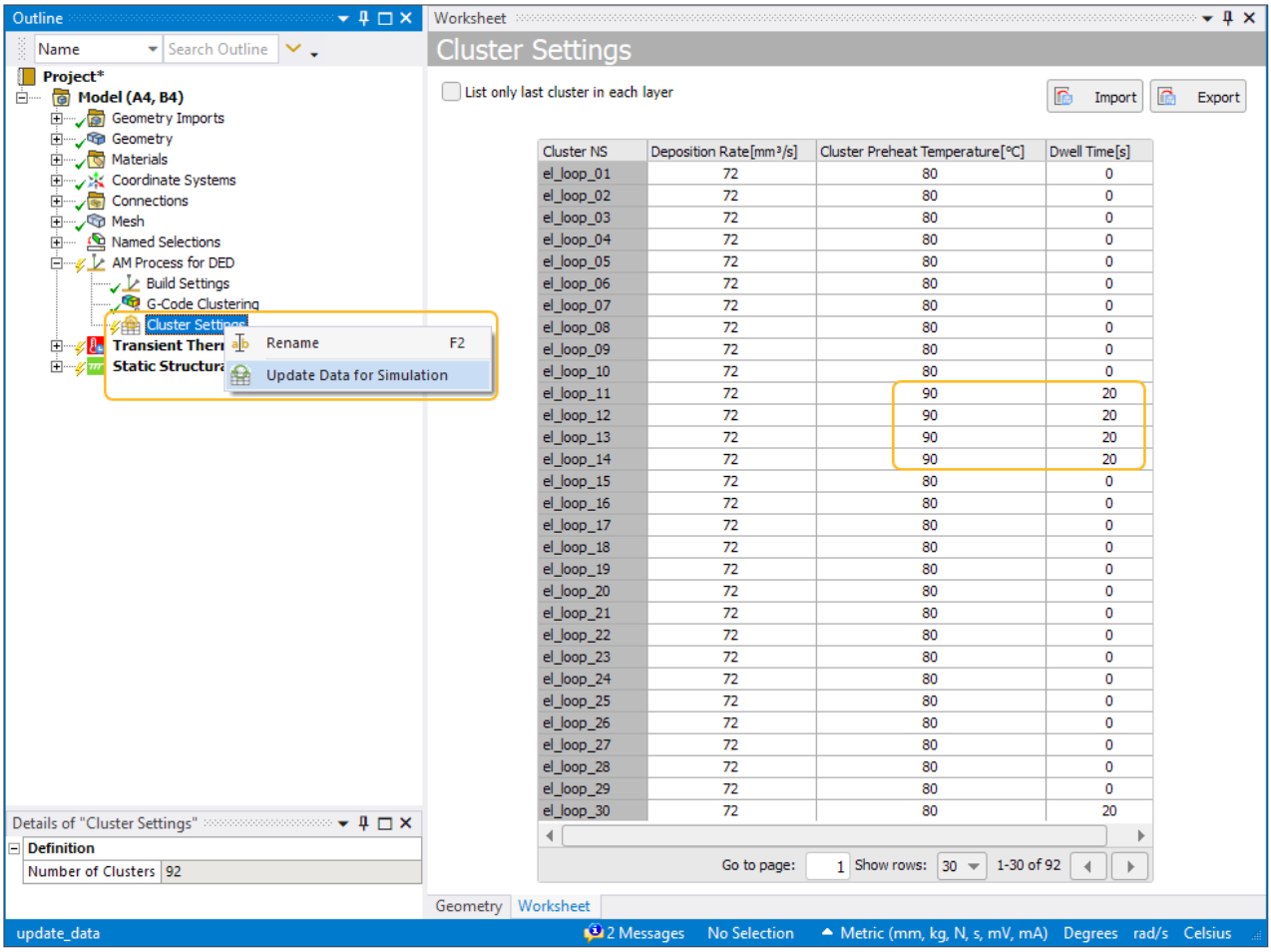
Data Format – Dots versus Commas

- Column titles, text strings, units, and numerical values in the Cluster Settings table always appear in English. Numerical values appear with decimal points (dots) rather than commas. For example, if your localization setting is set to French or German and you enter 93,33 for Preheat Temperature in the Build Settings details pane, the table shows 93.33.
- When entering numerical values into the table, use dots rather than commas.
- With .csv import, commas are interpreted as column separators, so use dots rather than commas as decimal separators in numerical inputs in your .csv file.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

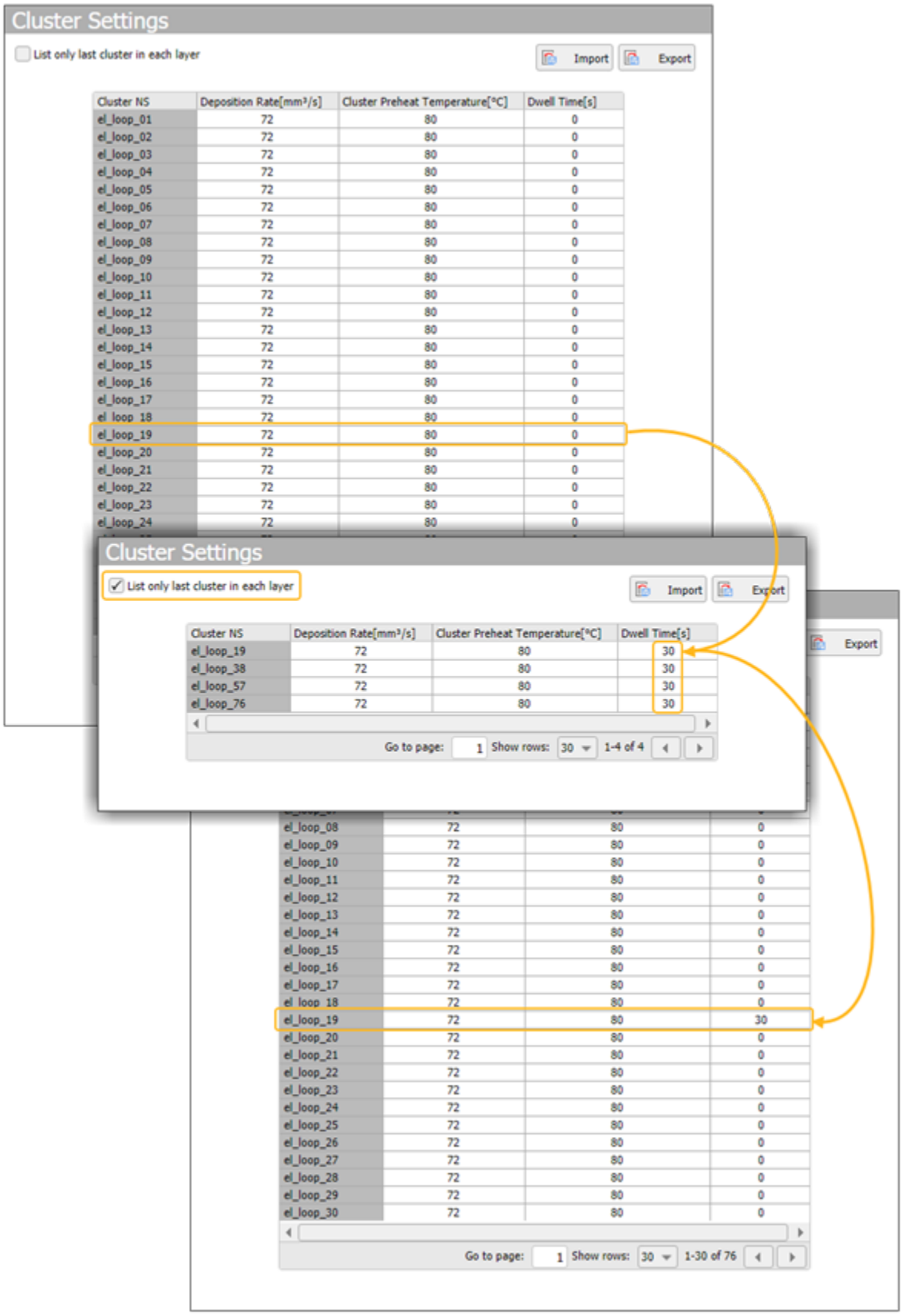
5.8.3.1.2. Modifying Table Data

You can manually change data in the table or import data from a file. Once you change any data, right-click **Cluster Settings** and select **Update Data for Simulation**.



You can filter the data shown in the table to list only the last cluster in each layer by clicking the checkbox in the upper, left corner above the table. This is especially convenient when you have a large model to identify the last cluster of every layer so you can adjust the Dwell Time before starting the next layer, as shown in the following example. By clicking **List only last cluster in each layer**, we see that clusters 19, 38, 57, and 76 are the last clusters in each of the four layers so we modified the Dwell Time just for those clusters. We followed this by right-clicking the Cluster Settings object and choosing **Update Data for Simulation**.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Note the following regarding valid data input:

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

- A string (non-numeric value) is invalid and is interpreted as a 0.
- For Deposition Rate, a value of 0 is invalid. The value must be greater than 0.
- For Cluster Preheat Temperature, a value of 0 is interpreted as Off.
- For Dwell Time, a value of 0 *is* a valid value.

5.8.3.1.3. Importing and Exporting Table Data

- Use the **Import** button to import data from an existing .csv file. Navigate to the desired folder and click **Open**. With .csv import, commas are interpreted as column separators, so use dots rather than commas as decimal separators in numerical inputs in your .csv file.

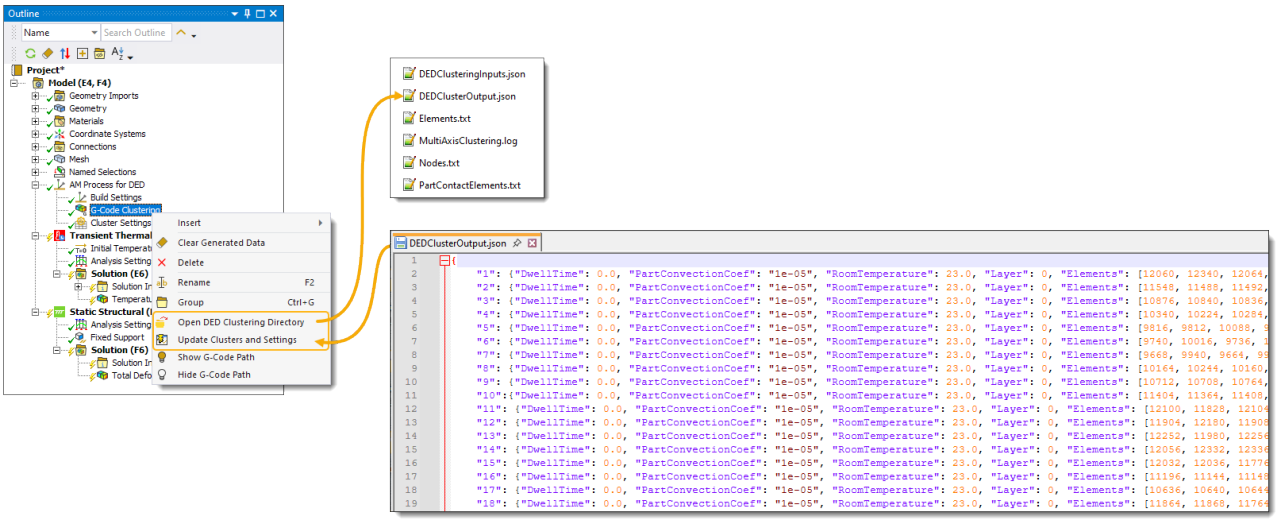
  If you try to import a .csv file that does not have the same number of clusters as the existing model in the tree, the import will not be successful and an error message will include the names of missing or extra clusters.

- Use the **Export** button to export data from the table to a .csv file. For clusters generated via G-Code, the export operation *does not* overwrite or affect the G-Code file in any way.

5.8.3.2. Editing the Cluster Output File Directly

As an alternative to using the Cluster Settings table (p. 84), you can edit the output file, `DEDClusterOutput.json`, from the cluster generation process directly. Change the build settings or add, remove, or move elements among clusters.

Use the right-click options on the G-Code Clustering object (p. 116) to do this. Right-click the G-Code Clustering object and choose **Open DED Clustering Directory**. From there, edit the `DEDClusterOutput.json` file as you wish. This is the output file from the cluster generation process. After editing, right-click the G-Code Clustering object again and choose **Update Clusters and Settings**. This action regenerates the Named Selections and updates cluster settings but will not rerun the clustering algorithm.



Files in the DED Clustering Directory include:

- The `Elements.txt`, `Nodes.txt`, and `PartContactElements.txt` files store the Named Selections defined earlier (p. 40).
- The `DEDClusteringInputs.json` file contains the inputs for the clustering algorithm.
- The `DEDClusterOutput.json` file contains the outputs from the clustering algorithm. You can edit this file as described above.
- The `MultiAxisClustering.log` file contains timestamps, statistics, and other log-type information from the clustering algorithm if you have a multi-axis simulation.

5.8.3.3. Controlling Heating and Cooling During the Build via Parameters (Command Snippets)

You can control the additional build settings (p. 111) (that is, those that are not in the Cluster Settings table (p. 84)) of power, for power based heating method, or process temperature, for temperature based heating method, and radiation emissivity and gas convection coefficients during the build for both the part(s) and the base using command snippets (Commands object) with specific Ansys-defined DED parameters. This may be useful to lower power in areas where there may be overheating, say, or to adjust build conditions for different materials. Note that these parameters have no effect on the cooldown process.

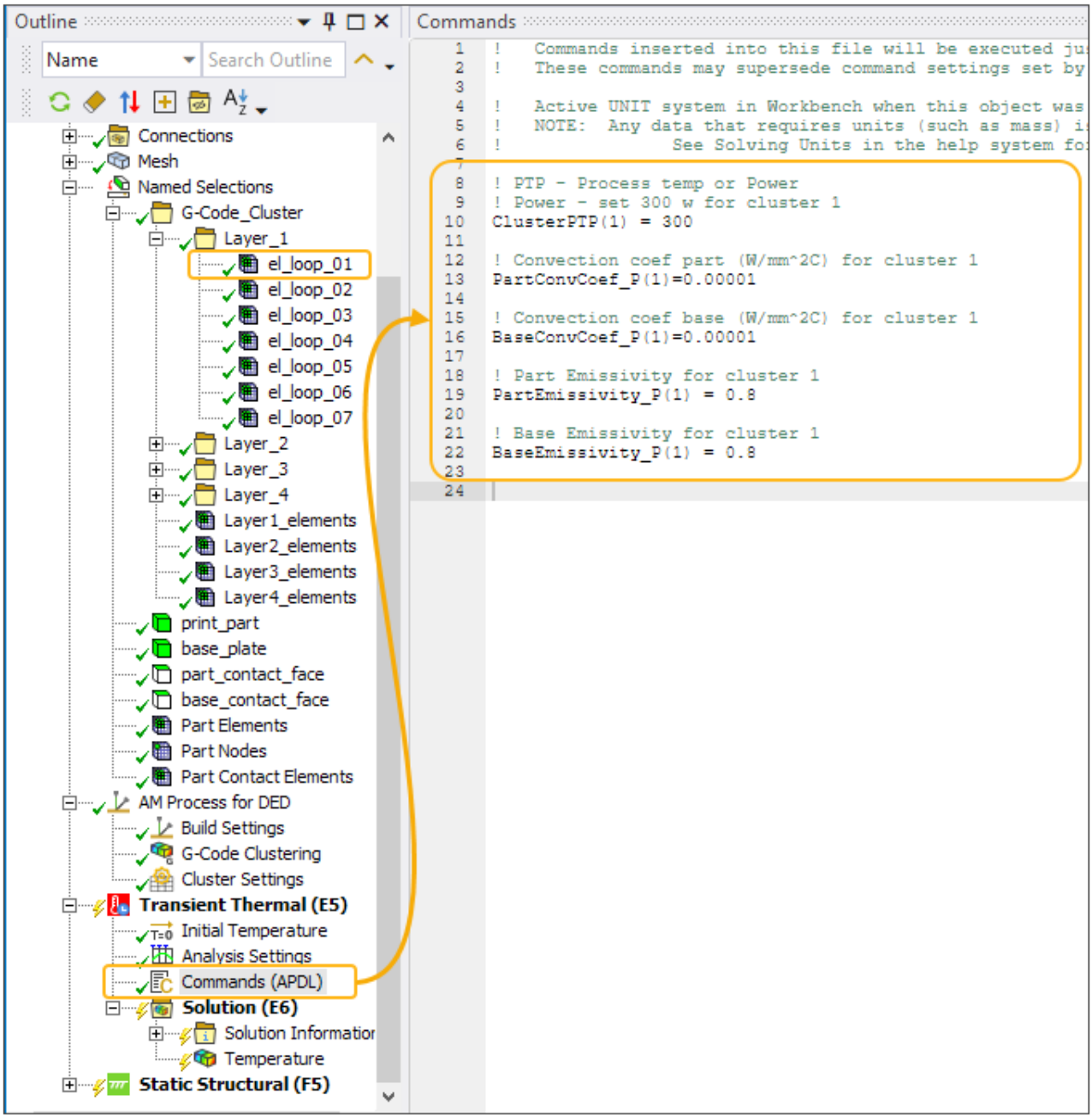
The following table shows the DED parameters that can be used to more precisely control the heating and cooling per cluster during the build process. The parameters must be named exactly as they appear in the table. For all the parameters, the index# is the cluster number, which can be found under Named Selections, G-Code_Cluster, Layer_1, el_loop_0**1**, or el_loop_0**2**, el_loop_0**3**, and so on.

Name of Parameter	Definition
ClusterPTP(*index#*)	Process Temperature or Power. This can be one or the other, as set by Heating Method on the Build Settings object (p. 111).
PartConvCoef_P(*index#*)	Convection coefficient for the part
BaseConvCoef_P(*index#*)	Convection coefficient for the base
PartEmissivity_P(*index#*)	Radiation emissivity for the part
BaseEmissivity_P(*index#*)	Radiation emissivity for the part

Insert a Commands object into the Transient Thermal environment in the project tree. In the Commands object, define the parameters for process temperature or power, and convection coefficients and radiation emissivity, as needed. Upon solution, the original build settings will get overwritten by those defined in the parameters.

In the following hypothetical example, the build settings are changed for cluster 1.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



5.9. Define Connections

Without shared topology between bodies, connections are the mechanism to ensure that the part and the base plate bodies in the simulation are aware of each other and can share data (temperatures and displacements) across boundaries. The connection type that is typically used in DED process simulations is a special case of bonded contact between the element faces on the bottom of the build and the element faces on the top of the base. **The name of this contact must be DED_Contact.**

In the project tree, a Contacts object is automatically added under Connections with a Contact Region object underneath it when using the DED Process Add-on. The Contact Side is defined as the lower element faces of the print part. The Target Side is defined as the upper element faces of the base plate.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

You must rename the Contact Region object to DED_Contact. Note that the DED_Contact name is automatically assigned when using the DED Process Wizard.

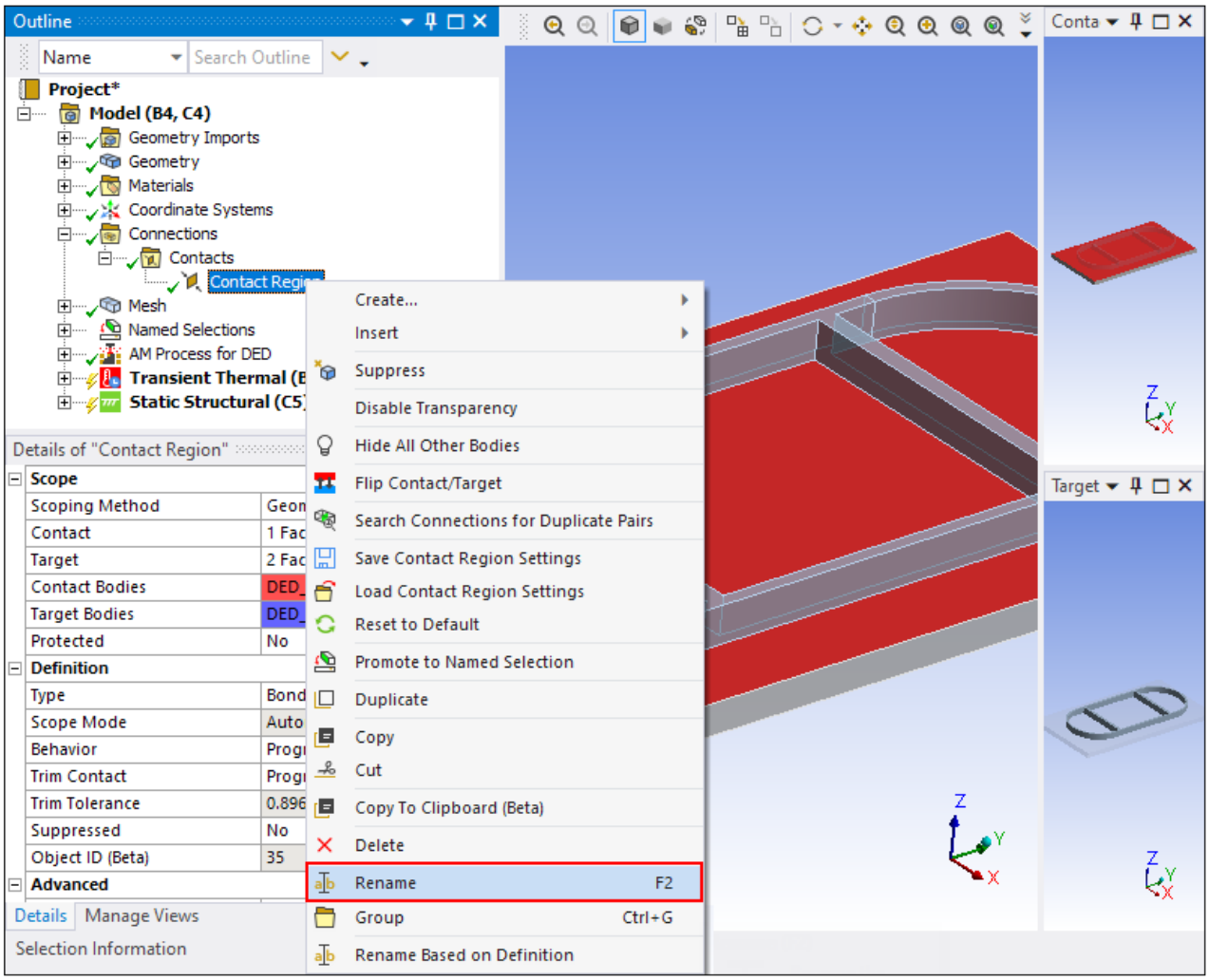
Important:

Even if there is shared topology between the part and the base plate (recommended for simulations with curved bases), you must still include the DED_Contact object in the project tree, as this is required for solution.

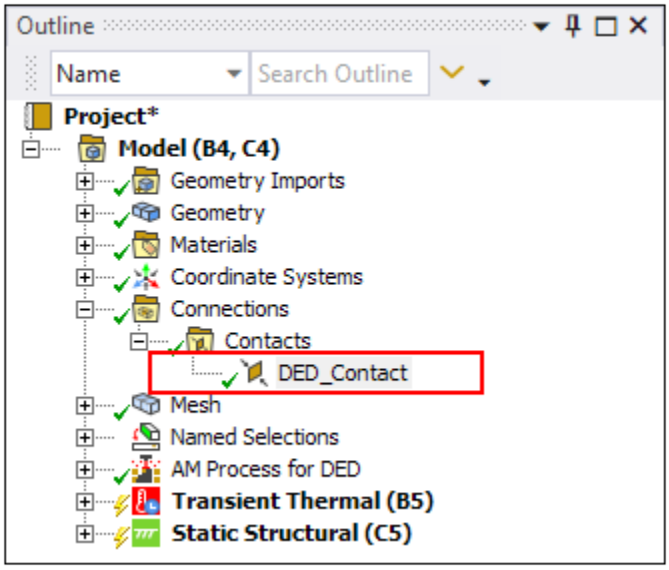
Procedural Steps

Rename to DED_Contact object and review options as follows:

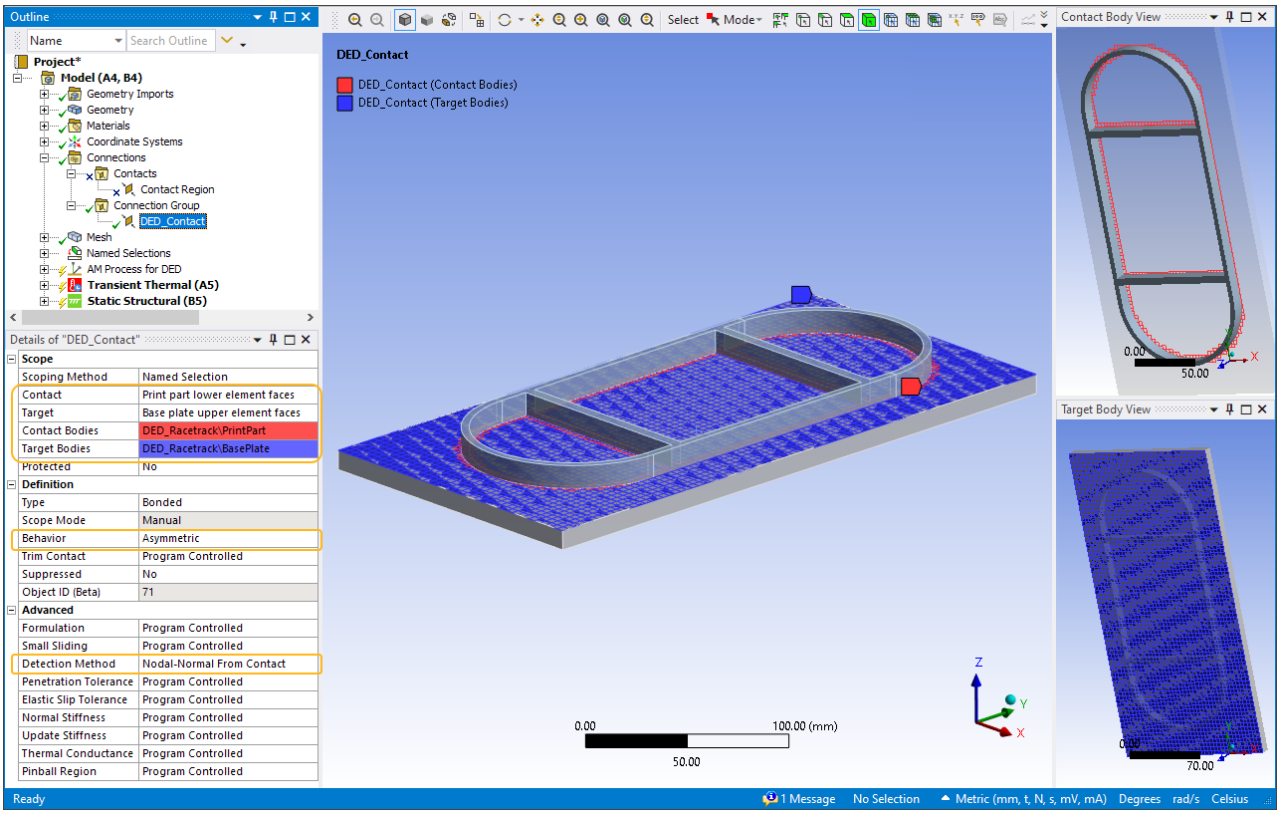
1. Expand the Contacts object and right-click **Contact Region**. Choose **Rename** and enter **DED_Contact** as the new name. This is required for solution of the DED simulation.



Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



2. The recommended settings (by default since release 2023 R1) for the DED_Contact can be seen below; the *Contact* surface is the lower element faces of the print part and the *Target* surface is the upper element faces of the base plate. Two Named Selections (p. 40) defined earlier are used to specify the contact and target. The Behavior is recommended to be *Asymmetric* with *Nodal-Normal from Contact* as the Detection Method. This setting is seen to be helpful to eliminate or reduce the negative temperature issue due to thermal shock found in previous releases. Defining contact between build and base can be challenging sometimes. Recommended settings may not work for your case. For more information, refer to the Contact Technology Guide.



Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

5.10. Define DED Process Steps

A DED process simulation consists of the build process and a cooldown process. The duration of the build process depends on G-Code data or user defined named selections for the definition of the weld track, and hence input parameters such as material deposition rate cannot be set by the user. The duration of the cooldown process can be defined as a parameter.

After cooldown, the build parts can be removed from the base plate. In this case, removal constraint nodes need to be defined. See Apply Structural Boundary Conditions (p. 98).

5.11. Establish Thermal Analysis Settings

Thermal analysis settings allow customization of various options during the transient thermal solution.

The transient thermal analysis will determine the temperature history during the build process. These temperatures will then be used in a static structural analysis to determine the build distortions and stresses. The Mechanical application will automatically determine all the steps and times needed for time integration in the simulation.

Procedural Steps

Click **Analysis Settings** under **Transient Thermal** and, in Details, review the available options. It is usually appropriate to leave most analysis settings set to program-controlled. These settings are determined when you insert the DED Process object into the project tree.

In a DED simulation, your results file will grow in size very quickly, so we recommend you keep the default **Output Controls** that will suppress calculation of thermal flux, nodal forces, Euler angles, volume and energy, and other miscellaneous items. Nodal temperatures are stored at all time points by default but you can change that option (under Store Results At) so that temperatures are stored at the last time point only, at equally spaced time points, or at a specified recurrence rate.

5.12. Apply Thermal Boundary Conditions

In the Build Settings (p. 57) step of our workflow, we have already accounted for convection between the part to the surrounding gas in the chamber in both the Build Step and the Cooldown Step. There is one other area of heat transfer to consider and that is associated with the base plate.

During the printing process, the plate is typically heated on the bottom to maintain a constant, slightly elevated temperature. You may insert a temperature constraint or convection surface to simulate this during the build step. If the plate is not heated, perhaps apply a room condition convection surface, or set the surface boundary condition to adiabatic.

After the print process is complete, the base plate heating is removed and the built part cools to room temperature. This is simulated by a room-temperature convection surface applied to the bottom of the base plate in the cooldown step.

The time duration of the cooldown step is estimated based on the average part temperature at the end of the build, its volume, and material properties. Using the convection values and room temperature, the Mechanical application solves the simple heat transfer equation to get an estimate of the cool down time. It is only an estimate so you can extend it if required or put in a preferred time in analysis settings.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

One final step is performed to force all the temperatures to room temperature for the subsequent distortion and stress calculation.

Procedural Steps

To apply a temperature to the bottom surface of the base plate for the build step:

1. Right-click the **Transient Thermal** object and then select **Insert** > **Temperature**.
2. Select the bottom face of the base plate (geometry selection) and hit **Apply** in the Details view.
3. In Details view, under Definition, **Temperature**, enter a temperature value for **Magnitude**. (In our example we use 80°.)

To apply room-temperature convection to the bottom surface of the base plate for the cooldown step:

1. Right-click the **Transient Thermal** object and then select **Insert** > **Convection**.
2. Select the bottom face of the base plate (geometry selection) and hit **Apply** in the Details view.
3. In Details view, under Definition, **Film Coefficient**, enter a value for the convection coefficient.
4. Under **Ambient Temperature**, enter a value of 22°. (This may be the default.)

Important:

Keep the DED_Loads command snippet object in the tree. This is an empty placeholder object that is required for the solution. It was are automatically inserted by the AM Process for DED object.

5.13. Solve the Transient Thermal Analysis

Some users prefer to solve the thermal analysis first to observe how the simulation is progressing while others prefer to set up everything and run both the thermal and structural analyses at once. Either way is perfectly acceptable. To run them both at once, simply execute a solve from the static structural side and the transient thermal analysis will be run first automatically.

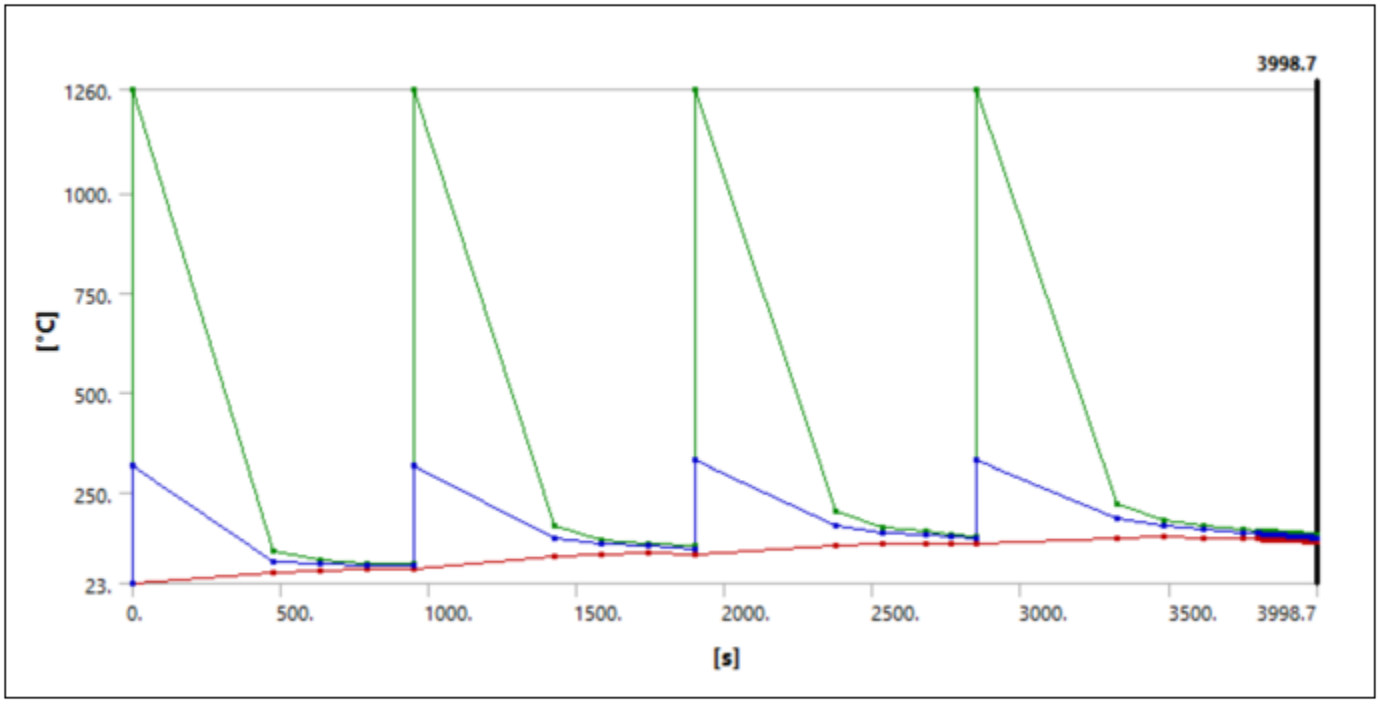
Procedural Steps

1. Review the items in the project tree. Be sure there are green check marks next to all the tree objects, including AM Process for DED (p. 109), indicating they are ready to solve. Be sure the DED_Contact object is under the Connections folder.
2. To initiate the solution, under Transient Thermal, highlight the **Solution** object, right-click and select **Solve**. It is safe to ignore the warning message: *"The model has no temperature, convection, or radiation conditions specified..."*, as the DED Process Add-on applies the appropriate boundary conditions automatically with information from the Build Settings step (p. 57).

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

What Happens During the Transient Thermal Solution?

To understand the thermal solution in a DED simulation, it's helpful to examine a temperature plot over time. The plot below shows maximum (green), average (blue), and minimum (red) temperatures over time for a four-cluster model. You can see four temperature peaks, each corresponding to the (nearly) instantaneous application of the Process Temperature as each cluster is activated (made "alive") during the build.

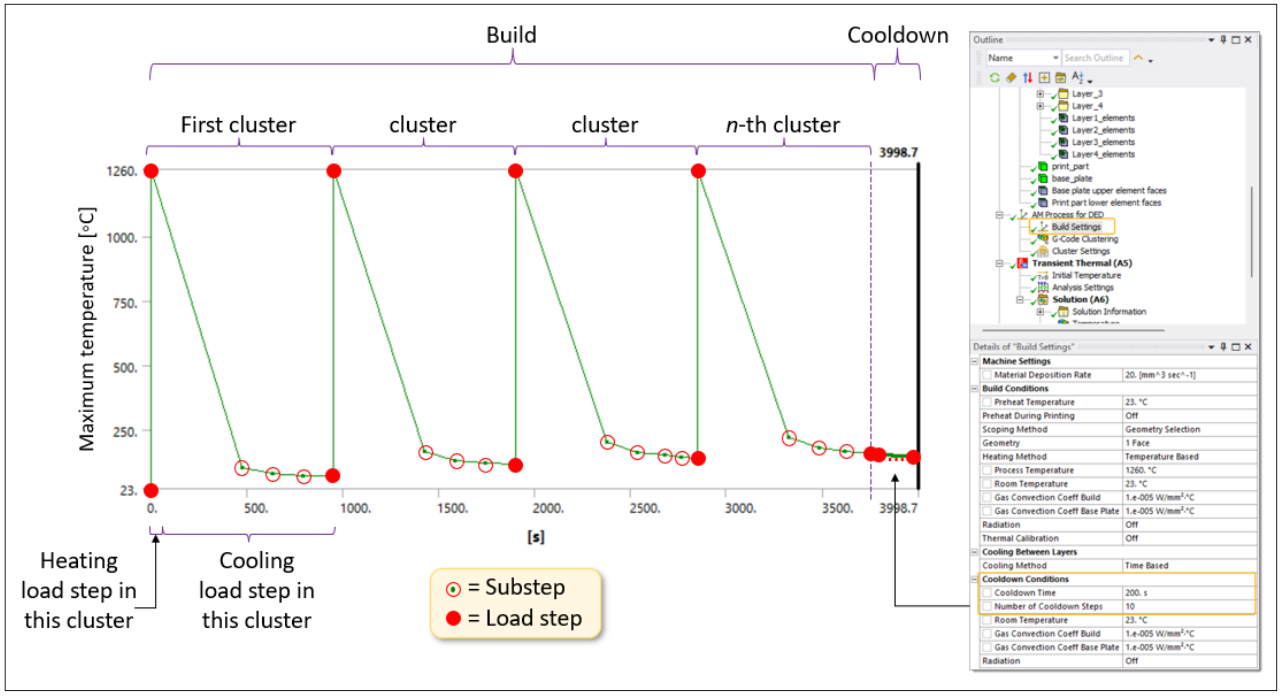


The enlarged, annotated image below shows only the maximum temperature (green) for clarity. All four clusters are labeled. Each cluster undergoes two load steps: one for heating and one for cooling (i.e., between cluster activations). In this example, the **Heating Method** is Temperature Based. For the first cluster, the simulation begins at time = 0, starting from room temperature. During the heating load step, the temperature rapidly rises to the process temperature of 1260°C in a single substep. The cooling load step then follows, consisting of multiple substeps.

Substeps are intermediate solution points within a load step. The time between two substeps is referred to as a *time step*. These time steps are determined automatically using automatic time stepping, which calculates an optimal time step at the end of each substep based on the system's response to the applied loads.

The total time span for both the heating and cooling load steps in each cluster is calculated as: **Cluster Volume** (from the Clustering object) divided by **Material Deposition Rate** (from the Build Settings object).

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Once the first cluster finishes, the next cluster is activated. It follows the same pattern: heating load step, then cooling load step. This process continues sequentially through all clusters following the weld track, completing the build step of the simulation.

After the build step, the cooldown step begins. Its duration is defined by the **Cooldown Time** field in the Build Settings object (200 seconds in this example, though this value has been shortened for demonstration purposes). The **Number of Cooldown Steps** defines how many load steps make up the cooldown phase. Each of these load steps will also have multiple substeps determined by automatic time stepping.

How Thermal History is Passed to the Structural Solution

The thermal history passed to the structural solution is the temperature at the end of every load step, not at every substep. This includes:

- Two temperatures per cluster from the build step (one from the end of the heating step and one from the end of the cooling step).
- One temperature per cooldown load step, as defined by the Number of Cooldown Steps (defaults to 10).

Passing only end-of-load-step temperatures ensures the structural simulation remains computationally efficient while still capturing essential thermal behavior.

5.14. Establish Structural Analysis Settings

Structural analysis settings allow customization of various options during the static structural solution. By default for a DED simulation, Large Deflection is turned on (under Solver Controls). Consider turning on Line Search (under Nonlinear Controls) if you have convergence issues.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

We recommend you think about using options to reduce the amount of data stored on the results file. You should suppress the calculation of stresses and strains if you are only interested in distortion, as these data will easily increase the size of the results file and it may become unmanageable.

Procedural Steps

Click **Analysis Settings** under **Static Structural** and, in Details, review the available options. As with the thermal analysis, many analysis settings in the structural analysis are program controlled.

Review the **Output Controls** and suppress items not of interest. Turn off calculations of Stress and Strain by changing those options from Yes to No. (Hint: Simply double-clicking in the Yes box changes it to No.)

5.15. Apply Structural Boundary Conditions

Unless you specified a base removal step, the only boundary condition that must be applied is a fixed support boundary condition on the build plate. If you want to remove the base, you need to apply fixed boundary conditions on three nodes of the part to prevent rigid body motion when the base is removed.

Procedural Steps

1. To fix the bottom surface of the base plate, right-click the **Static Structural** object and then select **Insert** > **Fixed Support**. Select the bottom face of the base plate (geometry selection) and hit **Apply** in the Details view. If you have a more detailed geometry for the base, such as one with bolt holes that you want to explicitly simulate, apply the boundary conditions to affix the plate accordingly.

2. If you specified a base removal step, you must constrain the part at three nodes to prevent rigid body motion. Be aware that you are constraining the nodes to their *displaced position at the end of the build*, rather than constraining them to 0. (For example, if the node displaced 0.1 mm during the simulation, the constraint will keep the node at 0.1 mm.) Use a Commands object to do this.

    a. You may want to hide the base body during this process. Right-click the base body and select **Hide Body**.

    b. Zoom in to the area of the part where you will apply constraints. Switch to Node mode of selection.

    c. Create nodal named selections: Right-click **Model** in the project tree, and select **Insert** > **Named Selection**. Select a node on the bottom of the part, perhaps toward the middle of the part. You should choose nodes that are unimportant to your results. Perhaps spread them out over the bottom of the part but *the nodes must not be co-linear*. Your node selection will affect displacement results but not stresses or strains. Click **Apply** in the Details view. Right-click the newly created **Selection** in the project tree (under Named Selections) and select **Rename**. Name it to DED_UX_Remove (for constrained node 1). Do the same step two more times and rename them to DED_UY_Remove and DED_UZ_Remove.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

d. Remember to make the base plate visible again if you hid the body. Right-click the base body and select **Show Body**.

Important:

Keep the DED_Loads command snippet object in the tree. This is an empty placeholder object that is required for the solution. It was are automatically inserted by the AM Process for DED object.

5.16. Solve the Static Structural Analysis

You initiate the structural solution in this step. Watch convergence plots or a live tracker of the total deformation as the solution progresses.

Procedural Steps

1. To initiate the solution, under Static Structural, highlight the **Solution** object, right-click and select **Solve**.

2. To observe convergence plots during solution, click **Solution Information** and in the Details view, under **Solution Output**, choose from among the selections in the drop-down list, such as **Force Convergence**.

5.17. Review Results

Once a solution is available you can view contours of the results or animate the results. You will have solutions at many time points and you can display the variation of the result item at a location over the build history.

Typical result items of significance are temperatures and residual "hotspots" in the Transient Thermal solution and displacements and equivalent stress in the Static Structural solution.

Procedural Steps

Recall that you used Output Controls under Analysis Options to control which items are solved for in the simulation. You also may have set up results trackers in the solution step to observe the results being processed during the solution. When reviewing results *after* the solution, you will generally need to *evaluate* results to obtain them for viewing. Evaluating results is a way to retrieve them from the data that was stored during solution. Only data items that were solved for may be retrieved in this way.

There are many different ways to view results but some of the ones most commonly used for additive manufacturing are described here:

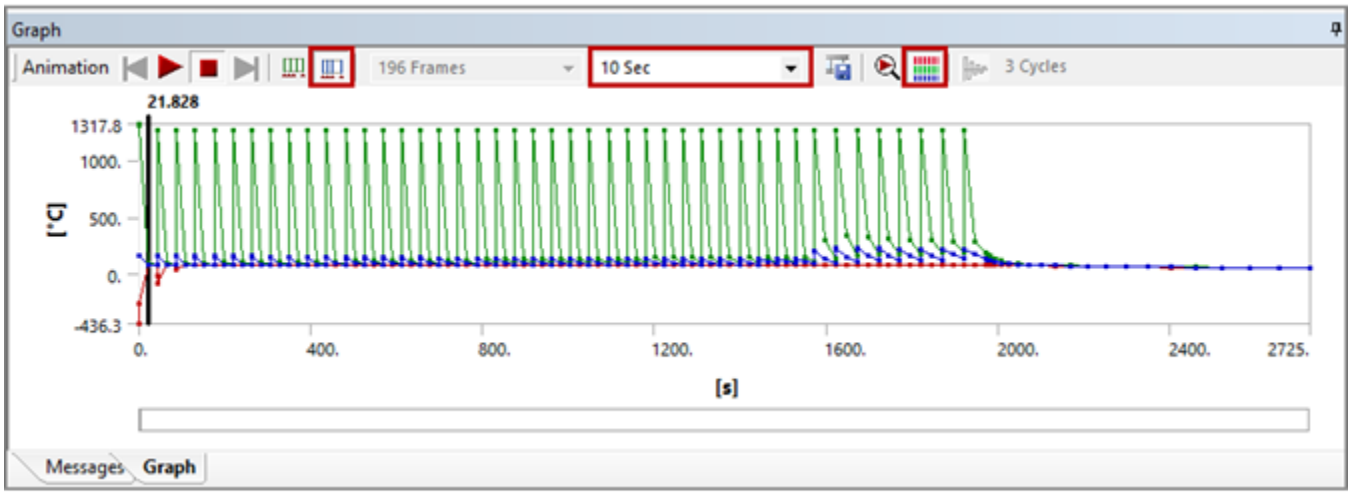
Animate Thermal Results - Temperatures

1. In the project tree under Transient Thermal, select the **Solution** object and then, in the context menu, click **Thermal** and then **Temperature**. Or right-click **Insert** > **Thermal** > **Temperature**. A Temperature object is created and becomes active.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

2. Right-click **Temperature** and select **Evaluate All Results**. This retrieves the temperature data and displays it in both graph and tabular form.

3. Use the animation controls at the top of the Graph window. Click the **Result Sets** button  and also the **Update Contour Range at Each Animation Frame** button  . Adjust the number of seconds for the animation and click **Play**. For more information about animation controls, see Animation in the *Mechanical User's Guide*.

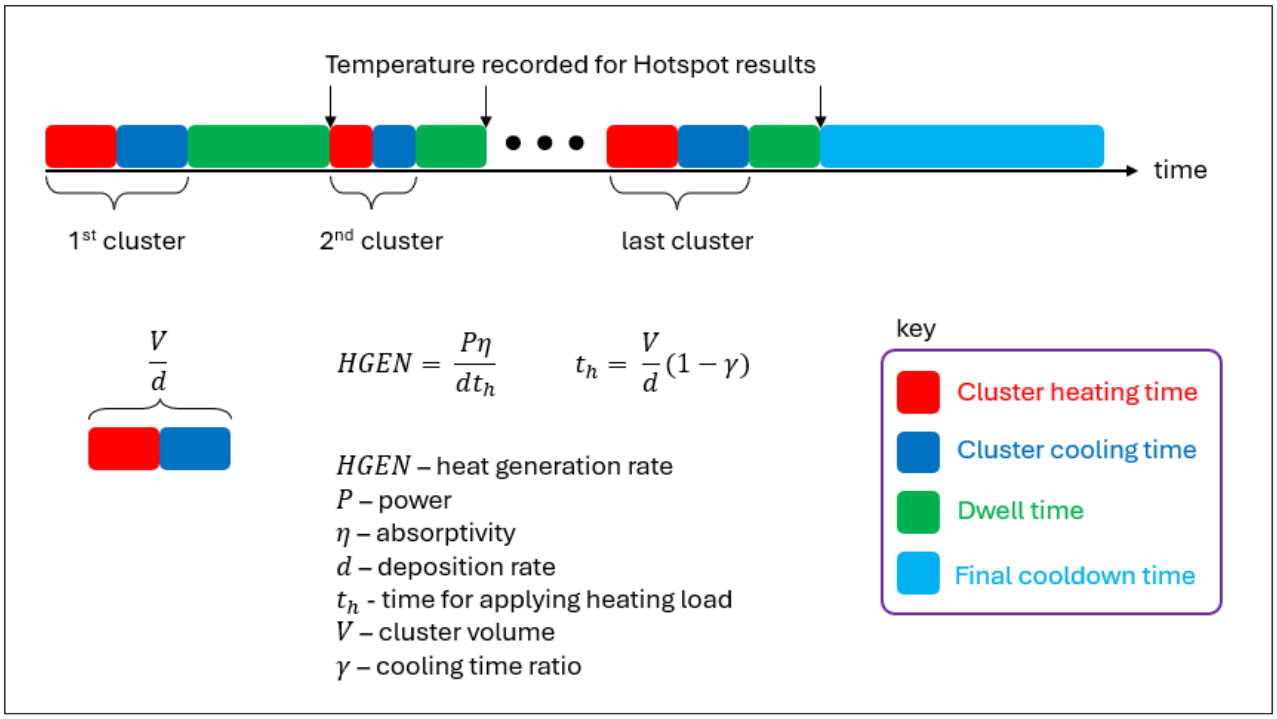


Examine Hotspots

Click **Hotspot** from the DED Process ribbon to add a Hotspot result item to the tree. Alternatively, right-click the Solution object, and choose Insert > DED Hotspot.

The Hotspot result shows the maximum temperature in each depositioned cluster of elements immediately before the next cluster is activated.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



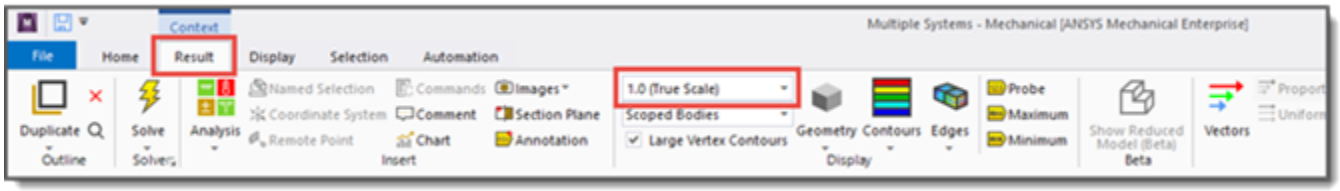
Animate Structural Results - Total Deformation

1. In the project tree under Static Structural, select the **Solution** object and then, in the context menu, click **Deformation** and then **Total**. A Total Deformation object is created and becomes active.

2. Right-click **Total Deformation** and select **Evaluate All Results**. This retrieves the deformation data and displays it in both graph and tabular form.

3. Use the animation controls at the top of the Graph window.

4. It is important to change the magnification scale of the display in the geometry window so you can get a true sense (or an exaggerated sense) of the deformation in your build. With Total Deformation highlighted in the project tree, from the **Result** context tab, select the **True Scale** option from the drop-down menu. Animate the results using the animation controls. Select other scales from the drop-down menu to see exaggerated results.



Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information
of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Chapter 6: Supported G-Code Commands

Within a DED process simulation, we use G-Code commands to create clusters of elements that represent the building sequence of the part. Since G-Code can contain extensions and variations specific to machine tool manufacturers and control manufacturers, we focus on basic commands for linear movement (G00 and G01) and circular movement (CIP), and assume a planar tool path in the X-Y, X-Z, or Y-Z plane, normal to print direction. The following table shows supported G-Code commands:

Command	Function and Parameter	Visualization	UI Options	Examples
Arbitrary command	Switch Laser On	N/A	Laser On Commands to specify one or more commands, comma separated	G01 M24
Arbitrary command	Switch Laser Off	N/A	Laser Off Commands to specify one or more commands, comma separated	G00 M28
Arbitrary command	Movement X0.0 Y0.0 Z0.0	If previous command is Laser Off: Blue line from last position If previous command is Laser On: Green line from last position	Move Commands to specify one or more commands, comma separated Extrusion Tag option to On if E-Parameter is defined Show Positioning Lines option to Off to suppress visualization	G00 X1.0 Y0.0 Z0.0 G01 X1.0 Y0.0 G01 X1.0 Y0.0 E1.0
CIP	Circle defined by intermediate position (I, J, K) and end position (X, Y, Z) (p. 107)	Green line segments	Show Positioning Lines option to Off to suppress visualization	CIP I1=1.0 J1=1.0 K1=0.0 X2.0 Y0.0 Z0.0
X	Linear movement X0.0	If previous command is Laser Off: Blue line from last position If previous command is	Show Positioning Lines option to Off to suppress visualization	X1.0 Y0.0

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Command	Function and Parameter	Visualization	UI Options	Examples
		Laser On: Green line from last position		
Y	Linear movement Y0.0	If previous command is Laser Off: Blue line from last position If previous command is Laser On: Green line from last position	Show Positioning Lines option to Off to suppress visualization	Y1.0
Z	Linear movement Z0.0	If previous command is Laser Off: Blue line from last position If previous command is Laser On: Green line from last position	Show Positioning Lines option to Off to suppress visualization	Z1.0
G90	Switch to absolute positioning for the following commands	N/A	Default	G90 G01 X1.0 G01 X2.0 G90 G01 X1.0 Z1.0
G91	Switch to relative positioning for the following commands	N/A	Default	G91 G01 X1.0 G01 X2.0 G91 G01 X1.0 Z1.0
F	Feed Rate	N/A		F200
T	Preheat Temperature	N/A		

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Command	Function and Parameter	Visualization	UI Options	Examples
G4	Dwell Time The following variations, as used in various machines, are supported: • G4 F*nnn* (*nnn* in seconds) • G4 X*nnn* (*nnn* in seconds) • G4 U*nnn* (*nnn* in seconds) • G4 S*nnn* (*nnn* in seconds) • G4 P*nnn* (*nnn* in milliseconds)	N/A		G4 P700

A G-Code file must be in ASCII format. Following is an example of a G-Code text file from the racetrack example used throughout this document.

```
G00 X-100.0 Y-83.5 Z0.0
; First layer
G00 X-54.0 Y-83.5 Z3.0
G01 X-49.9 Y-104.2 ; Circle 1
G01 X-38.2 Y-121.7
G01 X-20.7 Y-133.4
G01 X0.0 Y-137.5
G01 X20.7 Y-133.4
G01 X38.2 Y-121.7
G01 X49.9 Y-104.2
G01 X54.0 Y-83.5
G01 X54.0 Y80.2 ; Y-Line 1
G01 X49.9 Y100.9 ; Circle 2
G01 X38.2 Y118.4
G01 X20.7 Y130.1
G01 X0.0 Y134.2
G01 X-20.7 Y130.1
G01 X-38.2 Y118.4
G01 X-49.9 Y100.9
G01 X-54.0 Y80.2
G01 X-54.0 Y-83.5 ; Y-Line 2
G00 X-54.0 Y-64.6
G01 X54.0 Y-64.6 ; X-Line 1
G00 X-54.0 Y63.4
G01 X54.0 Y63.4 ; X-Line 2
; Next layer
G00 X-54.0 Y-83.5 Z6.0
```

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

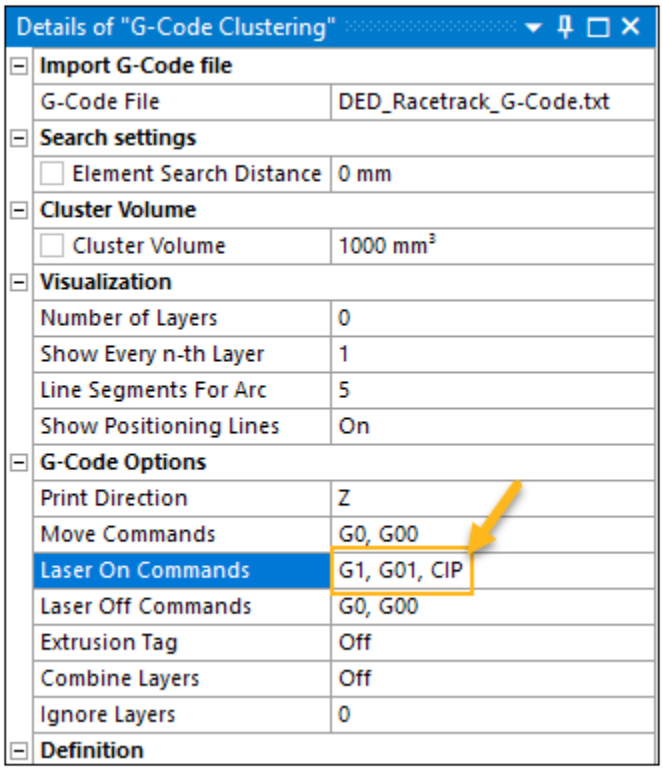
```
G01 X-49.9 Y-104.2 ; Circle 1
G01 X-38.2 Y-121.7
G01 X-20.7 Y-133.4
G01 X0.0 Y-137.5
G01 X20.7 Y-133.4
G01 X38.2 Y-121.7
G01 X49.9 Y-104.2
G01 X54.0 Y-83.5
G01 X54.0 Y80.2 ; Y-Line 1
G01 X49.9 Y100.9 ; Circle 2
G01 X38.2 Y118.4
G01 X20.7 Y130.1
G01 X0.0 Y134.2
G01 X-20.7 Y130.1
G01 X-38.2 Y118.4
G01 X-49.9 Y100.9
G01 X-54.0 Y80.2
G01 X-54.0 Y-83.5 ; Y-Line 2
G00 X-54.0 Y-64.6
G01 X54.0 Y-64.6 ; X-Line 1
G00 X-54.0 Y63.4
G01 X54.0 Y63.4 ; X-Line 2
; Next layer
G00 X-54.0 Y-83.5 Z9.0
G01 X-49.9 Y-104.2 ; Circle 1
G01 X-38.2 Y-121.7
G01 X-20.7 Y-133.4
G01 X0.0 Y-137.5
G01 X20.7 Y-133.4
G01 X38.2 Y-121.7
G01 X49.9 Y-104.2
G01 X54.0 Y-83.5
G01 X54.0 Y80.2 ; Y-Line 1
G01 X49.9 Y100.9 ; Circle 2
G01 X38.2 Y118.4
G01 X20.7 Y130.1
G01 X0.0 Y134.2
G01 X-20.7 Y130.1
G01 X-38.2 Y118.4
G01 X-49.9 Y100.9
G01 X-54.0 Y80.2
G01 X-54.0 Y-83.5 ; Y-Line 2
G00 X-54.0 Y-64.6
G01 X54.0 Y-64.6 ; X-Line 1
G00 X-54.0 Y63.4
G01 X54.0 Y63.4 ; X-Line 2
; Next layer
G00 X-54.0 Y-83.5 Z12.0
G01 X-49.9 Y-104.2 ; Circle 1
G01 X-38.2 Y-121.7
G01 X-20.7 Y-133.4
G01 X0.0 Y-137.5
G01 X20.7 Y-133.4
G01 X38.2 Y-121.7
G01 X49.9 Y-104.2
G01 X54.0 Y-83.5
G01 X54.0 Y80.2 ; Y-Line 1
G01 X49.9 Y100.9 ; Circle 2
G01 X38.2 Y118.4
G01 X20.7 Y130.1
G01 X0.0 Y134.2
G01 X-20.7 Y130.1
G01 X-38.2 Y118.4
G01 X-49.9 Y100.9
G01 X-54.0 Y80.2
G01 X-54.0 Y-83.5 ; Y-Line 2
G00 X-54.0 Y-64.6
G01 X54.0 Y-64.6 ; X-Line 1
G00 X-54.0 Y63.4
G01 X54.0 Y63.4 ; X-Line 2
```


Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information
of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Circular Movement (CIP)

When using the CIP command for circular movement, two things are required:

1. CIP must be added to the Laser On Commands option (p. 117) in the G-Code Clustering object (p. 116).



2. The circular movement must be defined correctly in the G-Code itself. Any arc can be defined by three points. Within the G-Code file, the following points define an arc:

   - Starting point—obtained from the last G-Code command position (that is, the line above CIP)
   - Intermediate point—indicated by (I,J,K) coordinates on the CIP command line
   - End point—indicated by (X,Y,Z) coordinates on the CIP command line

In the following example, the laser is off while moving to position (0,0,2), the laser is on while moving to (1,0,2), and the laser is on while moving in a circular movement from (1,0,2) up to an inflection point at (0,1,2) and to an end point of (-1,0,2).

```
G0 X0.0 Y0.0 Z2.0
G1 X1.0 Y0.0 Z2.0
CIP I1=0.0 J1=1.0 K1=2.0 X-1.0 Y0.0 Z2.0
```

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information
of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Chapter 7: Object Reference

Specifications are provided for the following Mechanical application objects that are unique to the DED Process Add-on:

7.1. AM Process for DED

7.2. Build Settings

7.3. G-Code Clustering

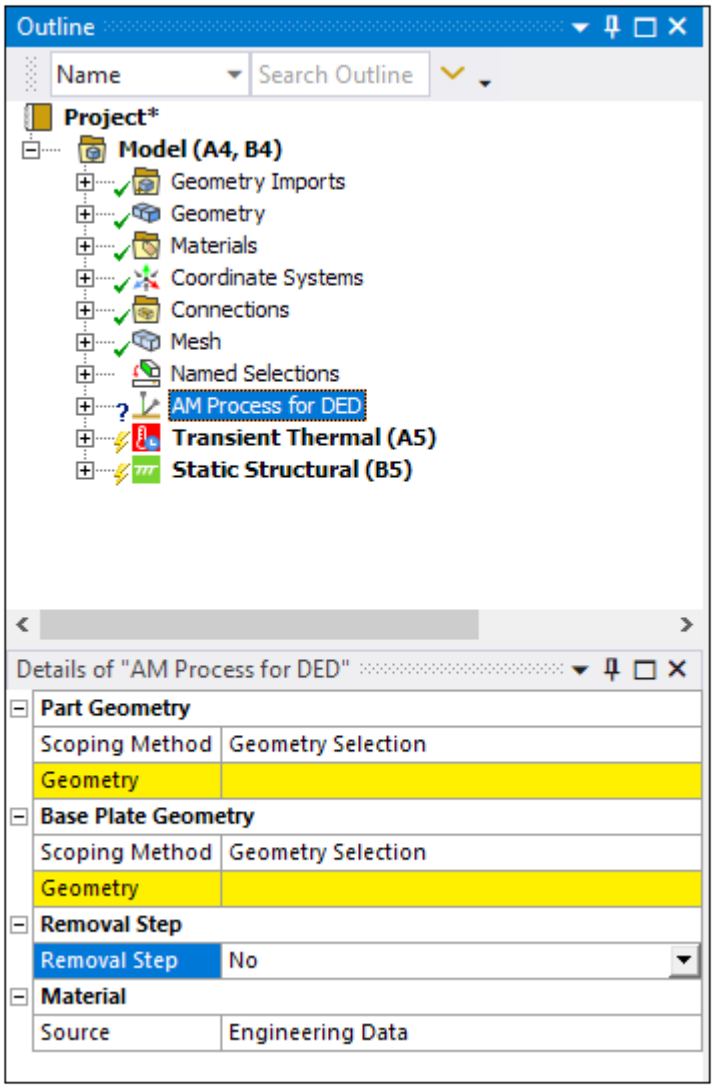
7.4. Manual Clustering

7.5. Cluster Settings

7.1. AM Process for DED

Identifies the geometries, source of material data, and base removal options in a DED process simulation. Also sets global options for all AM-related objects.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Details (p. 110)

Tree Dependencies (p. 111)

Insertion Methods (p. 111)

Right-click Options (p. 111)

Additional Related Information (p. 111)

7.1.1. Details

Category	Description and Settings
Build Geometry	The part to be printed. **Scoping Method**: Choose Geometry Selection or Named Selection to identify the part geometry.
Base Plate Geometry	The base plate upon which the part rests. **Scoping Method**: Choose Geometry Selection or Named Selection to identify the base plate geometry.
Removal Step	Should the simulation include the removal of the part from the base plate?

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

	Removal Step: No or Yes - If No (default), simulation does not include the removal of the part from the baseplate. If Yes, simulation includes the removal of the part from the base plate. **Scoping Method**: Choose Geometry Selection or Named Selection to identify the base plate geometry.
Material	The deposition material. **Source**: Engineering Data/Input file - Choose Engineering Data to use a predefined material. Or choose Input file and then specify **Input File Print Material** and **Input File Base Material**. Files must be in ASCII format with a `.inp` extension.

7.1.2. Tree Dependencies

- **Valid Parent Tree Object**: Model
- **Valid Child Tree Objects**: Build Settings (p. 111), G-Code Clustering (p. 116), Manual Clustering (p. 121)

7.1.3. Insertion Methods

- Inserted under the Model object in the project tree by default when using the DED Process Wizard.
- Select the **DED Process** tab and then click the **DED Process** button in the ribbon.

7.1.4. Right-click Options

Relevant right-click options include:

- **Generate**: Generates element clusters using the path defined in either the Manual Clustering or G-Code Clustering child object.
- **Clear Generated Data**: Clears clusters that have already been generated.

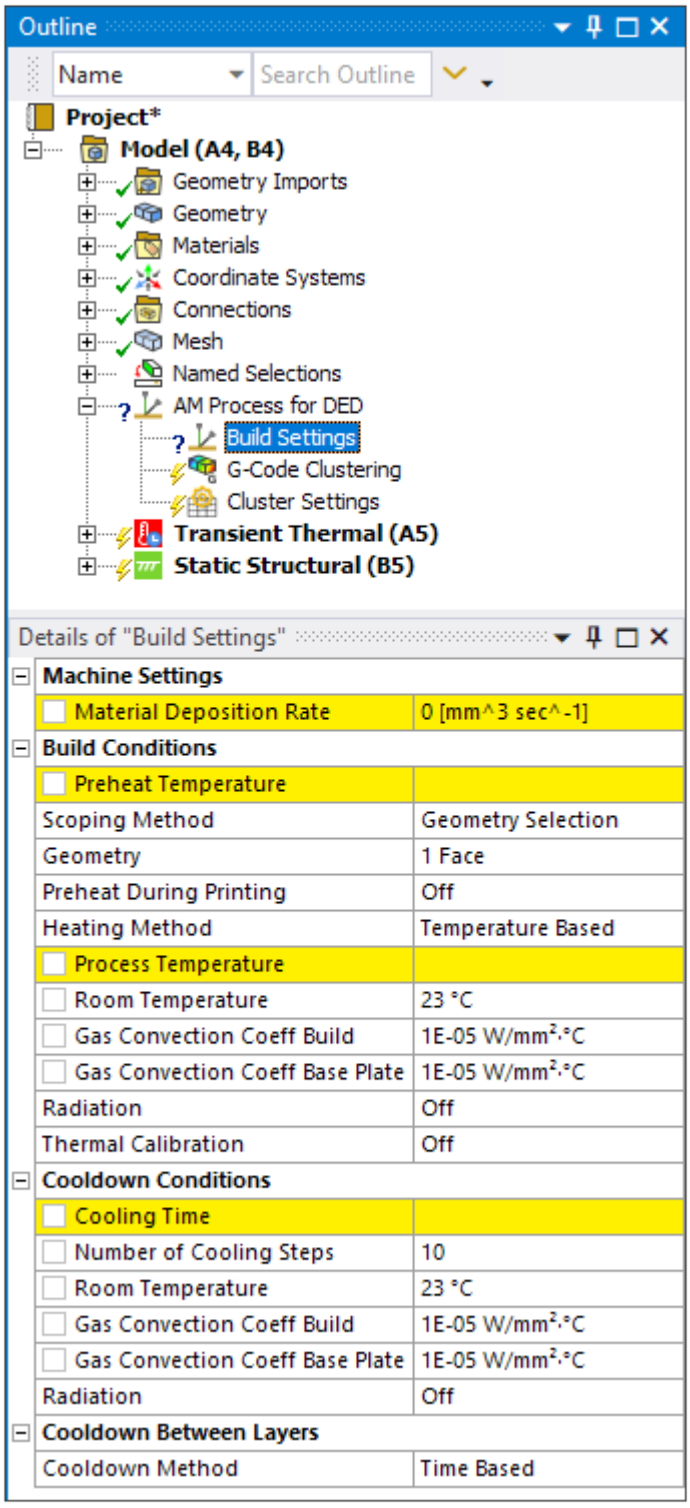
7.1.5. Additional Related Information

Insert AM Process for DED and Identify Geometries (p. 30)

7.2. Build Settings

Establishes settings and conditions related to a DED additive manufacturing process and machine.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Details (p. 113)

Tree Dependencies (p. 116)

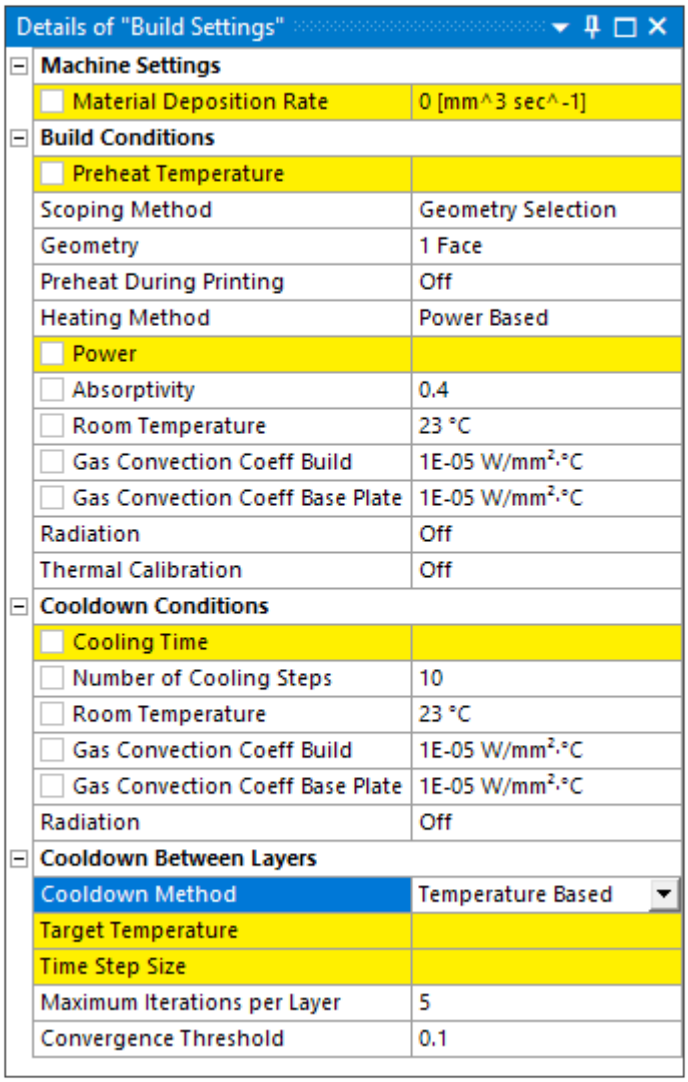
Insertion Methods (p. 116)

Right-click Options (p. 116)

Additional Related Information (p. 116)

Showing alternative heating and cooling methods:

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



7.2.1. Details

Category	Description and Settings
Machine Settings	Process parameters that vary for each DED machine and deposition material. **Material Deposition Rate**: The feeding rate of the melted material, in mm^3/sec. This value can be determined by multiplying *layer thickness* (mm) x *weld width* (mm) x *deposition speed* (mm/sec).
Build Conditions	The environment in the build chamber around the part as it is being printed, including the preheat temperature. **Preheat Temperature**: The starting temperature of the base plate. **Scoping Method**: Choose Geometry Selection or Named Selection to identify the bottom face of the base plate that is preheated. **Preheat During Printing**: Off/On - If set to Off (default), the preheat temperature is applied in only the first load step. On maintains the preheat temperature throughout the complete build process.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Heating Method: The method used to heat up the material. Choose Power Based or Temperature Based (default).

Power Based: Heating is based on laser power and material absorptivity.

- **Power**: The power of the laser in Watts.
- **Absorptivity**: The average fraction of energy that is absorbed by the deposited material and contributes to the heating process. Value must be between 0 and 1. Defaults to 0.4, or 40%.

Temperature Based: Heating is based on the process temperature.

- **Process Temperature**: The temperature of the feed material as it is exiting the nozzle—usually melting temperature or above. When using the DED Process wizard, this value automatically defaults to the build material's melting temperature.

Room Temperature: Temperature of the gas in the build chamber during the build.

Gas Convection Coefficient Build: Convection coefficient from the part to the surrounding gas in the chamber during printing. Convection is applied to the element surfaces of every element cluster, which can be especially relevant in the case of a bulky structure.

Gas Convection Coefficient Base Plate: Convection coefficient from the base plate to the surrounding gas in the chamber during printing.

Radiation: Off/On - If set to Off (default), the radiation effect between the build/base and surrounding environment is not considered during printing. Setting the option to On enables the radiation effect between the build/base and surrounding environment during printing.

- **Emissivity Build**: The emissivity of the part during printing.
- **Emissivity Base Plate**: The emissivity of the base surface during printing.

Thermal Calibration: Off/On - If set to Off, default heating and cooling behavior is used for cluster deposition. Setting the option to On allows you to change the Cluster Cooling Time Ratio, essentially using it as a calibration factor.

- **Cluster Cooling Time Ratio**: A value between 0 and 1 (default).

  The total deposition time, t, for each cluster is determined by the volume of the cluster and the deposition rate. By default, a very tiny fraction of this total time (1e-6 seconds) is used to raise the temperature to the Process Temperature, and the remainder is used for cooling. Use the Cluster Cooling Time Ratio to adjust the heating and cooling distribution. Lowering this value from the default of 1 has the effect of lowering cooling time, thereby increasing heating time. For example, a value of 0.8 means 20% of time, t, is used for heating and 80% for cooling (during printing). The lower the number, the higher the temperature spikes that will exist in the thermal history across the part.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Cooldown Conditions	The environment in the build chamber around the part in the cooldown step after the last layer is printed. **Cooling Time**: The duration of the cooldown process, in seconds. **Number of Cooling Steps**: The number of substeps within the cooldown calculation. Defaults to 10. **Room Temperature**: Temperature of the gas in the build chamber during cooldown. **Gas Convection Coefficient Build**: Convection coefficient from the part to the surrounding gas in the chamber during cooldown. **Gas Convection Coefficient Base Plate**: Convection coefficient from the base plate to the surrounding gas in the chamber during cooldown. **Radiation**: Off/On - If set to Off (default), the radiation effect between the build/base and surrounding environment is not considered during cooldown. Setting the option to On enables the radiation effect between the build/base and surrounding environment during cooldown. • **Emissivity Build**: The emissivity of the part during cooldown. • **Emissivity Base Plate**: The emissivity of the base surface during cooldown.
Cooldown Between Layers	Options to automatically adjust the dwell time between layers as measured after the last cluster in a layer is deposited. **Cooldown Method**: Choose Time Based or Temperature Based. **Time Based**: The dwell time between layers is read from the Cluster Settings table (p. 84). **Temperature Based**: Uses a Target Temperature together with time step and convergence options to automatically adjust dwell time. As soon as one of the convergence options is achieved, deposition of the next layer is started. When this method is chosen, the Dwell Time column is set to Off in the Cluster Settings table (p. 84) before and during the transient thermal solution. Once the thermal solution is completed, the table is updated to reflect the calculated dwell times. • **Target Temperature**: Temperature to cool down to before printing the next layer. • **Time Step Size**: The time increment, in seconds, before each solve command during this iterative cooling process. • **Maximum Iterations per Layer**: The maximum number of solve commands/iterations per layer. Defaults to 5. • **Convergence Threshold**: Percent change in maximum temperature from this iteration to the previous iteration. If the change in maximum temperature in the layer is less than or equal to this value, the layer cooldown is considered converged. This is another way of controlling the number of iterations per layer. Defaults to 0.1, or 10%.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

7.2.2. Tree Dependencies

- **Valid Parent Tree Object**: AM Process for DED (p. 109)
- **Valid Child Tree Objects**: None

7.2.3. Insertion Methods

- Inserted automatically when the AM Process for DED (p. 109) object is added.

7.2.4. Right-click Options

Relevant right-click options include:

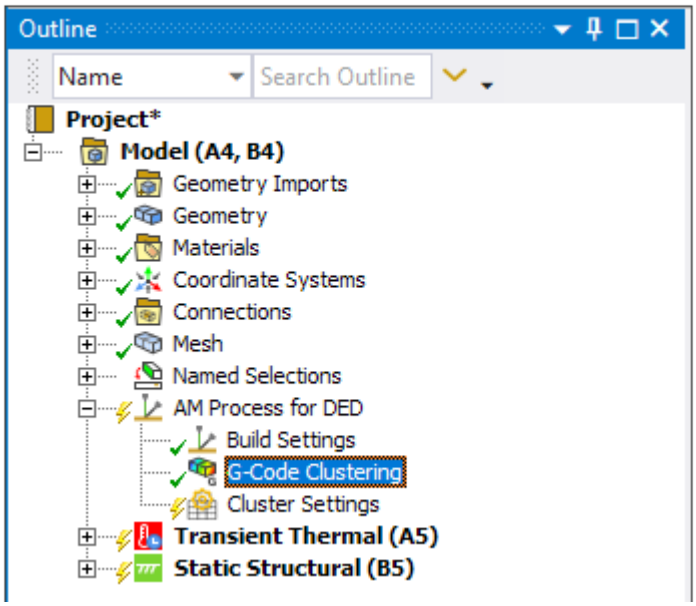
- **Save Build Settings**: Saves the configuration of your build settings as an XML file. The property values are always saved in Consistent MKS Unit System.
- **Load Build Settings**: Imports a build settings XML file. Once loaded, the settings populate the properties of the Details pane. Sample build settings XML files are available in: [Ansys Installation Directory]\aisol\DesignSpace\DSPages\SampleData\AdditiveManufacturing.
- **Reset to Default**: Resets build settings to default values.

7.2.5. Additional Related Information

Define Build Settings (p. 57)

7.3. G-Code Clustering

Controls element clustering using a G-Code file in a DED process simulation.



Details (p. 117)

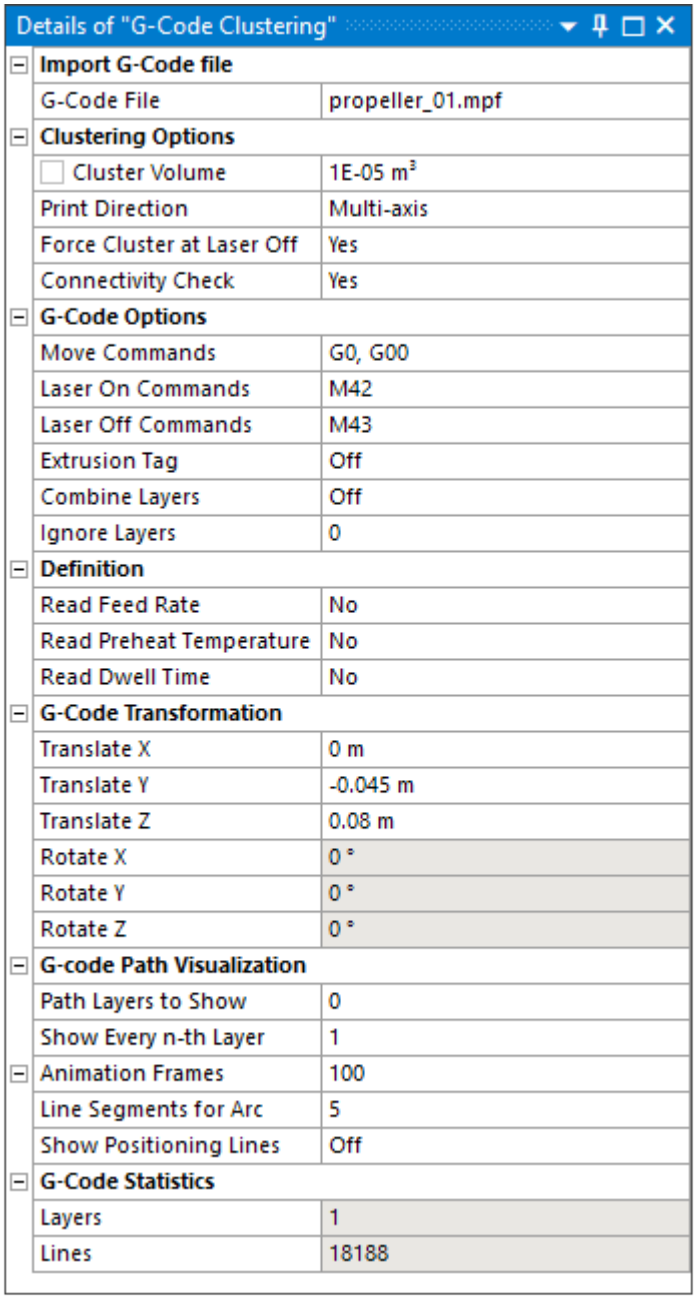
Tree Dependencies (p. 120)

Insertion Methods (p. 120)

Right-click Options (p. 120)

Additional Related Information (p. 121)

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Details of "G-Code Clustering"	
Import G-Code file	
G-Code File	propeller_01.mpf
Clustering Options	
Cluster Volume	1E-05 m^3
Print Direction	Multi-axis
Force Cluster at Laser Off	Yes
Connectivity Check	Yes
G-Code Options	
Move Commands	G0, G00
Laser On Commands	M42
Laser Off Commands	M43
Extrusion Tag	Off
Combine Layers	Off
Ignore Layers	0
Definition	
Read Feed Rate	No
Read Preheat Temperature	No
Read Dwell Time	No
G-Code Transformation	
Translate X	0 m
Translate Y	-0.045 m
Translate Z	0.08 m
Rotate X	0 °
Rotate Y	0 °
Rotate Z	0 °
G-code Path Visualization	
Path Layers to Show	0
Show Every n-th Layer	1
Animation Frames	100
Line Segments for Arc	5
Show Positioning Lines	Off
G-Code Statistics	
Layers	1
Lines	18188

7.3.1. Details

Category	Description and Settings
Import G-Code File	Identification of G-Code file. **G-Code File**: Name of the file containing the G-Code. The G-Code file must be a flat file with ASCII characters.
Clustering Options	Options to control the size of the element clusters, print direction, and clustering technique. These options have an influence on simulation time and accuracy. **Cluster Volume**: Volume, in mm^3, of each cluster. This value determines how many elements are activated per load step, and the time for this load step is

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

	then determined by volume/deposition rate. A smaller cluster volume tends to provide a more accurate result. **Print Direction**: The direction of increasing layers in which the part is to be built. Choose X, Y, Z, -X, -Y, -Z, or Multi-axis for the print direction. Defaults to Z direction. See Selecting the Right Print Direction (p. 66) for guidelines. If Multi-axis, additional options are: • **Force Cluster at Laser Off**: Yes (default) or No - Controls whether or not a cluster is forced at the end of each layer end. If No, clusters are generated based on volume only. • **Connectivity Check**: Yes (default) or No - Checks for and corrects badly connected elements if they exist. A badly connected element is one that: a) is not face-connected with any other elements of the current cluster and b) is not face-connected to any activated elements and c) is not in contact with the base. The connectivity check occurs during clustering and may cause the operation to take longer. Consider setting it to No to save time when working on preliminary simulation setup. Set Connectivity Check back to Yes for your final (non preliminary) simulation.
G-Code Options	Controls for interpreting the G-Code. **Move Commands**: Defines the G-Code commands for pure movement. Multiple commands can be specified using comma separation, such as *G00, G01*. **Laser On Commands**: Defines the G-Code command(s) for switching the laser to on (Laser On). All following move commands are used for element clustering until a Laser Off command is found. Multiple commands can be specified using comma separation, such as *M24, M174*. **Laser Off Commands**: Defines the G-Code command(s) for switching the laser to off (Laser Off). All following move commands are not used for element clustering until a Laser On command is found. Multiple commands can be specified using comma separation, such as *M28, M175*. **Extrusion Tag**: On/Off - G-Code commands may be extended by an extrusion tag "E." This can be used to additionally control if material is deposited or not. With the Extrusion Tag option set to On, only commands with the "E"-tag are considered for subsequent element clustering. Default setting is Off. **Combine Layers**: On/Off - For multi-body parts, named selection folders from the different bodies that represent clusters at the same build-direction height can be combined. Useful for reducing the number of named selections but does not affect the element clustering itself. If Off (default), layers are not combined. **Ignore Layers**: Layers to be excluded from element clustering. Multiple layer numbers can be excluded using comma separated values. For example, *2,3* ignores the layers two and three from subsequent element clustering, so the clusters are divided into two instead of four layers. The default is 0, which includes all layers for element clustering (that is, 0 exclusions).
Definition	Sets advanced controls for Feed Rate, Preheat Temperature, and Dwell Time.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Read Feed Rate: Yes or No. If Yes (default), feed rate commands will be read from the G-Code file and used in the simulation.

- **Feed Rate Command**: Defines the G-Code command for switching to a new feed rate. Defaults to F.

Read Preheat Temperature: Yes or No. If Yes (default), preheat temperature commands will be read from the G-Code file and used in the simulation.

- **Preheat Temperature Command**: Defines the G-Code command for switching to a new preheat temperature. Defaults to T.

Read Dwell Time: Yes or No. If Yes (default), dwell time commands will be read from the G-Code file and used in the simulation.

- **Dwell Time Command**: Defines the G-Code command for switching to a new dwell time. Defaults to G4.

G-Code Transformation	The G-Code path must be correctly aligned on top of the build part to ensure proper element cluster generation. Use these translation and rotation offset options to position the G-Code path in relation to the build part, if necessary. Rotation is allowed only in the printing direction, so once the Print Direction field is specified, the rotation options for directions other than Print Direction are grayed out. See more details about G-Code alignment (p. 67). **Translate X**: Defines the offset in the print direction with respect to the actual X, Y, or Z positions defined in the G-Code file. **Translate Y**: Offset in the Y direction. **Translate Z**: Offset in the Z direction. **Rotate X**: Defines the rotation about the X axis positions defined in the G-Code file. Rotation is only allowed in the printing direction. **Rotate Y**: Defines the rotation about the Y axis positions defined in the G-Code file. Rotation is only allowed in the printing direction **Rotate Z**: Defines the rotation about the Z axis positions defined in the G-Code file. Rotation is only allowed in the printing direction
G-Code Path Visualization	Controls the appearance of the G-Code path in the geometry window when the Show Path button  in the DED Process ribbon is on/pressed. The path is redrawn any time one of the options is changed. Visualization settings have no influence on element clustering itself. **Path Layers to Show**: Path layers from the G-Code file to display. Enter a range of values with a hyphen and multiple values separated by commas. For example, *1-10, 25-30* displays layers 1 through 10 and 25 through 30. The value 0 displays all layers and is the default setting. **Show Every n-th Layer**: Displays every *n*-th path layer of the designated Path Layers to Show. For example, the value 2 displays every second layer, 3 displays every third layer, and so on. The value 1 displays all layers designated in Path Layers to Show and is the default setting.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

	Animation Frames: Animation frames in which the designated Path Layers to Show are rendered. Click the Show Path button  to render the animation. The value 1 displays all Path Layers to Show in a single animation frame, that is, all at once. The value 2 divides the Path Layers to Show across two animation frames, rendering half of the layers first and then the other half. The greater the amount of Animation Frames, the clearer the G-Code path progression will be. The default setting is 10. Values greater than the total number of layers in Path Layers to Show are rendered with Animation Frames equal to the total number of layers in Path Layers to Show. Because the animation rate is fast, this option is most useful when visualizing large models and is less useful when visualizing smaller models. **Line Segments for Arc**: The graphical display of an arc path is approximated by a number of lines in the geometry window. The number of lines per arc can be modified, a useful option for G-Code files with many arcs in order to speed up the graphical display. The default number of line segments is 5. **Show Positioning Lines**: On/Off - On (default) displays the rapid movement commands in the G-Code. Off suppresses the display of rapid movement commands.
G-Code Statistics	Read-only indications after generation of element clusters. **Layers**: After generation of element clusters, shows the number of path layers in the part. This number includes the rapid movements without material deposition. **Lines**: After generation of element clusters, shows the number of lines in the weld path.

7.3.2. Tree Dependencies

- **Valid Parent Tree Object**: AM Process for DED (p. 109)
- **Valid Child Tree Objects**: None

7.3.3. Insertion Methods

- Inserted under the AM Process for DED (p. 109) object in the project tree automatically by the DED Process Wizard if G-Code is selected.
- Select the **DED Process** tab and then click the **G-Code Clustering** button in the ribbon.

7.3.4. Right-click Options

Relevant right-click options include:

- **Generate**: Generates the element clusters using the path defined in the G-Code file.
- **Clear Generated Data**: Clears any generated clusters.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

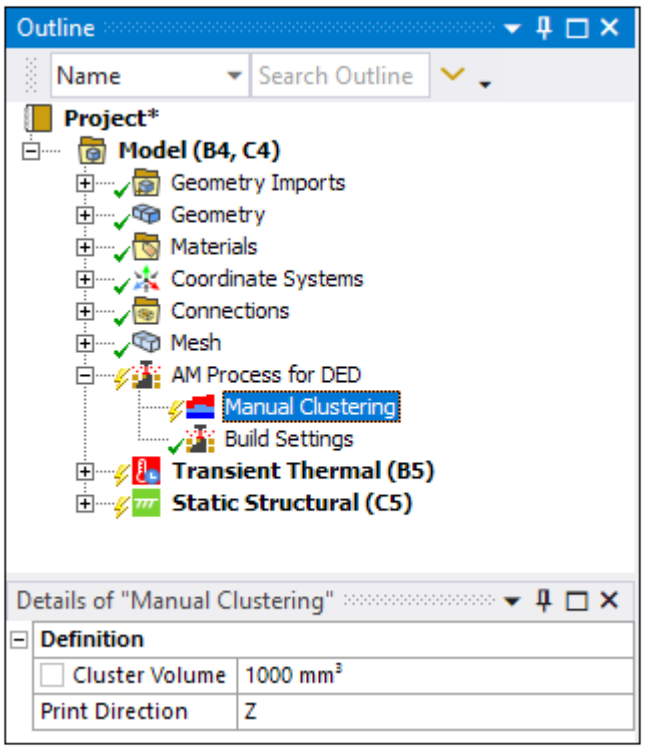
- **Open DED Clustering Directory**: Opens the file directory where G-Code cluster-related files are stored. You can edit the `DEDClusterOutput.json` file, which is the output file from the cluster generation process. After editing, click Update Clusters and Settings.
- **Update Clusters and Settings**: Regenerates the Named Selections and updates cluster settings but does not rerun the clustering algorithm.
- **Show G-Code Path**: Displays the G-Code path (p. 63) on the geometry.
- **Clear G-Code Path**: Hides the display of the G-Code path on the geometry.

7.3.5. Additional Related Information

Element Clustering: G-Code Approach (p. 62)

7.4. Manual Clustering

Controls element clustering using a manual clustering technique in a DED process simulation.



Details (p. 121)

Tree Dependencies (p. 122)

Insertion Methods (p. 122)

Right-click Options (p. 122)

Additional Related Information (p. 122)

7.4.1. Details

Category	Description and Settings
Definition	Controls the size and direction of the element clusters. **Cluster Volume**: Volume, in mm^3, of each cluster. This value determines how many elements are activated per load step, and the time for this load step is then

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

determined by volume/deposition rate. A smaller cluster volume tends to provide a more accurate result.

Print Direction: The direction of increasing layers in which the part is to be built.

7.4.2. Tree Dependencies

- **Valid Parent Tree Object**: AM Process for DED (p. 109)
- **Valid Child Tree Objects**: None

7.4.3. Insertion Methods

- Inserted under the AM Process for DED (p. 109) object in the project tree automatically by the DED Process Wizard if Manual Clustering is selected.
- Select the **DED Process** tab and then click the **Manual Clustering** button in the ribbon.

7.4.4. Right-click Options

Relevant right-click options include:

- **Generate**: Generates element clusters.
- **Clear Generated Data**: Clears any generated clusters.

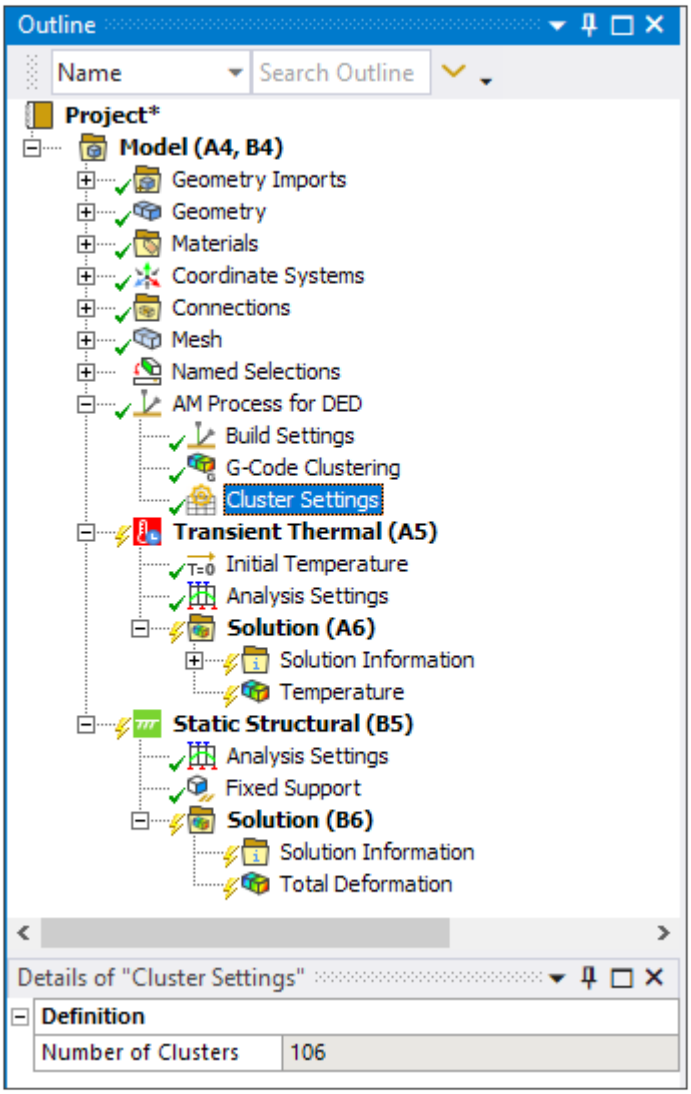
7.4.5. Additional Related Information

Element Clustering: Manual Approach (p. 60)

7.5. Cluster Settings

Table for importing, exporting, and modifying machine parameters for clusters in a DED process simulation.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



Details (p. 124)

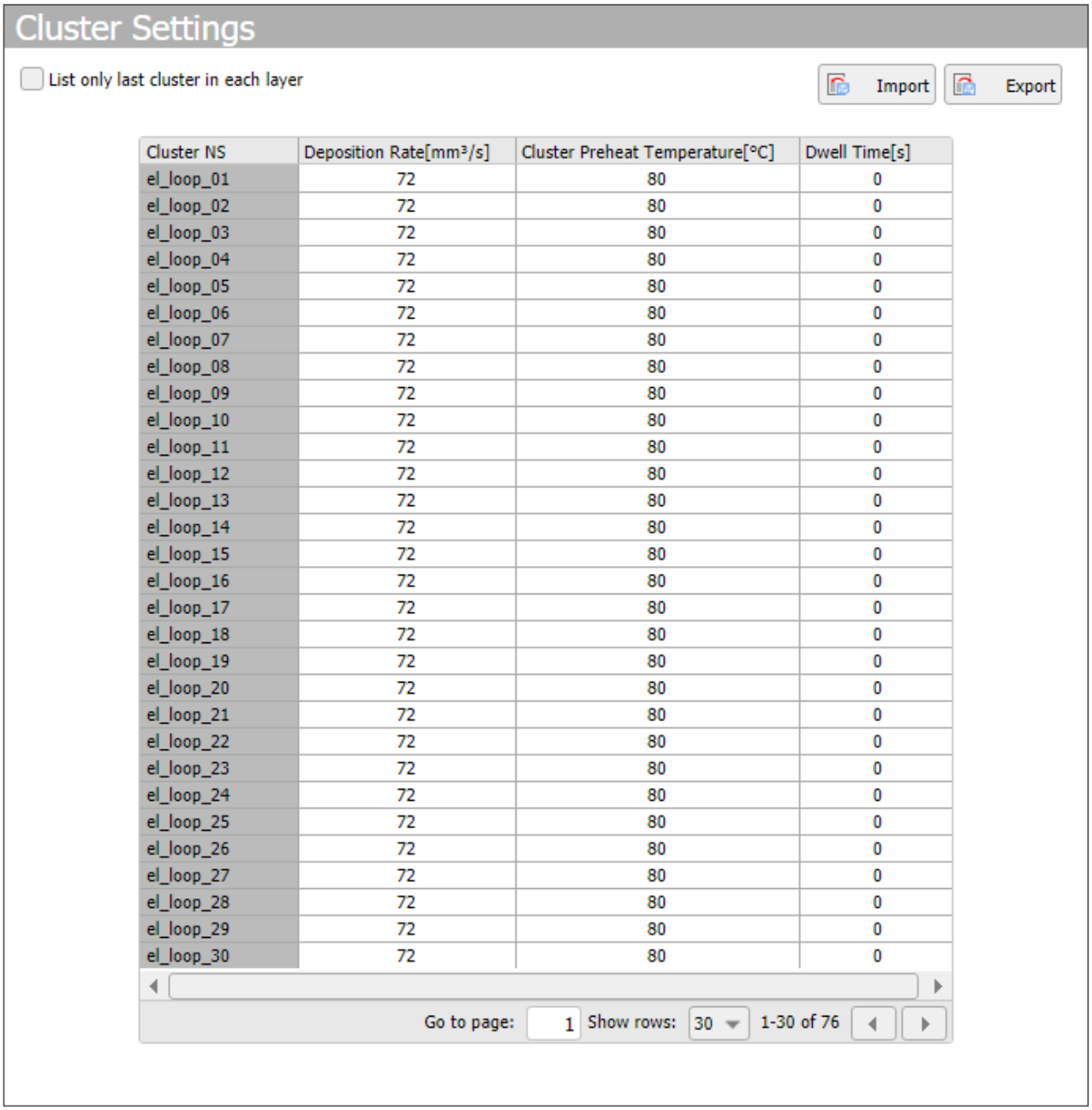
Tree
Dependencies (p. 124)

Insertion
Methods (p. 125)
Right-click
Options (p. 125)

Additional
Related
Information (p. 125)

The Cluster Settings table below appears in the Worksheet area of the UI when the Cluster Setting object is selected.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.



7.5.1. Details

Category	Description and Settings
Definition	**Number of Clusters**: Read-only indication of the number of clusters in the model.

7.5.2. Tree Dependencies

- **Valid Parent Tree Object**: AM Process for DED (p. 109)
- **Valid Child Tree Objects**: None

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

7.5.3. Insertion Methods

- Inserted under the AM Process for DED (p. 109) object in the project tree automatically when either the Manual Clustering (p. 121) or G-Code Clustering (p. 116) object is inserted. (Manual Clustering and G-Code Clustering are inserted either by the DED Setup Wizard or by manual insertion.)

7.5.4. Right-click Options

Relevant right-click options include:

- **Update Data for Simulation**: Applies any modifications from the Cluster Settings table to the simulation.

7.5.5. Additional Related Information

Using the Cluster Settings Table (p. 84)

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.

Release 2026 R1 - © ANSYS, Inc. All rights reserved. - Contains proprietary and confidential information
of Synopsys, Inc., ANSYS, Inc., subsidiaries and affiliates.